

>SRS014923.103535-NL *Faecalibacterium* sp. CAG:74

MSDRTFATHQIRRAVPVPTLWNFAALDTPVPARLPVPSAWELVPALHNYRGRAHYTTIRIRCGGNIRLS
FGGVSFRRARVLLDGTEIAQHYGAFTAFETVVRRLPEGEHTLTVEVDNRFGEDSALHVPNDYYSYGGITRP
VVIEQLPDVDFVERVGITTKRAADGWTADISANIRNLADEADASADVTLTLAGQTFTQTAHIPADSTAIICI
SDAHYADVTAWTPDTPALYALRAEIREGNQVRDDLIDRVGFREISISGMDLLLNGEKLRLMGFNRHEEY
AFGCAVPLQAMAQDILMMKDMGCNCVTRCHYPNAPRFLDLCEMGLLVWEEAHARGLQEEQMRNPFPMPQ
TRQCVREMVAQHRNHPSIFIWGCNECADNCDYGADCYREYVALLHDLDASRPMTAALLERPGSRVYGDS
DVVSVNIYPQWYHNTTVAESLAQKLKEIREHGGAGKPVIISEIGAGGIYGYHDPLGESKWSEERQCTILR
DQVEAVLKNPACSGVFLWQFADCRVTEEWAMHRPKTHNNKGVVDEYRRPKRSYAVVKELFQKQKG

>SRS051031.100709-NL *Clostridium* sp. CAG:75

MIRTFKTHEVRMQEELTGSLWEFEPCEGDYAGTKFQMATPGCFETNPLFANYRGEAYFRKKITAGGNIRI
ECKGVSHSTATVLLDGEQIAQHYNAYTPFSVIKKDLTEGTHLLEIKVDNRFTEKSALHIPNDYMSYGGVSR
PVVLEKIGALYFEYVHVKTYHKNGKWTAGVKVAVHILDQKRLIGENVVISLQIGEVIYDWQLLSDKGDRQ
ILSTELIFEDVQEWSPDSPKLYEVKLQAKVGEIVMDDLIDRVGFREIKVQGKEILLNEKKFRIKGVCRHE
DHPDYGCALPYAAICHDLMLIQQMGANSIRTVHYPNDEIFDLDCDELGILVWEENHARGIEEPEMRHPLF
ETQAENVIREMIMNHYNHPSIYIWGILNECASETEFGRSYQKQFALIRQLDATRPCTFASCKFFQDICF
DLPDVISCNIYPRWYVDKPVQQYLAEVCWDIKEDGKGKDKPFIVSEIGAGALYGCHNSYHGKWTEEYQAE
ALAEQLTACLESSECMGVYIWQFCDVRVSSEWFAGRPREMNNKGIVDEYRRPKLAYEKVKEIFQKY

>SRS052697.294668-NL *Roseburia* sp. CAG:100

MIRLFEQHNIRKQRELDGIWSFITEEGAERYIPVPGCWEQNPELLNRYRGKGTYTRKVYIQEKTNIRLEFK
GVSHATADVYFDEFIVHHYNAFTAFCGIIMNVEPGEHVIRVEVDNSFSEKSSLHVPNDYTYGGFNRTVV
LEEIKDVYLKNIHFEPHKDKDGWKAKISIDVENLSDEEKNAEVTITLADKIFHVNGLLLEGKSI AILELVE
SFENVREWFVLKPELYLLKAEKCDGVVMDLTDVRVGFREVEVKGRLLLNKGKSVFLKGFNRHEDYGTG
SAVFFQVMMQDLDMREMGANAVRTHYPNDERFLDLCEKGMVLVWEENHARGFDLEKMMNPHFDEQCRD
CIDEMIYQHFNHPSIYIWGILNECASETEEGRQKYKMQYEQIKSLDTSRPTTSATCRHFKDICLDLPDIV
SFNMYSGWYEDISVKEKHEKEMAWIDA AVEKEKPVIVSEFGAAAIYGFRDRGHCKWSEERQADVIRENLE
VYKEDDRLTGTFIWQFADCKVTEEEWFSTRARCHNNKGVVDEYRRPKAYDVVKEYFNQMPGYDEL

>SRS012273.107272-NL *Fusicatenibacter*

MIRTFETHKIRKTAELSSALWNFHTIGTQGEAAVIAQAPVPGCWENYPDTVSYRGQASYSREFEAKGNIRL
EFKGVSHATASVLVDGKPVGSHYNAYTPFDVVLKDIRPGIHQLEVIADNSFGPDSALHVPNDYQSYGGISR
GVVLEELGEAYLSWIHFTPFRLKDGWYGKAEICVRNLSSGRLDGVSVEVEIGKNSFAVLPIVLEGEEEKSF
STEELPCPWAECWSPSPVLYLITAVLRTADGAADDIDRVGFREIRTEGKDILLNGRKLRIKGFRCRHE
HPQFGCALPFSAMQHDLMMLIKDLGANSIRTVHYPNDELFLDLCEQILVWEENHARGLSEENMRNPHEK
QQCGDCIREMITAHYNHPSIYIWGILNECASDTEYGRECYSEQYELIKSLDPYRPRSSASCRTDCLG
YPEVVSYNIYPKWYHDPVEDYLDLEYQWIQNESEGTGKPFILITEIGAGAIYGYRTPAHVKWSEEYQVQA
LKEQLQAVFSREGCSGVYIWQFCDVRVCDSWFGSRPRTMNNKGIVDEYRRPKLAYEVVKDSYRSLGNYF

>SRS013687.96337-NL *Butyrivibrio* sp. CAG:318

MIRVFSTHNIRECKELTGRIWNLRLFNGTDDGKNYSVFTPCCWESIPDLAAYRGKGEFSTSIFYAGGNIRL
EFKGVSHATATVYVDDCEVVTHYNAYTGFDVILKGLDEGIHTLTVVADNSFSKQSALHIPNDYKSYGGVTR
PVVLEKLNLDLYIRQMHVRTENDGTGRWMAHVEIKVQNLSDRKQAALFISIDKDEKIVGCENEISIEAG
QEITIVRDLSCGVSEWTQSSPELYMINAELYVNGKAEDDLTDVRVGFREIVVEGSKILLNGKPVRIKGV
RHEDHPQYGCALPCEAMAYDLTLIKDLGANSVRTTHYPNDERFLDMCEMGLVWEENHARGLNEESMMN
PFFEPQAEQVITEMIESHYNHPCIIYIWGILNECASETEYGRKCYAAQFELIRKMDTTRPCSFACRIKND
ICLDLPDVVSFNIYPQWYLDIPVKEYLDDLYEWIQDDTPGTGKPFILITEIGAGAIYGYRTPAHVKWSEEYQVQA
QAYALEKQLTSILANS GCMGVYIWQFCDNRVSEECFAGRPRTMNNKGIVDEFRRHKLSYDVVKRIYQSDI

>SRS023526.251759-NL *Alistipes senagalensis*

QDLIANVYGREMQLNGKWDVAVDLYDQGRKNKIYLNKPEGKTDIFYEYAFVNGLRDLVPSDWNSQTPEL
KYYEGTVWYARRFDAPRQDGRRLFLHFGAVSYRCRVYLNNGEEIGSHEGGFTPFQIEVTGKLKPADNFLAV
EVNNTRTPD AIPAMAFDWWNYGGITRDVSLVCVPEVFVEDYFIRLDKDSADRISADVRLSGAPAGTSVLV
EIPELKVRETASADERGAAALSFRVKNLKRWSPEDPKLYRVIVSAGRDRVEEMIGFRNVRVEGEDIYLN
APVFLKSVSFHEEIPQRMGRAFSESDAVMLLSEARALGCNMIRLAHY PQNEYIVRLAEKMGFMLWEEIPI
WQGIDFANDATRQKAGRMIREMVGDRNRNRCALTFWGVANETQPSERNEFLRYLISCCRELDDTRLIVAA
FDLVRFRDRDRQLFVMDDPFIEALDVAVNKYMGWYHPWPVTPDRAVWDVARGKPLIISEFGGEALYGQHG
AADTASSWSEYQEALYRDNLRMFENIPNLRGVSPWILFDFRSPFRFHTNQEGWNRKGLVSDRGYRKA
WYLMKAYYDKR

>SRS011405.40805-NL *Bacteroides intestinalis* CAG:564
QFAMTNVYGRKYVLLNGKWDVIIDPYLHGKDSIYKDKPLKSKSDFKEYAFEGGVRLNVPGDWNSQYPEL
KYYEGTVWYARHFSIKRSSDENVFLYFGAVNYQCTVYLNNGTKVGEHEGGFTPFQINITDRIVSGDNFLVL
EVNNYRNKDAIPALLYDWWNYGGITRDVMIKTPQLYIDHYFIQLDKYKPDLVHVQVQLSEKRVNENIRI
TIPALNVLRQLSTDSTGMAQTSIKVKRLQRWSPQTPQLYLVQLATRSDTIKEELGFRNLYVKGKQIYLN
SPVFMKGISFHEEIAQRQGRAFSQQDAVALLSEAKALGANLIRLAHYPQNEYIVRLAEKMGFMLWEEIPI
WQNIDFTSKATLQKALTMHSEMIMRDRNRCALTFWGIANETSLSKSRNSFLKEIINNCCQKMDTTRLLTAA
FDKVKWNGDTQTFELEDTLTEYLDVVSINKYMGWYHPWPIKPISEISWNICPDKPLFISEFGGEALYQSG
DATVASSWSEEYQERLYKDNLEMFISIPNLGTGIAPWILFDFRSPYRFHPLNQNGWNRKGLISDQGYRKA
WYVINEFYKNINF

>SRS015663.263152-NL *Bacteroides finegoldii*
QNLITNVYGRDIRSLNGKWNVIIDLYDQGRGMKVYRNQSPKGATDFYEYSFKGGLRLNVPGDWNSQMP
KYYEGTVWYARHFDKRLVDKRQFLYFGAVSYRCRVYLNNGTEIGSHEGGFTPFQIEVTDLLNEGDNFIVV
EVNNRRTKDAIPAMSFDDWWNYGGVTRDVLVLTTPQTYLEDYFIQLDKESPDQMIKVLHSDKKPGEKITI
AIPELKTSIDMQTDAEGKAEGKFNVKKLQRWSPENPKLYEVISSAGDCVKEQIGFRNITVKGTDIYLN
KPTFMCSISFHEEIPQRMGRAFSEADAAMLLDEAKALGVNMIRLAHYPQNEYTVRLAEKMGFILWQEIP
WQIDFTDNTRKKAQRMLTEMIKRDQNRCAVGWYWGIANETQPSKERNEFLTSLLETGKKLDTTRLYVAA
FDLVRFNREKKCFVMEDSFTSQLDVAVNKYMGWYHPWPIEPENAVWEVDPDKPLIISEFGGEALYQSG
DENVASSWSEEYQVRLYRDNIRMFNDIPNLRGVSPWILFDFRSPFRFHTNQGWNRRKGLISDQGMRRKA
WYLMKDYKTKAGE

>SRS022137.27111-NL *Bacteroides multispecies*
QNLITNVYGRDIRSLNGKWNVIIDLYDQGRGMKVYRNQSPKGNTDFYEYSFQGGGLRLNVPGDWNSQTPEL
KYYEGTVWYARHFDKRLTHKRQFLYFGAVSYRCRVYLNNGAEIGSHEGGFTPFQIEVTDLLNEGENFIAI
EVNNRRTKDAIPAMSFDDWWNYGGITRDVLVLTTPQTYLEDYFIQLDKESPNRMIKVALSDKKAGEKITV
SIPELKTSIDMLTDAEGKAETVFNKKLERWSSNPPLYEVIVSSANDRVEEQIGFRNITVKGTDIYLN
KPTFMCSISFHEEIPQRMGRAFSEADAAMLLNEAKALGVNMIRLAHYPQNEYTVRLAEKMGFILWQEIPV
WQIDFTNNTRKKAQRMLSEMIKRDQNRCAVGWYWGIANETQPSKARNEFLTSLLETGKQLDTTRLYVAA
FDLVRFNREKKRFVMEDSFTSQLDVAVNKYMGWYHPWPIEPENAVWEVDPDKPLIISEFGGEALYQSG
DENVASSWSEEYQARLYRDNIRMFNDIPNLRGVSPWILFDFRSPFRFHTNQGWNRRKGLVSDQGIRKKA
WYLMREYKTKFGE

>SRS016203.126554-NL *Bacteroides ovatus*
QELITNVYGRNIHSLNGKWNVIIDLYDQGRQMKIYENRQPEGNIDFYEYAFEGGLRLNVPGDWNSQSPEL
KYYEGTVWYARHFDKRLADKRQFLYFGAVSYRSKVYLNNGKEIAEHEGGFTPFQVEVTDLLKGDGDNFLAV
EVNNRRTKDAIPAMAFDDWWNYGGITRDVLVKTPTPTFIKDYFIQLDKNTPDRIIARVRLSDKKAGEKVT
AIPELKINAELTTDAEGKAETVLNAKRLQRWSPPEEPKLYGVTVSSADRVEEQIGFRNITVKGTDIYLN
KPTFMCCISFHEEIPQRMGRAFSEADAAMLLNEAKALGVNMIRLAHYPQNEYTVRLAEKMGFLLWQEIP
WQIDFTDNTRKKAQRMLSEMIKRDQNRCAVGWYWGIANETQPSKERNEFLTSLLETGKQLDTTRLYVAA
FDLVHFNSEKQRFVMEDSFTSQLDVVAINKYMGWYHPWPVEPKDAIWEVTDKPLIISEFGGEALYQSG
DENVVSSWSEEYQARLYRDNIRMFNDIPNLRGVSPWILFDFRSPFRFHTNQGWNRRKGLISDQGMRRKA
WYLMRDYYMKRRNR

>SRS024132.96578-NL *Bacteroides multispecies*
QDLSNVYGRDYQLLNGKWGAIVDLYDQGERMQVYKNRKPQDNKQFYEYSFDGGLVLNVPADFNSQMP
KFYEGTVWYGRSFDTPKTDGKRLFLYFGAVSYRSKIYLNNGEKIASHEGGFTPFQVEITDKVKQKDNFLV
KVNNRRTTDAIPAMSFDDWWNYGGITRDVMLVSVPSYIKDHFQILDKVQSNLIRAKVTLSEAKPGINVQV
AIPELGIRQNMTTDANGIAQATIKTKKLQRWSPDAPKLYDVTITSESDRLNEQIGFRNLYVKGEDYIYLN
QPIFLRSISFHEEIPQRRGRAFSEADATMLLSEAKELGANMIRLAHYPQNEYTVRLAERMGFLLWEEIPI
WQIDFKNTATREKAGRMKEMVMRDKNRCSVAFWGIANETATSAPRNEFLKHKMQCCLDIDSTRLITAA
FDLVRFNRESRDFEMNDSIINILDMVAINKYMGWYHWDPLAPDQAVWNVAPGKPLFISEFGGEALYKKG
DAEVKSSWSEDYQAQLYKDNLEMFKHIPNLRGTSWPWILLDFRSPFRFHPHQGGEWNRKGLVSDQGGRRKA
WYIMRDYYNQKTEQ

>SRS063985.97426-NL *Alistipes shahii*
AQELITNVYGRDVQSLNGKWDAIVDLYDQGRKNKIYLNRRPEGKTDFFEYSFDNGLRLDVPDWNQMP
LKYYEGTVWYARRFDAPEQTGRQFLYFGAVSYRCRVYLNNGEEIGAHEGGFTPFQIEVTGRLRPEGNFLA
VEVNNRRTTDAIPAMAFDDWWNYGGITRDVMLVSVPAFIRDYFIRLGKDSDDRIDADVQLAGAAAGTAVR
IEIPELVNDRVTAGATGTASASFRVRLQRWSPADPKLYRVIVSTDRDRVEESIGFRNIRTEGEEIVLN

GSPVFLKSISFHEEIPQRMGRAHSEADAVMLLSEAKALGCNMIRLAHYPQNEHIVRLAEKMGFLLWEEIP
IWQGIDFANDPTREKAGRMIREMVTDRKNCALTFWGVANETQPSGPRNAFLRHLIACCREIDTRLIVA
AFDLVRFDQRQLFVMDPPFIGELDVVAINKYMGWYHPWPLTPDKAVWDVARGKPLIISEFGGEALYGQH
GPADVASSWSEEYQEALYRDNLRMFENIPNLRGVSPWILFDFRSPFRFHPANQEGWNRKGLVSDQGGRKK
AWYLMKAYYDKIKRR

>SRS051031.173469-NL *Clostridium* sp. CAG:253

MIRTFRTHNIRKQQELTGRYWEFSACSGEYAGEKFNVLTPCCWESHKPFAGYRGEGEYTTTFETGGNIRI
EFKGVSHTAEVYLDGKKITEHYNAYTPFDTVCHNLPYGEHTLMVKADNRFSKKSALHIPNDYMSYGGVSR
PVVLENISDVYIKNVHVTPYKENDKWRAKIEIYIENIGVAGETENS SDDIMKSKTEEIDKNKSENINIE
INTAGKHIILENIDTLRQNKVTTEIEFESVREWMQETPNLYEIEVKLLRDKNADFDDFIDRFGFREIKIAG
KEIHLNGRKIRIKGVCRHEDHPQFGCALPMAAIAADIELIKDMGANSIRTSHPNDEIFLDICDEQGILV
WEENHARGLSEADMNPNFEPQAEKVIEEMIPAHYNHPSIYIWGILNECASETEYGKECYKKQFDIIRKL
DTSRPCSFSACKFKTDVCLGLPDVVSYNIYPLWYHDTAPDEYLDALYKWWQNETEGAGKPFMITEIGAGG
LYGYRNNDYDSKWTEEYQAKALEKQLTAVLDYKDCIGVYIWQFCDIRISDEWFGGRPRTMNNKGIVDEFRR
KKLAYNVVKKIFGEKM

>SRS019910.125985-NL *Prevotella* sp. CAG:732

ASTYLANVYGRQYELLNGKWNAPDLYERGEEMKLYLDQKPQGKTDFFEYSFAGGLRLNVPGDWNSQNAE
LKYFEGSVWYARHFNVNKQGMKRQFLCFGGVSNRCKVYLNKGKIGEHEGAFTPFQMEVTNALRQGDNTLI
LEVNNRRMKSAPALSFDWWNYGGITRDVMLVSVLPQLYVSDYFIQLDKHRADLIHVTVNLSDALANQAVS
VEIPELKKKLNLTNADGEAKGMLKVSHLQWRNPENPKLYQVRIKSGSDEVTEIDIGFRNIEVKGTKVYLN
GKQTFILRSISFHEEIPQRMGRACTEGDADMLLAEAKALGVNMIRLAHYQQNEYIVRKAEQMGMLWQEIP
VWQAIDFSNQETLQKAGMLTETIKRDKNRCADCFWGIANETRPSEARNAFLSELLQTGKMDTTRLFVA
AFDNVYFKEDSQLFEMEDSFVNQLDMIGVNKYMGWYAPWPLKPAECVWKVAVNKPLLISEFGGEALYGQH
GDETIASSWSEEYQAKLYRDNLRMFENIPNLVGVSPWVLFDFRSPFRFHPNQGDNWRKGLVSDQGGRKK
AWYVMHDYHQQKQKEYHN

>SRS015794.75472-NL *Prevotella* sp. CAG:386

AETYISNVYGRQYESLNGKWNAPDLYERGEEMKLYLNQTPQGKSDFYEYAFDGGGLRLNVPGDWNSQSSE
LKYFEGSVWYARHFHLHALDAKRRFLYFAGVSYRCNVYLNKGKIGEHEGSFTPFQMEVTDALVQGDNSLV
VEVNNRRQKSAIPALSFDWWNYGGITRDVMLVTLPKQYISDYFVQLDKHRADLVHVKVFLSEGVAHQDVC
VEIPELKRRLLLKTASGMATGDLKISKIQRWSPDFPKLYQVNIKTQCDEVTEMIGFRNVEVKGTKVHLN
GKPVFLRSISFHEEIPQRMGRACESDAEMLLSEARALGVNMVRLAHYQQNEYIVRKAEEKMGIMLWQEIP
VWQAIDFSNKETLNKAKRMLKETIRRDQNRACDCEWGIANETRPDARNALFTDLQLAGKDVDTRLFVA
AFDNVYFKENSQFLFEMEDSFVNQLDMIGVNKYMGWYAPWPLEPAKCVWKVAVHKPLLISEFGGEALFQGH
GDDTTASSWSEEYQAKLYRDNLRMFENIPNLVGVSPWVLFDFRSPFRFHPNQGDNWRKGLVSDQGGRKK
AWYVMHDYHQQKQKEYKNIK

>SRS015190.3291-NL *Bacteroides uniformis*

QQVLQIMNVGARQTSLDGQWKTIVDPFENGYYDYRLKPYDGGYAQDKTYSDKTQLQEYDFETDKLLFVP
GDWNTQRPQLYYYEGTVWYRKHFYSLQPGKRLFLNFGAVNYEAIIVWLNGKRLGRHIGGFTPFNFETGL
LREGTNSLVVKVDNKRHPEAVPTVNADWWNFGGITRPTVTLIETPSTYIRDYYVQLKKNKDKTVEGWVQLD
GSAKEQKIVLDIPELKVKKEVVTDSNGYASFQVKAKPVLWTPEKPKLYVNLSSSETDKVSDEIGFRTIRT
EGTKILLNDEEIFCRGISIHEETPYYSGRAYS RDHARTLLSWAKELGCNFVRLAHYPHNEEMVREAERMG
FLVWSEIPVYWTIHWENKDTYQNAEQQLCDMIARDKNRCNVIIWSIANETPLSEPRLAFLTKLANKVRSL
DNVRLVGAAMEKEEVRPGVMTVNDPLGELLDIISFNEYVGWYDGDSEKCDRVNWWFDARKPVFISELGGG
ALYGRHGSRKERFTEEYQEDLYIRHVNMLKRIPGLAGTTPWILKDFRSPRRHVPEIQDDFNKGLVSDKG
QKKKAFFVLQKWYRELEESYK

>SRS011529.42663-NL *Bacteroides uniformis*

QQAPQIMNV SARQTSLDGQWKTIVDPFENGYYDYRLKPYDGGYAQDKTYSDKTKLQEYDFETDKLLFVP
GDWNTQRPQLYYYEGTVWYRKHFYSLQPGKRLFLNFGAVNYEAIIVWLNGKRLGRHIGGFTPFNFETNL
LKEGTNSLVVKVDNKRLEAVPTVNADWWNFGGITRPTVTLIEMPATYIRDYYVQLAKDDKNMIEGWVQLE
GSDKEQKITLDIPELKVKKEVTTDANGYASFLIKSKPILWTPENPKLYAVNLASETDKVSDEIGFRTIRT
EGIKILLNDKEIFCRGISIHEETPYYSGRAYS KDHAHTLLSWAKELGCNFVRLAHYPHNEEMVREAERMG
FLVWSEIPVYWTIHWENKDTYQNAEQQLCDMIARDKNRCNIIWSIANETPHSETRLTFLSNLANKARSL
DSVRLIGAAMEKEEVQPGVLTVNDPLGELLDIISFNEYVGWYDGDSEKCDRVNWTFTDQKPVFISELGGG
ALYGRHGSPKERFTEEYQEDLYIRHVNMLKRIPGLAGTTPWILKDFRSPRRHVPEIQDDFNKGLVSDKG
QKKKAFFVLQKWYKELTEAYK

>SRS015264.145530-NL *Coprobacter secundus*

KEIPLLSNIYGRSYISLNGKWNIVDPLENGYYGYRLTPKKNNGFFLNEKVETSSSNLIEYNFDTAQSM DI
PSDWN SKDES LFFYEGTVWFKRDFDTRKKAGMKYVLYFGAVNYDAKVYVNGRKVGEHEGGYTPFNFDVTK
YLV DGHNFVVVKVDNKRHP EGVPTVNMDWWNYGGITRDVLLAEVPEIYIEDYQIHLAKDMSGYISGWVKL
DTT LSLQNI EF EIPELRLKEFLRTDSSGKATFLLGAQPELWTPEKPKLYDIVIRYNKEVLKDEIGFRTIT
TKGRKIYLN GNPIFLRGVSIHEEAPFRQGRAWSKEDAVTLLSWAKEMGCNYVRLAHYPHNEHMIRVAEKM
GLMVWSEIPVYWTIQWKNKKTYENAKRQLHDMIYRDKNRCCVI IWSIANETPHGKDRDYFLSSLSKYARE
LDNSRLISMAMEVTKAVDNVNELKDNMHKYVDILSFNQYIGWYRGTNESCRDMKWKIPFEKPVIVSEFGG
GALQ GKYGDKTERWTE DYQEELYIQNLKMLDRIEGLSGLSPWILVDFRSPRRQLPGIQDFFNRKGLISSW
GIKKKAYFVMQRY YKNKAGLY

>SRS017307.68457-NL *Alistipes senagalensis*

QRTEGLDPLIQHVYGREIRSLNGYWN YFADPLETGYDYRRRDPNGFFKDKQVDNVTTWKEYDFDLSPV
MAVPGDWNTQVYQLFHYEGMVWLRHKFPAAREAGKRQYLYFGASNYLTHVFLNGEHVGDHEGGYTPFNFD
VTDLLRDGENTLVVAVDNKRLPERVPTIICDWFNYGGITRDVMLVSVPERFVESYTVRLKPGTTDTVEAV
VRMNGKEAGQHVGFAIPELKIDLRGTTDADGVARFEKRARIGLWSPESPKLYAVEIASDDDRVEDEIGFR
TIETRGREILLNGKPIFLRGISIHEEAAFRNGRIAKVEEDRTLLGWARELGCNYVRLAHYPHNEQMVREA
ERMGLLVWSEIPVYWTILWEREDVYANALNQLDEMIERDQNR CNVI IWSIANETPHGDARDRFLSSLAAY
AREKDPTRLISMAMERNDKSPEVLSVQDNMNEYVDIISLNEYVGWYDGTNEKIDRVRWEIDYEKPVMISE
FGG GAPYGKRGEATEIWTEEYQADLYRKTLMIEERMPPVSGISPWVLKDFRSARLLPQTQDGFNRKGV
ISDRGERKQAFYVLQDY YKTKR

>SRS012273.106850-NL *Fusicatenibacter*

MIRTFQTTTIRTQQELTEHLWEFEALEGDKAGEKRKVALPSCVECLPGYENYRGKAAFRTDFTVPGESGN
IRLCFKGISHTAEIYLDGAPAGSHYNAYTPFELVINGLAPGSIHRLEVTADNSYSEKSALHIPNDYESYL
GITRPVLLESVSDCFMEWVHITPIWEAEGWKARIEASLRNTAASPFCGKLEVSLCDACFSIPDLTVPSGE
CRVVGLQTD FLRAEAWSPENPCLTTVRTVLYRDGKAVDDCLERTGFRTVEVKENQIILLNGRPLYIKGFCR
HEDHGLFGCAIPYTAMDYDLNLLSDLGANSVRTSHYPNDERFLDLCDEKGILVWEENHARGLSEEQMRNP
FFERQCEDCIREMITAHYNHPSIYIWGILNECANHTPYGASCYEKQFALIRKLDASRPRTTASCRFKLDL
TIQTPEVKAYNIYPEWYHDDRTEEFAADLYDWIREKCGGEVPFLISEIGAGAIYGWRDNRRTKWSEDYQA
EALEKQIRGVFGLECCNGLYLWQFCDTRVSEEWFGPRPRTMNNKGVVDELRRKKLSYDVVKHLFKEKGN Y
RDARQEKTAEVPGSTEGNRNRNENY

>SRS017821.190781-NL *Lachnospiraceae bacterium* TF01-11

MVRTFNTHNIRKQQELTGKYWDFSPCVGKYAGEHFKVATPCCWESH PKFAGYRGEGEYTTTFTKTGGNIRL
EFKGISHTAEVWLDGRKIAKHYNAYTIFDAVCTNLSYGEHTLTVRADNRFSEKSALHIPNDYMSYGGVSR
PVVLEQISDVYIRNVHVTYPYKADGKWKAKIEIYLENTKDNKYGDKDCFKINETENISKNNRYVPNGENDI
KNSENRSVDVLVNVAGENLI IKNIDA EKN NIVKAEMEFDNIN EWTQETPELYYIEVKLLRNEKAFDDLID
RFGFREIKASGKEILLNGEKIRIKGVCRHEDHPQFGCALPFAAIAADIELIKDMGANSIRTSHPNDEIF
LDICDEQGILVWEENHARGLSEENMRNP NFEPQAEKVIEEMIPAHYNHPSIYIWGILNECASDTEYGKEC
YKKQFDIIRELDTSRPCSFASCKFKTDICFELPDVVSYNIYPLWYHDTPEYLDLDLYKWVQNETKGVGK
PFMITEIGAGGLYGYRNNDYDSKWTEEYQAEALRKQLTAVLGYKDCIGVYIWQFCDIRISDEWFGTRPRTM
NNKGIVDEFRRKKLAYNVVKEIFNK

>SRS049995.49420-NL *Alistipes senagalensis*

GEEAPAPQIVNVAGRKT VSLDGLWKTIVDPYENGYYDYRRKPLKDNMSYFADDDDFDKDRTKLFEYNFNTD
RTLLVPGDWNTQRPQLYYE GTVWYRNLF DYEPRADRRTFLYFGAANYETIVGLNGRKIGKHVGGYTPFN
FEITGKVREGRNSLVVKVDNKR LAEGVPTLNSDWWNYGGLTRSVVIVETPLTFIRDYSVQLKKGEPQVIA
GWVQLDGAEAEQTVTVEIPELKISKVTTDADGYAAFEVKAKPVYWSPENPKLYAVRIA AETDSVDDRIG
FRTIETRGT KILLNGREVFCRGVSIHEEMPYGM SGRAFS AEHARILLGWAKEMGCNFVRLAHYPHNEAMV
RAAEEMGLMVWSEIPVYWTIQWENPATYANAE AQLADMITDRNRANVVVWSVANETPHGEARLKFLRSL
IAKARSMDGTRLVSAAMEKTRLKPDLLTVKDDLADLVDLISFNQYVGWYDGESEKCDRVNWTFDLEKPVF
ISEFGGGAVYGRHGARTERFTEEYMEDLYERSVNMFRMPGLAGTTPWVLKDFRSPRRQMVGIQDDFNRK
GLISDQGGKKKAYFVMQRWYDELEKRYR

>SRS017191.147240-NL *Bacteroides multispecies*

QELLTNVYGRQYTF LNGKWNVIIDPYQQGRRTSIYKNRPLRNKADFKEYEFEGGLRLNVPDWNSQLPEL
KYYEGTVWYARKFKVNKNQDENLFLYFGAVSYRCRVYLNKKEIGSHEGGFTPFQFNITDLIEGENFLAV
EVNNTRTVDAIPALSFDWWNYGGITRDVMLVHTPKVYISNYFIQLDKYKPDYIHAILQLSERKAGQKVRI
EIPELKIAS EVQTDGQGI AKTSFRAKNLERWSPQKPKLYQVTVSSATDHVKENIGFRNLSVKGTKIY LND

APIFMKGISFHEEIAQRQGRAFSEQDAVALLSEAKELGANLVRLAHYPQNEYIVRLAEKMGIMLWEEIPI
WQGIDFKNAGTRMKAQRMYTEMVMRDRNRCALAFWGVANETAPSEARNFLKSLVEHCHEMDTTRLITAA
FDLPKLNPEKAFEMNDSFIEELDVSINKYMGWYHAWPSDPSEVCWNVAPGKPLIFSEFGGEALYQSG
DSTVTSSWSEEYQEKLFDRDNLEMFNNVKNLAGISPWILFDFRSPYRFHPTNQEGWNRKGLISDQGRKKA
WYVMKEFYNNK

>SRS050422.152741-NL *Bacteroides stercorisoris*

QELLTNVYGRQYTFNLNGKWNAIIDPYQQGRRTSIYKNRPLRNKADFKEYEFEGGLRLNVPDWSNQLPEL
KYYEGTVWYARKFEVSKNQDENLFLYFGAVSYRCRVYLNKEIGSHEGGFTPFQFNITDLVKAGENFLAV
EVNNTRTVDAIPALSFDDWNYGGITRDVMLVVRTPKVYISNYFIQLDKYKPDYIHATLQLSEQKAGQKVRI
EIPELKIASSEVQTDGQGMAKISFCAKKLERWSPQKPKLYQVTSTATDHVKENIGFRNLSVGKTKIYLND
TPVFMKGISFHEEIAQRQGRAFSEQDAVALLSEAKELGANLVRLAHYPQNEYIVRLAEKMGIMLWEEIPI
WQGIDFKNAETRMKAQKMYTEMVMRDRNRCALTFWGVANETAPSEARNAFLKSLVEHCHEMDTTRLITAA
FDLPKLNPEKAFEMNDSFIEELDVSINKYMGWYHAWPSDPSEVRWNVALGKPLIFSEFGGEALYQSG
DSTLTSSWSEEYQEKLFDRDNLEMFNRVENLAGISPWILFDFRSPYRFHPTNQEGWNRKGLISDQGRKKA
WYVMKEFYNNK

>SRS050422.79744-NL *Fusicatenibacter saccharivorans*

MEAKKEKKYMSDIHLEDYTEQYETGFATVDTMIFEGGRREELLNGGWHYAVDQYDTCRLQKWYKERYRDE
KGFTVPIDYSFDEWPVMQLPCSWNITIDPMYLLYEGSMVFTRKFSYIAEREETVFLKVGAANYLCRVFLNG
KYVGMHRGGSTPAFWNITEYLKAENRIVLAVDGTTRPEQVPTENTDWFNYCGVYRDIALIRVPKCHIKTF
KIALVPDGTFGHVIAKATLSEKITAKAELVIEELGVSRKIQLENGAGEVVFDAKPELWTPEKPKLYDVRV
TCGTDTVSDRVGFREIRVNGRDILLNGEPVFLRGISCHEDSVENGKGLTREERIEINIRIAKELGCNFMRL
AHYPHNEEMAKLADELGLLLWEEIPVYWAIRFEREKTYEDAQNQLRELINRDWNRASVIWSVGNENADT
DERLKFMSALAECAHREDETRMVSAACLVNAANKIEDRLMEYLDIIGINEYCGWYTPDFAMLPALMENS
QPDKPVIVTEFGADALLHQHGTISDKGTEECQADVYEKQIATLRNIDYIKGMTPWILYDFRCPRRTSLIQ
KYYNRKGLLSEDKKYRKPAFYVLQKFYEELKRKEQ

>SRS017701.44399-NL *Bacteroides* sp. CAG:709

GDKETYGNPDLMVNVYGRDFQMLNGKWDIIDLVDQGRKKVWENRKPAAKADFEFFEYSFDGGLRLDVPDGD
FNSQVPQLQYYEGTVWYARHFSSDGVPSGKKLYLYFGAVSYRCTVYLNKRIASHEGGFTPFQVDVTGNV
VAGDNFLVETNNRRSPDAIPAMSFDDWNYGGITRDVMLVSVPEYIKDYKVQLDKNEADRINAYVRMSD
AVPEGSAVTLSIPELNVSKTLELDKDGFGELSFRVKKLQRWSPESPPLYDVSLTYAPAGGSAGDAVSEQ
IGFRNINVDGTRILVNGKPTFMCCISFHEEIPQRMGRAYSEADAAMLLSEAKALGVNMVRLAHYPQNEYI
VRMAEKMGIIWLQEIIPVWQGIDFENAATMDKALRMFSEMILYRDKNRCVGFWGVANETRPSPARNAFLRR
ELAFARSIDSTRLTAADFNVYFKKEYGEFRIEDDTELDVVSFNKYMGWYAPWPVSPSECRWNVAPGK
PVIISSEFGCEAQYGQHGDAEWASSWSEEFQADQYRKNIEMFGQIPNLAGVSPWILFDFRSPFRFHPVNQD
GWNRKGLVSDKGMKQAWYIMHDYYNAQKHIYDNEK

>SRS049900.6409-NL *Bacteroides ovatus*

DISMQNVSGREKQSLNGVWNILIDAYDEGLKKKWIILRDSNNGGLNELCYEGNLTLRVPGDWNHQLPELD
LYEGHIWYQRRFDSKILPNKRMFLHFGAVSNNCIVFLNGEKIGEHKGGFTPFQFEVTGKLKEKDNYIILR
VDNIRGENSIPALSFDDWNYGGITRNVDLVYVPENYIQDYFIRLDNDDTGKLDITVELNGENVSGKEIVV
ELNEINKKITLKTDENGIAHSSKKISLELWSPEKPKLYDVAITSGNDKITDRIGFRSIVKDGKLLLNGE
PIYLRGINFHEEIANEKRRGYSKADAFLVNSALEMNCNFIRLAHYPQSEYTVRLAEKGIILMWEEIPLW
QKINFASEDVCKLAETMADEMIEDKNRCGVIIWSISNETSKFSKPRNKFLAELVTRVKSLDNTRLVSSA
LHQSQAVKEGEWTVMKLEDPLIEFLDVVGYNKYLGWYQKFPCEPENLKWEAYGKPLIISSEFGAECVQGND
LGDTTNLNSWSETYMENYKKDLISFDNSEDIVGLSPWILFDFRVPRRSHALFQKGWNRKGLISPEYRKK
RAWFVMKDYYDKKSK

>SRS015264.139063-NL *Faecalibacterium prausnitzii* L2-6

MRSIAKLMQDWQFTGPDGKTTLVALPHTWNAKDGQDGGNDYWRGTCTYSTTFAAPAFDAASQEVWLQFEG
VNSSAKVLLNGRNICHTDGGYSTFRVHVSLLAKDNQLTVEVDNSINDRVYPQKADFTFYGGIYRDISLM
VVSРНHIALGHFGDTGVKITPALKDGKADIRVETIVEGEGAVSVELQDAAGSIVARAEGADAQLHLDAPH
LWDGVKDPYLYTCVVRLSSDGTVVDEVSTRVGLRTFSVDAKKGFFLNRPYPLHGVSRHQDRKGVGNAIT
REMHDEDMALIRELGANTVRLAHYQHDQYFYDLCDQYGMVVWAEIPYISEHLPNGRANTISQMKELIYQN
YNHPCIVCWGSNEITISTRDKADMLDNHRELNDLCHKMDSTRLTTLACYAMCGPFNPVAHITDLVSWNL
YLGWYVPGFLFNDLWMDFFHLVYPGRPLGFSEYGAEGMPNLHSSKPRRGDHTTEYQAKYHEYMLRCFDRH
KWMWATHVWNMFDAADARDQGGEPGMNHKGLVTFDRKTRKDSFYLYKAWWSDENFVHICSKRFTDRTES
EMEVKVYSNQKSVALYVNGEKVGEQTGEHVFSFRVPLTGEIEVKAVAGDCTDTASFRHVDTPNPSYKLVK

TKSKSANWV

>SRS017307.32848-NL *Faecalibacterium prausnitzii*

MRSIAKLMKDWQFTGPNVTTAVELPHTWNARDGQDGGNDYWRGCCTYCTRFAAPVYDPAQQQVWLQFEG
VNASAAVLLNGQQQLCTHDGGYSTFRAEVELLQAEENQLTVRVDNSVNDRVYPQKADFTFYGGIYRDVSL
VVKSHIALGHFGDTGVKITPALQENSADIQVETLVEGSGTVTVELLDAEGRTVATGEGENAFHLHLDAPH
LWNGVKDPYLYACVVRLLQDQGPVDEVTTTRVGLRSFSVDAKKGFFLNGRPYPLHGVSRHQDRKGLGNAIT
REMHDEDMALIRELGANTVRLAHYQHDQYFYDLCQYGMVVWAEIPYISEHLPNGRENTISQMKELIHQN
YNHPSIVCWGVSNEITISTKDKADMLDNHRVLNELCHKMDPTRLTTLACYAMCGPFNPVAHITDLVSWNL
YLGWYVPGFLFLNDLWMNFFHLVYPNRPLGFSEYGAEGMPNLHSAHPRRGDHTTEYQAKYHEYMLKCFDRH
PWLWATHVWNMFDAADARDQGGEPGMNHKGLVTFDRKTKKDSFYLYKAWWSEENFVHICSKRFTDRTEK
EIEVKVYSNQNSVALYADGKKLAEQTEGHI FRFRVPLHGKVELKAVAGDCIDTASFCSVAEPNPSYKLVK
TKSKSANWV

>SRS017521.57073-NL *Faecalibacterium* sp. CAG:82

MRKIVRLMKNWQFIGPEGNTTAVDLPHTWNAIDGQDGGNDYWRGTCVYRTRFAAPAFEEVSQQQVWLQFEG
VNASAKVILNGTEVCVHDGGYSTFRANITTLREENILTVEVDNTVNDHVYPQKADFTFYGGIYRDVSL
VVHKDHIALGHYGDGTGVKVTPTLCKDHTADIRVETRLEGEGETVRVTLNDAEGNTVAAAEGKDVMTLAHP
HLWDGVKDPYLYTCVVELEQSGEVVDTVSTRVGVSFRVDPKKGFFLNGRSYPLHGVSRHQDRKGLGNAI
SREMHDEDMTLIRELGANTIRLAHYQHDQYFYDLCDEYGMVVWAEIPYISEHLPNGRQNTIDQMEELIYQ
NYNHPCIVCWGVSNEITISTKDKADMLDNHMLNDLCHKMDPTRLTTLACYAMCGPFNPVAHITDLVSWN
LYLGWYVPGFLFLNDLWLNFFHLVYPNRALGFSEYGAEGMPNLHSEHPRRGDHTTEYQAKYHEYMLRCFDR
HPWMWSTHVWNMFDAADARDQGGEPGMNHKGLVTFDRKTKKDSFYLYKAWWSDEAFVHICSKRFTDRTA
SEIEVKVYSNQKTVALFADGQKIAEQTEGHEVFTFKVPLNGEVKLQAVAGECIDTASFRKVDTPNPGYKLV
KTKSKSANWV

>SRS057717.13088-NL *Ruminococcus gnavus*

MRREIELKKGWVFTGRDGKAVKVDLPHTWNAVDGQDGGNDYRGTCHYSTTFAAPDFDSKTQRVYLEFQG
VNSSAEVELNGQKVMSHDGGYSTFRADVTEFLTKEKLEVAVDNSVNDRVYPQKADFTFYGGIYRDVLLR
IVSNEHFEMDYGGDGMVMTPTVEGKDGKVHVESFHKAENAKVAVTLIDAQGNPVAKGEGKDVTVTIPDV
HLWDGVEDPYLYTAVVELMKDGEVKDDVRI PFGVRTFSVDPKKGFFLNGRSYPLHGVSRHQDRKGIGNAL
TKEHHREDMEFIREIGANTVRLAHYQHDQYFYDLCQYGMIVWAEIPYISEHMPNGRENTISQMKELIVQ
NYNHPSIVTWGISNEITISTKDNKMDMLDNHRELNDLCHKMDSIRPTTLACYAVCGPFNKVAHLSDLVSWN
LYLGWYVPGFLFLNDWWISFFHRVYPNRCLGYSEYGCEGMPNLHSSRPRRGDHTTEYQSLYHEFMLKCFER
HPYMWSTHVWNMFDAADARDQGGEPGMNHKGLMTFDRKVKKDSFYIYKAYWSKEPFVHLCGSRYVDRTE
AVTTVKVYSNQTKVALYANGKLLEEKTGNKVFEFRVPLNGEVRL EAVSGQAKDTSTIRRVSQPNPDYKLV
KSKGKSQNWV

>SRS015578.35375-NL *Roseburia inulinivorans*

MRRIQNLMEQWLFIGPDNRQEQVNLPHWTNAKDGQDGGNDYRGTCHYEKTFEKPEFDPEAEVYLEFLG
VNASSDVTVNGKELAHHDGGYSTFRINITEVLEDNRNKLEVTVDNSVNGKVYPQKADFTFYGGIYRDVRL
IVSKEHFDLDYLGSGNGIKITPEVKENEANVHIETFTNTEEAQVNVQIKDQDGKVVAEGNGCDVNLNIQNP
IRWHGRKNPYLYTAVATLTVNGEVKDEITNRFGIRTFVSDPKKGFFLNGEHYPLHGVSRHQDFKALGNAI
STKNHDEDMMICELGANTIRLAHYQHDQYFYDLCDERGMVVWAEIPYISEHMPGGRENTISQMRRELIVQ
NYNHPSIVVWGVSNIEITISTKDKKDMLDNHVLNDLCHKMDPTRLTTLACYAMCGPFNRSAHITDLVGWN
LYLGWYVPGFFLNDLFLRFFHLVFPNRPIGYSEYGAEGMPNLHSRRPRRGDHTTEYQAKYHEYMIKCFER
HPFMWSTYVWNMFDAADARNQGGEPGMNHKGLVTFDRKTKKDSFFLYKAYWSDEPFVHLCGSRRSDRSE
DTTITKVYTNQPVVSLYNNGELIEQKSGNHI FTFLPMEKENQIEAVAGSVKDAMVLHHTKENPDYKLV
KEAKKSENWV

>SRS023176.71291-NL *Lachnospiraceae bacterium 7_1_58FAA*

MTRAYISLNGEWTFHGPNGSVEQVNI PHTWNGKDGQDGGNDYWRGTCTYEKNFRQPEFDSEKQRVYLEFQ
GVNSTADVEINGNLAVHHDGGYSTFRADITEMLQEEENRLTVRVDNSTNDRVYPQKADFTFYGGIYRDVAL
IVVSQNHFDLDYFGGKGKIKITPKLEKGCCKIHVETFTVTGTGTVEISLLDAKGNQVASGEGQDIILDIPNA
HLWNGLEDPYLYRAVARLVKSSEVVDQVEDHFGVRVFEVDPKKGFFLNGRSYPLRGVCRHQDRPGIGNAI
SQAIHEEDMALIREIGANTIRLAHYQHDQYFYDLCQYGLIVWAEIPYISEHMPSGRENTISQMTRELIVQ
NYNHPSIVCWGVSNEITISTKNKRDMLDNHRLNDLCHRLDPTRFTTLACYAMCGPFNRSAHITDVVSWN
LYLGWYVPGFLWLNMDMWVKKFFHTVYPNRCLGFSEYGCAMPNLHSHPRRGDHTTEYQCVYHEYMLRFFDR
NPWMWGTHVWNMFDAADARDQGGEPGMNHKGLVTFDRKTRKDSFYAYKAWWSKEPFVHLCGKRYVDRPE
SETMVKVYSNQPSVSLYVNGKLFSEQRGDKVFTFRIPLGEGETRLEVHSGELTDSAVLRRVDEPNPAYTL

VKHKKSQNWV

>SRS014923.34218-NL *Faecalibacterium* sp. CAG:74

MRHIIELKDGWQFTNLGQTSEVSIPHTWNAIDGQDGGDDYKRGTCYTLALQKPEFDPDTQEVWIEFDG
VNASAQVSLNGQDVCSHDGGYSTFRANLTGLLADENELVVKADNSKNDRVYPQMA DFTFYGGIYRMVRL
VVNKAHFALDYLSSGIRVTATPEGAGAKVRVQSRVTTGDVTVTLIDRKGRVVGTTGLDTTIEMAKARR
WHGVEDPYLYTARCEVTVDGSVTDRL EIPFGVRSFRVDEKHGFILNDEPYDLHGVSRRHQDRKGIGNAITR
EMHDEDMALIREIGATTLR LAHYQHDQYFYDLCDRNLIVWAEIPYISKHMPNGRANTISQMEELIEQNY
NHASIVCWGVSNEIT IATKDKQDMLDNHRL LNDMIHQTDPTFRFTTLACYAMCGPFNKVAHITDVVSWNLY
LGWYVPGFLFLNDLWIGFWHLCPKRL LGYSEYGAEGMPNLHSDHPRRNDHTEEYQAVYHEYMLRCFRRHP
WMWANHVWNMFDFAADARDQGGEPGMNHKGLVTFDRKTKKDSFYLYKAWWSSDRFVHICSKRFVDRCGKK
ATIKVYSTDDTVTLMVNGKTVGKKTSSDHVFTFTVPLDGDLDVEAFSGLKKDTAHFHKVDKPNPAYKLVK
NKNAQKSNWVDN

>SRS024625.15175-NL *Bacteroides uniformis*

VEVIPLNEGWNFHRGFQAPASEIVKVTLPHTWNAGDGMFGNADYYRGLCNYSRKLSVPAHYRGKRFFLKV
MAAQTVADIFIDHFFVMQHKGGYTAFVAELTD FLVPGTESVLEIRVSNAQTMDVAPICGDFNMFGGLYRG
VELLVTEDVCIEPAYYASSGVFFTQSDVSEKKAHLKIEVLLSARETTVTGCEVEFRLYDKEKLLCRVASA
EVGEDKKVMMDIVLERPHLWDGVKNPYLYKGVVILRKDGKEIDRREEEIGFRYFHADA EKGFLLNGKPYR
LNGVNRHQDRAERASAFYPQDHDEDLDMQEMGCNAVRLCHYPQAKYMHQQMDKRG LVAWAEIPFVN VYV
NNPAYDENLRLQLKELILQNYNHPCILFWGLFNEINSGWLD RPSRMAAELHALARQLDPSRPTMGASNQD
DDFNGFTDLIAFNKYFGWYGDNMDDMGRWIDREHAAH PERKMGISEYGAGACVFQQEDSLRHPEPWGQWH
PENWQTYHYHVENWKQLQEREF LWCNFIWCMFDFS AAGRREGSIMGRNDKGLVTYDRKIKKDAFYFYKANW
NQEDKFVYIAGKRLVNRTRKTVDVQVFSNSGA AKLYINGKAYGTAKPDSVNVLEWKGI VLDTGENRIEVR
NKYGADCCVWICPF

>SRS016056.7615-NL *Bacteroides uniformis*

VEVIPLNEGWNFHRGFQAPASEIVKVTLPHTWNAGDGMFGNADYYRGLCNYSRKLSVPAHYRGKRFFLKV
MAAQTVADIFIDHFFVMQHKGGYTAFVAELTD FLVPGTESVLEIRVSNAQTMDIAPICGDFNMFGGLYRG
VELLVTEDVCIEPAYYASSGVFFTQSDVSEKKAHLKIEALLSARETTVTGCEVEFQLYDKEKLLCRVASA
EVGEDKKVVMVMVLERPHLWDGVKDPYLYKGVVILRKDGKEIDRREEEIGFRYFHADA EKGFLLNGKPYR
LNGVNRHQDRAERASAFYPQDHDEDLDMQEMGCNAVRLCHYPQAKYMHQQMDKRG LVAWVEIPFVN VYV
NNPAYDENLRLQLKELILQNYNHPCILFWGLFNEINSGWLD RPSRMAAELHALARQLDSSRPTMGASNQD
DDFNGFTDLIAFNKYFGWYGDNMDDMGRWIDREHAAH PERKMGISEYGAGACVFQQADSLRHPEPWGQWH
PENWQTYHYHVENWKQLQEREF LWCNFIWCMFDFS AAGRREGTIMGRNDKGLVTYDRKIKKDAFYFYKANW
NQEDKFVYIAGKRLVNRTRKTVDVQVFSNSGA AKLYINGKAYGTAKPDSVNVLEWKGI VLDTGENRIEVR
NKYGADCCVWICPF

>SRS011529.47957-NL *Paraprevotella clara*

GQLRQQSFLSEKWDFRLDRDEADV EQVDVPHTWNATDGTSPDYR DGTYRYT LSVSPAMMKKRIFLR
FEAASQKAEVKLNGQEVGEHVGGFNAFCFEITPYVHAGENLLEVAVSNRKELDIAPLEGDFTVFGGIYRP
VSLLLLPKACITPLDYASSGVYVSQQVGERRADLSVVT KVDNGLKRVKELSVKTDVFD AAGKLVATATTS
QATEGQARLDFTNRLTVDNPM LWDGRKSPYLYRVFVELKRGKEVLD T LSYVGLRSFRVDAEKGFIQNGK
PYKIKGVNRHQDRPDKGWAISSADHREDMSLIKEIGANGIRLAHYPHSDEFYTLCDREGMLVWAEIPLVG
KALDTP EFTENMKTELTELIRQNFNHPSIFCWSL CNELSKGDVKGLISELNDVSHREDPGRLTGLAANVE
KRPENAVPDIMAFNTYPGWYWAEPSEM KSTLERWNRQCGNKGI AVSEYGAGANVWHHWQTGKKT PKADGV
FHPEEWQATVHEQTYAIAKRSGFAWGSFVWNMFDFASASRNEGDMPGMNDKGLVSYDRKIRKDAFYFYQS
EWSEEPVVHIASRRDVARDSLTDVKVYTNCETVRLKVNGKDCGSPERNGSVCIWRKVGLKHGENQVEAV
GLRGGQEFSDRCKWVLF

>SRS011586.24273-NL *Bacteroides nordii*

QLRQQYILKDNWEFRSDKSGADAIWEKVTVPHCWN AEDGITADYYRGTTQYRYHLSILPEMLKKRVFLRF
EAVSQVADIFINGIKVGTHKGAFNAFCFEITSL LKKGDNLLEVKVSNSPDNDIAPLAGDFTVFGGIYRPV
SLLLLPKTCITPLDYASSGIYIHQDQVDSKRAELSIVTKVNSSQPGKALSVRTSVFNAQGKAIGSTTTTN
KSYVKGETAKFINKLNIDRPILWDGCKSPYLYRVFVEVLKGD AVVD T LSYQTGLRYYTVDAEKGFFLNGK
PYKIKGVNRHQDREGKGWAI SNEDHNEDMDLIKEIGATGIRLAHYPHADYFYSLCDREGMLVWAEIPLIG
KANTSSAFTENIKLQ LTELIRQNFNHPSIFCWSL FNELSEGEVKNLIGELNDLAHQEDPGRLTVA AANIE
NRPENKVTDIMAYNTYPGWYWAQPSMMKSSVAHWNRLVGNKGIGVSEYGAGANVWHHWQDVK SAPRTDGI
FHPEEWQSIVHEQNYQAISDCNFVWGSFVWNMFDFASASRNEGNLPGRNDKGLVTYDRKIKKDAFYFYKS
AWVESPVLHITSKR DVARRDSVTDVKVYSNCENIRLKVNGQECGNPIKRQNIYIWQIGL RKGENQIEVC

GDRDRHVLSEKCNWILL

>SRS050422.97742-NL *Bacteroides salyersiae*

QLRQKLLLDQEWFRSDKDGNNLTWEKVTVPHWTNATDGLTPDYIRGTGEYRYHLNLIPEMLKQRAFLRF
EAVSQVADVFNIGIPVGNHKGFSNAFCFEITSSLLKKGDNLEIEVKVSNRLDNDIAPLSGDFTVFGGIYRPV
SLLVLPKTCVTPLDYASSGVYIHQNKVDSKQAELSVVTKVNSNKA EKDL SVKTSVFNAGQLVGNCITRK
ASGKDETVEFVNKLNIKQPVLWNGKKSPLYRVFIEVMKGNTVVDTLSQYIGLRYYSVDAQKGFFLNNGNP
YKIKGVNRHQDRAGKWAITNKEHNEDMELIKEIGANGIRLAHYPHSDFFYSLCDREGMLVWAEIPLIGK
ASPTPAFTENIKLQLTELIRQNFNHPISFCWSLFLNELCEGDVKKLIQELNEVAHKEDPGRLTVAAANVEN
RPENVITDIMAYNTYPGWYWAQPSMTKSSVAHWNRLVGNKGICVSEYGAGANIWHHQQDVQSAPKTDGIF
HPEEWQAIVHEQNYRGISSDFVWGSFVWNMFDFASASRNEGDI LGLNDKGLVTYDRKVKKDAFYFYQSA
WSESPVLHITSKRDIARREPATDIKIYSNCDHIHLKVNGQDCGHPDREDNILIWKVNSLKKGENRIEASG
KKGTVELTDQCKWVLLDKKK

>SRS018656.34945-NL *Candidatus bacteroides timonensis*

RDIVPFNDKWFFKKDSLSTEVTLPHTWNATDMQIKAKAFYEGSASRYKSCLPESWKGKRLFLRFEGVGS
CAEVYVNGKLTGSHKGGYSAFTCEIQNEIKFGEENEIKVLVDNTARPDVVPVNHNLFGIYGGIYRPVWLI
ITDSCNIAVTDHASSGVYISQKNVSRQSAEISVKVKLDNNGLYPVPVILENVIYDVTGKQIAAERQSFDL
QFQGMQARTSHFKLKRPHLWQGREDPYLYKVTSRLYQNGKVIDEISQPLGLRNYEIIAGKGFFLNKKYP
MYGVARHQDWETGAALSNEQHDTDLAFIMDIGATTVRFAHYQQSEYLYSRCDSLGLIIWAEIPFVNRVT
GKEWENAHNQLRELIRQSFNHPSIYVWGLHNEVYHPHAYTTQLTRSLHDLAKTEDPGRYTVAVNGYGHAE
HPVNMQADIQGMNRYFGWYEKKIRIDDAWVEHLETEYEMKMLMTEYGADANLQHQTFLGESLNWTKEF
YPETFQTKTLEYQWSVIAKHPYIIASYVWNMFDFACPMWERRGGVPARNMKGLVTFDRKTLKDSYFWYKAN
WSKNPVLVLTQRRNDQRETRTTSVTVYSNLGTPKVFLNGKALDNIRKGYTAVHYIIDNVTLANGENKLQA
VITDRNGKEYADEIIWHYQGEQRRSTDNRENREHSGF

>SRS052697.111918-NL *Tannerella* sp. CAG:51

ALIIQSSAQLRDRISLSGEWQFFKGDTEISSQVEHVTVPHSWNAQDGTADYIRGKGTYRYNFDLPKNTK
KKRVFLRFEAVSQEADVFLNGQYLGHNHKGAFNAFCFEITPYLKKKDNELIVKANNILNPDIAPLSGDFTI
FGGIYRPVSLILPQTCITPLDYASSGIYIKQTEIQDHKATIDITTKIDGDISGNKKLRLRTSVFDPTGK
LVTTSENEEIRASENTVAIDQRIYLDNPILWNGKAAPNQYRFFCELLQGGKVIDTSLSVYTGRLYFKVDPQ
KGFFLNKGSYQIRGVNRHQDRPGKGWAISEKDHQEDMQLIKEIGANGVRLAHYPHSDYFYSLCDKEGLLV
WAEIPFIENGTESIAFEENLKQQLTELIRQNFNHPISFCWSLFLNELAKGNPEKLVAELNDLAHKEDPTL
TVAAANIEGRPENTITDIMAFNTYPGWYWAEPETMGASIDWKKDEKRGVCISEYGAGASIKHHQORMKKA
PKCEGDHFPEEWQAIVHEGNYKEIDKRDFVWGSFVWNMFDFASGGRKEGDTYGMNDKGLVTYDRKTPKDA
FYFYKANWSEKPVHLITSKRHIERTEPETDIKIYSNCS DITLYVNGMKYAGNSGDHNIFIWKNIRLQPGK
NKIRVTGTDFKETITDQCEWILKN

>SRS011405.12981-NL *Bacteroides thetaiotaomicron*

REVTSFNEGWLFRGPFQSQDPVKVAAQWNADWETVNLPHWTNAKDMQVKADAFYEGVGYYRKTQFFGNEL
KGKRVFLRFEGVGANTEVYVNGKLVGMHRGAYS AFAFEIGTALKLGAENEIMVKADNASRPDVI PVNHNL
FGVYGGIYRPVWLIITEQNNITVTD CASPGVYITQRNVSKRSADITVKVKLDNGSLTPADLVLENTIYTQ
NGKKVTSRHLPLVLT PQGTQTYISTFKLNKPHLWQGRKDPYLYKVISRLLADGKVIDEVVQPLGVRRYEI
VPGKGFYLNGEKYPMYGVTRHQDWWGLGSALTNKEHDFDLAQIMDIGATTVRFAHYQQSDYLYSRCDSLG
LVIWAEIPFVNRVTGYEAENAQS QLRELIRQSFNHPSIYVWGLHNEVYQPHEYTAGLTQTLHNLAKTEDP
DRYTVSVNGYGHAEHPVNQNADIQGMNRYFGWYEKKLQDIEPWIKGLEEKYPWQKMLMTEYGADANLDHQ
TEYLGDALNWGKSFPETFTQTKTHEYQWSVIAKHPYIIASYLWNMFDFGVPMWSRGGVPARNLKGLITFD
RKIKKDSYFWYKANWSKDPVLYLTQRRNIDRERKHTSVTVYSNIGTPQVYLNKGELTGIRQGYTDVHYVL
DDVTLEQGKNTIKTVATYNGKEYTDEIEWNYTGEKKRSADWSENKEEHAGW

>SRS011586.223194-NL *Bacteroides* sp. HPS0048

RETISFNQGWLFKKGPFSSPEPAKVVSQWNSKWEVEIPHTWNAKDMQVKVNSFYEGTGYYKKNYYADKSL
QGKRVFLRFEGVGTCTEVYVNGKLVGTHKSAYS AFAFEIGTALKYGEDNEILVKTDNTARPDVIVNHSL
FGVYGGMYRPVWLIVTEQSSITVMDCASPGVYITQKNVSHQSADITVKVKLDNGSLKPESLVLQNTIYTQ
EGKKVASDDQAIELTPQGTQSYTTFKLKKPHLWQGRKDPYLYKVVSRLADGKVIDEVTQPLGVRYEIE
IAGKGFYLNGEKYPMYGVTRHQDWWGLGSALKNEQHDFDLATIMDVGATTVRFAHYQQSDYLYSRCDSLG
LIIWAEIPFVNRVTGYEAENAQTQLRELIRQSFNHPSIYVWGLHNEVYQPHEYTASLTQALHDLAKTEDP
DRYTVSVNGYGHAEHPVNQNADIQGMNRYFGWYEKKVQDIKPWIENMEKEYPWQKMLMTEYGADANIEHQ
TEYLGDALNWTKPFYPETFTQTKTHEYQWSVIAEHPSIIASYLWNMFDFAVPMWGRGGVPARNLKGLVTFD
RKIKKDSYFWYKANWSKDPVLYLTQRRNTEREKKVTSVTIYSNIGTPKVYLNQELNGIRKGYTDVHYVF

DEVTLNNGKNTLTAIVRYKGKEYKDEIEWNYTGEKKREADSYVNKSVHAGW

>SRS024132.77963-NL *Bacteroides caccae*

REVTSFNEGWLFRGPFSEDPVKVVAQWEGKWETVSLPHTWNAKDMQVKAASFYEGVGYYKKKQFFNEEL
KGKRVFLRFEGVGANTEVYVNSKLVGTHKGGYSAFAFEIGTALKFGAENEIMVKADNTARPDI PVNHSL
FGVYGGIYRPVWLIITEQNNITVTD CASPGVYITQKNVSKRSADITVKVKLDNGSLTPANLVLENTLYTQ
EGKRVASHRLPLELTPQGTQTYFSTFKLNKPHLWQGRKDPYLYKVVSRLVSEGKVIDEVIQPLGVRKFEV
VAGKGFYLNDEKYPMYGVTRHQDWGWLGSALTNKEHDFDLAQIMDVGATTVRFAHYQQSDYLYSRCDSL
LVIWAEIPFVNRVTGYETENAQSQLRELIRQSFNHPSIYVWGLHNEVYQPHEYTAGLTQALHNLAKTEDP
DRYTVAVNGFGHAEHAVNQNTDIQGMNRYFGWYEEKLQDIEPWVKGLEEKYPWQKLMLTEYGADANLEHQ
TEYLGDALNWGKPFYPETFQTKTHEYQWSVISKHPYIIASYLWNMFDFAVPMWSRGGVPARNLKGLMTFD
RKIKKDSYFYWKANWSKEPVLYLTQRRNADRERKQTSVTVYSNIGTPIVYLNKGKELTGIRKGYTDVHYVF
DNVTLEQGKNVIRTVAATHNGKEYTDEIEWNYTGEKKRNADSRENKNEHAGW

>SRS019601.2605-NL *Bacteroides fragilis*

REVKSFYDGEFQRGKYPDETILATAKWDGKWQPVSI PHTWNNKDMQSGYDRFYQGEAYYRKQFDCPTL
KSKRVFLRFEGVGACAKVYVNRMLGTHKGAYS AFAIEISEALKYGEVNEIVVLANNEQRPDI PVNHRL
FGVYGGIYRPVQLIVTEKSNITVTDNASPGVYVSQTNVTKK SADISVNVKLDNADLHATSVSLETVIYTQ
EGKQVAKNVKEINLTPQGSQPFADFR LKNPHLWQGREDPYLYKVVARLVKD GKTIDEVVQPLGVRHYEM
VAGKGFYLNGEKYPMYGVC RHQDYLGLGSALT NKHHNDLADIMEIGATTVRFAHYQQSDYLYSRCDSL
LVIWAEIPFVNQVTGEEAENAKSQLREMIRQSYNHPSIYVWGLHNEVYSPTPYTAQLTAEHLDLAKTEDP
YRYTVSVNGYGHVEHSVNLNADIQAMNRYYGWYEGRIKIDIDKWVDGLTKNYSDLRLIFSEYGAENIEHH
TEILGESLNAWGNYPPEYQTKFHEYQWSVIANHPVILASYLWNMFDFCTPMNDRTDIPAQNTKGLITFD
RKIKKDSYFYWKANWSKEPTVYLTQRRNTQREKNTTVTVYSNIGTPKVYLNKGKELTGIRQGYTVVHFVF
DNVTLADGKNLTKAVAMKNGKEYVDEIEWNYNGERDRSPEHYLNEKEHVGL

>SRS011586.51365-NL *Bacteroides fragilis*

RQVTA FN TGWQFKKGPF AADPMQAASQWDGKWETVEI PHTWNAKDMQVQSCSFYEGVGYYRKTQFFPQDL
KDKRIFLRFEGVGACSEVHVNGKLVGTHKGGYSAFACEIGTVLQFGTENEIVVKADNKARPDVI PVNQNL
FGVYGGIYRPVWLIVTEQNNITVTD CASPGVYITQKNVSKKSADITVKVKLDNAGLQPAALTLENAIYTQ
DGRKVS VHSQSFDLTPQGTQTYTSTFKMKDLHLWHGRKDPYLYKVVC KLMADGKVIDE VVQPLGVRKYEI
VPGKGFFLNGEKYPMYGVTRHQDWGWLGSALKNEHHD FDLATIMDVGATTVRFAHYQQSDYLYSRCDTLG
LIIWAEIPCVNRVTGYEAENAQSQLRELIRQSFNHPSIYVWGLHNEVYQPHEYTAALTQSLHDLAKTEDP
DRYTVSVNGYGHMEHPVNLNADIQGMNRYFGWYEEKIQD IKPWVEQLEKEYPYQKLMLTEYGADANLAHQ
TEYLGDALNWGKPFYPETFQTKTHEYQWSVIKDHPIYIIASYLWNMFDFAVPMWTRGGVPARNMKGLVTFD
RKIKKDSYFYWKANWSNPVLYLTQRRNV DREKMTTVTVYSNIGTPKVYLNKGKELGGIRKGYTDVHYIF
DNVALADGKNVLKAVAAIKGKEYTDEIVWNYSGEKNREVD SYENKNEHSGF

>SRS018427.209365-NL *Bacteroides massiliensis*

RDVTSFN TGWQFKKGPF SADAMKAASQWEGKWETVEI PHTWNARDMQVRYNNFYEGVGYYKKNFFCPGDL
KDKRVFLRFEGVGACSEVYVNGRLAGTHKGGYSAFVVEIGPSLKP GSDNEIVVKADNQSRPDVI PVNHVL
FGVYGGIYRPVWLIVTGKSSIAVNDCASTGVYITQKNVSKRSADIAVKVKLDNAALHPVPVLENTIYTQ
EGRKVASQKQNIELTPHGIQACVANFKLNHPHLWQGREDPYLYKVVSRLMQDGQVIDE VVQPLGVRKYEI
IAGKGFYLNGEKYPMYGVC RHQDWGWLGSALKNENHDFDLAQIMDVGATTVRFAHYQQSDYLYSRCDSL
LIVWAEIPFVNRVTGYEAENAQSQLRELIRQNFNHPSIYVWGLHNEVYHPHEYTSQLTAALHDLAKTEDP
DRYTVSVNGYGHMEHPVNLNADIQGMNRYFGWYEEKIQD IDAWVEGLEEKYPYEKLMLSEYGADANIYHQ
TEYLGQSLNWT KPYYPETFQTKTHEYQWSVIARHPYIIASYLWNMFDFATPMADRGGIPGRNMKGLMTFD
RKTRKDSYFWYKANWSKEPVLHLTQRRNV DREKQETS VTVYSNIGMPKVFLNGRELQGV RKG YTDVHYVF
DHVMLGDGKNRLKAVVSRDGKEYTDEIEWNYSGEKNRGTEAYENKNEHFGL

>SRS015190.16570-NL *Bacteroides cellulosilyticus*

REVASFNDGEFQFKGPF SKEALQAVQKNSDWQEVSI PHTWNADDMQKQASAFYEGVGYYRKKFTFPENM
EGKRIFLRFEGVGSYAEVYINGYLAGTHKGAYS AFACEISSQVKFGKENEIIVKADNSSRPDIPTNHIL
FGVYGGIYRPVWLIVTETCSIIANDCASPGVYITQKNVSKKSAEVTVKVKVDNATLAPVPLELENAIYDR
NGKLVKKHSQNFELTPQGVQNYSSHFR LNNPHLWQGREDPYLYKVVSRLMQNGRVIDEVIQPLGIRKYEI
VAGKGFYLNKKYPMLGVTRHQDWGWLGSALTNKEHDFDLEQIMDIGATTVRFAHYQQSDYIYSRCDTLG
LVIWAEIPFVNRVSGQEWDNAHQQMRELIRQSFNHPSIYVWGIHNEVYHPHGYTAALTQSIHDLCKLEDP
DRYTVAVNGYGHADHPVNQNADIQGMNRYFGWYERKIQD IKPWIEKMEKEYPWQRLMLTEYGADANIEHQ
TEILGDALNWTSPFYPETFQTKTHEYQWSIIAQHPYIIASYLWNMFDFAVPTSKRGSVDARNMKGMMTFD
RRIKKDSYFWYKANWSKEPVLYLTQRRNAVREKKETSITVYSNIGTPKVYLNKGKELTGVRKGYTDVHYIF

DKVVLDDGKNILKASLLHQKEYTDEIEWNNGECNREADFLEHKKEHGSW

>SRS020869.17121-NL *Bacteroides intestinalis*

REVTSFNDGWFEKKGPFSSKETMQAVQKWN TDWQEV TI PHTWNADDMQKKVSAFYEGAAYYRKKCTFPESL
EGKRIFLRFEGVGSCAEVYVNGYLAGTHKGAYS AFVCEIGSQVKFGKENEIVVKADNTARLDVIPTNHVL
FGVYGGIYRPVWLIVTEPCNIVVNDHASSGVYITQKNVSKKSAEVTVRVKVDNATLAPVPLILENAVYDR
NGKLVKKHSQAFELTPQGVQNYSSHFKLNSPHLWQGREDPYLYKVVSRLMQDGQVIDEVVQPLGLRKYEV
VAGKGFFLNGKQYPMYGVTRHQDWGWLGSALTNKEHDFDLEQIMDIGATTVRFAHYQQSDHLYSRCDTLG
LIIWAEIPFVNRVSGQEWDNAHQQMRELIRQSFNHPSIYVWGIHNEVYHPHGYTASLTQSIHDLCKQEDP
DRYTVSVNGYGHVNHVPVNQNA DIQGMNRYFGWYERKIQDIKPWIEKMEQEYPWQRLMLTEYGADANLDHQ
TEILGDALNWTSPFY PETFQTKTHEYQWSVIAQHPIIASYLWNMFDFAVPTSKRGSDARNMKGLMTFD
RKIKKDSYFWYKANWSKEPVL YLTQRRNAVREKETS SVTVYSNMGAPKVYLNNQELTGARKGYTDVHYIF
DKVVLNDGKNMLKAVISYQGKEYSDEIEWNNGECNREAEFFEFKKEHGSW

>SRS015578.36951-NL *Bacteroides fragilis*

RQVTA FN TGWQFKKGPFATDPMRAASQWDGKWETVEIPHTWNAMDMQVQSGSFYEGAGYYRKTQFFPYDL
EGKRVFLRFEGVGACAEVYVNGKLAGTHKGGYSAFACEIGTALKLGAENEIIVKADNKARPDVIPVNQNL
FGVYGGIYRPVWLIVTEQNNITVTD CASPGVYITQKDVSKKSADITVKVKLDNAGLQPAAVTLENTIYTQ
EGQKVGTHSRSFDLSPQGTQTYLSTFKLKNPHLWQGRKDPYLYKVVCRLMADGKVIDEVVQPLGVRKYEI
VAGKGFFLNGEKYPMYGVTRHQDWGWLGSALKNEHDFDLAAIMDVGATTVRFAHYQQSDYLYSRCDTLG
LIIWAEIPCVNRVTGYETENASQSLRELIRQSFNHPSIYVWGLHNEVYQPHEYTAALTRSLHDLAKTEDP
DRYTVSVNGYGHMDHPVNLNADIQGMNRYFGWYEKKIQDIKPWVEQLEKDYPYQKMLTEYGADANLAHQ
TEYLG DALNWGKPFY PETFQTKTHEYQWSIIKDHPYIIASYLWNMFDFAVPMWTRGGVPARNMKGLITFD
RKTKKDSYFWYKANWSEEPVL YLTQRRNADREKRTTAVTVYSNIGTPKVYLNQELSGIRNGYTDVHYVF
DNVSLADGKNILKAVVSTKGKEYTDEIEWNNGEKNREIDSYENKNEHSGF

>SRS011529.202112-NL *Bacteroides coprophilus*

REVTPFNNGWFEKKGPFPTDAMQTAARNWADWEQVNVPH TWNADDMQKKTNSFYAGVAYYRKHVFPESL
EGKRLFLRFEGVGACAEVYVNGYLVGTHKGAYS AFVCEIGSQVRLGEENEIVKSDNVARPDVIPVNHAL
FGVYGGIYRPVWLIVTEPCNIVVND CASPGVYITQKNVSKQSADVSVRVKLDNATLAPAALELENAIYTR
EGKLVKVHRQFFELTPQGVQNYVSDFKISHPHLWQGREDPYLYKVVSRLKQDGQVIDEVIQPLGLRKYEA
VAGKGFFLNGKKYPMYGVTRHQDWGWMGSALTNREHDFDLEQIMEVGATTVRFAHYQQSDYIYSRCDTLG
LIIWAEIPFVNRVTGQEWDNVHQQMRELIRQSFNHPSIYVWGIHNEVYHPHGYTAALTQSVHDLCKLEDP
DRYTVAVNGYGHAHPVNGNTDIQGMNRYFGWYEKKIQDIKPWIEKMAKEYPWQRLMLTEYGADANIEHQ
TELLGDALNWGSDFY PETFQTKTHEYQWGVISQHPYIIASYLWNMFDFAVPQWKRGGVDARNMKGLITFD
RKIRKDAFYWKANWSKEPVL YLTQRRNTERRETS SVTVYSNLGTPRVFLNGREL DGVRKGYTDVHYIF
DQVTLAEERNVLRVAVISDQKEYTDEIWNNGEHRQADAYEHKKEHAGW

>SRS011239.209065-NL *Parabacteroides multispecies*

EGRKVESFNSGWSFKKAPAEKELAINAPKWDKGWSEVEIPHTWNAKDMQVQANSFYEGAAYYKKQYFFPA
ELKD KRVFLRFEGVGSCAEVFNGLM LATSHKGGYSAFACEISPLLKAGEENEIIVKADNKSRPDVIPVNH
NLFGVYGGIYRPVWLIVTEPCNISVTD CASPGVYVTQKNVSKKQADV KVKVKLDNGTLQPVPTLQNTIY
DQEGKQVATHSQSFELSAQGEQAYESSFTIKKPTLWQGRENPYLYKVVSRLIKDGQVIDEMVQPLGLRKY
EIVAGKG FYLNGEKYPMYGVTRHQDWGWLGSALKNENHDFDLATIMDVGATTVRFAHYQQSDYLYSRCDS
LGLIIWAEIPFVNRVTGQEAENCRNQLREMIRQSFNHPSIYVWGLHNEVYQPHQYTKELTQSLHDLAKTE
DPDRYTVSVNGYGHMEHPVNLVADIQGMNRYFGWYEKKIQDIKPWVENLEKEYPHQKMLTEYGADANLN
HQTEYLG DALNWTKEFY PETFATKTHEYQWSVIAAHPIIASYLWNTFDFCAPMWVRGGVPARNMKGLVT
FDRKIKKDSYFWYKANWSKEPVL YLTQRRNWDREKKETS SVTVYSNIGTPKVYLNKGELTGIREGYTPVHY
IIDNITLDMGKNIVKTVVVKDGKTYEDEIEWVNGEKKRDSQSVNKEEHAGF

>SRS011529.28992-NL *Parabacteroides merdae*

EGRIVEPFNFVGWVFKKAPAGKELAVNAPKWESGWTEVEIPHTWNAKDMQTQYDQFYEGPAYYKKHYFFPA
ELKD KRVFLRFEGVGAVAEVYVNGSLVTTHKGAYS AFSCIEGTALKPGADNEIIVKADNASRPDVIPI NH
VLFGVYGGMYRPVWLIVTEPYNICVTD CASSGVYITQKNVSKKQADV KVKVKLDNGTLQPAPLTLQNTIY
DQEGKL VATQSRSFDLTPQGVQSFEADFKIKNPTLWQGRKNPYLYKIVSRLIRNGKVIDEVVQPLGLRKY
EIVAGEGFY LNGEKYPMYGVTRHQDWGWLGSALKNENHDFDLATIMDIGATTVRFAHYQQSDYLYSRCDS
LGLIIWAEIPFVNRVSGQEAENARNQLRELIRQSFNHPSIYVWGLHNEVYHPHEYTKELTRSLHDLAKTE
DPDRYTVSVNGYGHMEHPVNLNADIQGMNRYFGWYEKKLQDIEPWVEGLEKNYPEQKMLTEYGADANIV
HQTEYLDNSL NWT KPFY PETFQTKTHEYQWSVIAKHPYIVASYLWNTFDFATPLANRGDL PARNMKGMVT
FDRKTKKDSYFWYKANWSEDPVL YLTQRRNVDRKKVTSITVYSNIGEPKVFLNGQELSGIHKGYTDVHY

IIDNVTLADGKNIVKTVASKDGKTYEDTIEWMYTGEKKREADSFSIKGVHAGFE
>SRS021484.217051-NL Parabacteroides johnsonii
EGRIVESFNSGWVFKKAPAGKELAVNAPKWESGWTEVEIPHTWNAKDMQVQFDNFYEGPAYYKKHYFFPV
EMKGKRVFLRFEGVGAVAEVYVNGSLVATHKGAYSASFCEIGTVLKPAGADNEIIVKADNASRPDVI PVNH
VLFGVYGGMYRPVWLIVTDPYNICVTDCASSGVYITQKNVSKKQADV KVKV KLDNGTLQAPLTLQNTIY
DQEGKQVATQSR SFDLAAQGVQSFEADFKIKNPTLWQGRKNPYLYKVVSR LVRD GKVIDEVVQPLGLRKY
EIVAGKGFYLNGEKYP MYGVTRHQDWWGLGSALKNENHDFDLATIMDIGATTVRFAHYQQSDFLYSRCDS
LGLIIWAEIPFVNRVSGQEAENARSQ LRELIRQSFNHPSIYVWGLHNEVYHPHEYTKELTKSLHDLAKTE
DPDRYTVSVNGYGHMEHPVNLNADIQGMNRYFGWYEKKLQDIEP WVEGLEKNYPDQK LMLTEYGADANLL
HQTEYLENSLNWTKPFYPETFQTKTHEYQWSVIAGHPYIVASYLWN TFD FATPLANRGDLPARNMKGLVT
FDRKTKKDSYFWYKANWSEDPVLYLTQRRNV DREKKVTSITVYSNIGEPKVFLNGQELSGIHTGYTDVHY
IIDNVTLKDGKNVVKAVVQSDGKTYEDTIEWMYTGEKKREADSFSIKGVHAGFE
>SRS024132.98377-NL Odoribacter laneus
EGRTIKAFNDNWQFKRGPFPEDPVKWQAVWEKGWKEVQLPHTWNAQDMQEQHNNFYAGPAYYRKRLSVSP
DWKGKNI FLRFEGIGQVAEIIYVNGEYAGNHLGGYSACAFDLTALLKS GEENEILVRVDN SSRPDVI PVNH
SLFGVYGGIYRPVWLIVTEPVHIAVTDYASPGVYITQKNVSARQAEVGIKVKIENRLGTVC RMEL ENAVY
DAEGKVCKLQREEIIIVLPQGTQEF RQQFRIKRPHLWQGRPD PYLYKVVTR LKREGQIWDEVIQPLGFRKY
EIIAGKGFYLNGEKYP MYGVTRHQDWWGLGSALGREQHADLAAILDIGATTIRLAHYQQSDYFY SRCDS
LGLIVWAEIPFVNRVSTQEGPNAKEQLRELIRQSFNHPSIYVWGLHNEVYVPTAYTTALTRQLHDVAKTE
DPDRYTVSVNGYGHMEHPVNRNADIQGMNRYFGWYEKNIQAIQPIEDLETRFPDYK LMLTEYGADANIA
HQTEYIEESLDWTKAFYPETFQTKTHEYQWGI IAHPIYIIASYLWNMFDFAVPMWKRGGVEARNMKGLIT
FDRQVKKDAYYWYKANWSKEPVVYITQRR LKEREKQQTEVTVYSNVGRPDLYVNGIRQSPARQGYTDVHF
IFENVRLNKGKNVLEAIVQQGDKTHTDQIEWNYTRERKAGEAVPLDKNTEHWGF
>SRS018427.198716-NL Bacteroides vulgatus
RDITSFNEGWSFKKGPFPAKDPLLSAGFDARWQDVTIPHTWNAQDMQTEMNNYYAGEAYYKKQYTPAAEL
KGKRVFLRFQGVGSVADVYINKTYAGNHKGAYS AFAIEISKLLKYGESNEILVKVDNSPRPNVIPVNN TL
FGVYGGIYRPVELIVTEPVNIAVTDYASPGIYISQKNVNRKSADVNIKIKLENKKAQAKKIVVSTTIYER
DGRVKAQQQFPVTVSPQGRQIYDQDFTIKNPHLWQGLEDPYLYKVVVNLLADGKVIDEVSQPLGLRHYEL
KAADGMYLNGKKVPMYGVCRHQDWWKLG SALT DKEHDTDLAI IREIGATTIRFAHYQQAE RIYAKCDSIG
FMVWAEIPFVNRVSTQESGNAKQQLVELIRQNYNHPSIYI WGLHNEVYAPYNYTVSLTTELNDLAKTEDP
YRLTASVNGYSVVNQESNLNADVQGINHYFGWYGGKIGID EKWVTGMEQNFPDYRVVFSEYGAEANI HQQ
EEAVGEVGRYFSQFY PETFSTKFHEIQWGVISKHPYLVASYIWN TFD FATPATAQGGVKARNMKGLVTFD
RQTKKDPFYWYKANWSKEPVLYITQRRATERENEVTPVTYCVNGVPRLFVNGTEVKT VSKGQTSVHYIF
ENVTLNKGENVIEARVDHQGKTLEDTIRWSYSPENKNRLNLSDPKEKTKEHVGL
>SRS011239.45377-NL Bacteroides coprocola CAG:162
AMKVESLDRGWFFNRGYESQAADKVTVDLPH TWNAADAMFGNVNYYRGMSTYSRFIVVPSS ENFKRVFLR
VKAAQTVADVYLDNHFVAQHKGGYTSFTVELTHLIRRGEKQKLDIRVSNAQTMEIAPICGDFNIWGG LNR
GVELLYTDDACIDPTFYGSSGVFFTQYNVSEKYAGLEMKALLSGTEMALNDCSVEFSLLD AKRKVVWCKE
TSVSGKEAVVKMELNPNHLWDGVNDPYLYTG VVVLKRGGKEIDRREEHVGF RFYSADPDKGFFLNGKPYS
IHGVNYHEDRAERASAFRPEDFESDLNLIQEMGCTAVRLAHYPHAQQ LHNLMDRRGLIAWAEIPFVNVFV
SHPLYREN LKQQLLELICQYYNHPCILTWGLFNEINPGWLENPNPMAEELNK LAKKWDISRSTTGASNQK
DPLNGIPDLIAFNRYFGWYGDDCKEMGEWIDKEHRAYPQRCMGISEYGAGADV FQQTDSL IHP EPWGW H
PENWQTYHYHIENWKQLASRPFLWCKFVWC MFDFS AVGRKEGTT FGRNDKGLV TYDRRIKKDAFYFYKANW
NKKEKVL YIASKR CQNR SNSSVDIQVFSNCGDAELFVNGKSMGKCTPDEV CVATWNNVPLQSGENQVTVR
ASGLSDNCVWYNN
>SRS019910.29965-NL Bacteroides caccae
RDVISFNTGWQFKKGPFSTDPMRASAQWDGKWEV TIPHTWNAKDMQVKADAFYEGVGYYRKKQFFGNEL
AGKRVFLRFEGVGACTEVYVNGKLVGTHKGGYS AFAFEIGTALKLGAENEIIVKADNTARPDVI PVNHSL
FGVYGGIYRPVWLITTEQNNITVTD CASPGVYITQKNVSRRSADITVKVKLDNGSLAPAQLVLENSIYTQ
EGKKVVSQRLPLELTPQGTQSYVSTFKLKNPHLWQGREDPYLYKVVSCLLANGEVIDEVTQPLGV RHYEI
VAGKGFYLNGEKYP MYGVTRHQDWWGLGSALT NREHDFDLAQIMDVGATTVRFAHYQQSEYLYSRCDTLG
LIIWAEIPFVNRVTGYESDNAQTQLRELIRQSFNHPSIYVWGLHNEVYQPHEYTASLTQSLHDLAKTEDP
DRYTVAVNGYGHANHPVNQNTDIQGMNRYFGWYEKKVQDIKPWVEGLEKDY PWQK LMLTEYGADANLDHQ
TEYLG DALNWGKPYYPETFQTKTHEYQWSVIAEHPIYIIASYLWNMFDFAVPMWSRGGVPARNLKG LMTFD
RKIKKDSY YWYKANWSKSPVLYLTQRRNTDRERKQTSVTVYSNIGTPKVYLN GKELTGIRKGYTDVHYIF

DQVTLEKGNKIKAVAVYNGKEYVDEIEWDYQSEKKRDADAHENKNEHGGVSKCAPSFF

>SRS063985.27039-NL *Bacteroides dorei*

QRVSYSLNDSWLFKQANVMTPTDNWQEVTLPHSWNMEDGQYSGKNRLDDNGHAQIQRGDTVSIVDPAI
KFGYYRGVGYARIINIPVEWEKKRIFVRFEAAGQVARVHVNGQMLGEHRGAFTAFCYELTPYIIIGKAN
ELRVEVDNIQRQDIPPLSGDFNVNGGLYRPAQLIVTDQVCVSPLDYASSGVYITTRKLDVDKATVEIRSL
VNNGVHAAMSNKPEVSAEDVVVETVIKDTAGNAVATKRTFRTIEPDCTKEIIQELKIENPHLWNGRKDPY
LYQVEVNVLRNEKIVDRIIQPLGLRIFRIDSALGFLNEQPYVPYGVGRHQDFKDKAWALTEENELDYS
IIKEMGATAIRFAHYPQSQNMHDIADREGFLVWDEIPLVNEIRQDYGARQNVWIMMKEMVKQLYNHPSVA
WWGIYNEIENLYTPPSEEFLIKLRDGIRALDPSCRIITGASDHGLRAHNLVPEATCFNNYPGWYHGNYPK
EEGYTGEHSQFSKWIQNRakeIGMRTAISEYGAGGDYNQHSEGKPYKPPAHGGPFQPEEWLAYVHEEDW
RIMKDNDNLWGTFLWAMFDFASCMRNEGSVPSINTKGIVSQDRTIRKDPYYMYKANWNNEPMVYIASRRA
IKRKLAVTDIKVYSNCQVITLKVNGETVGCMPDDIKVCFNNVLLKGSNQIEAIGSTVQGNELVDTC
WILE

>SRS017103.228639-NL *Ruminococcus* sp. CAG:77

MRNVIRINDGWLFTLGGKTETVNLPHWTNAFDGQDGTNGYLRTKATYSKELPKSDKKTFLGCRGVNSKAE
VRVNNELVAVHEGGYSAFRVDISDYIRYGCVRDITADNSPDESUYPTMADFTFYGGVYRDVNLIEVPDCR
FTMNDYGS DGIYITPRCVGEDGWELSIKALIDNADCSHKARFTLIDADGNEKASTIADLRPIISAKLPVE
DPVLWNGRKNPYLYTVKCEIFDSTTQEVTDNIDIRTGLREYHIDSHKGFFLNGEHIKLQGVSRHQDREN
GNAITEKQHAEDVKIILDVGANSVRLAHYQQNSYFYDLCDESGLLVWAEVVISRFS PKRNENAEQQLVE
LIKQNYNHPSIFCWGIENEITIAGSASKKLITFLKHMHKVARHLDPTRYTTCAQLTMCPVNSPLNHITDI
IGYNHYWGWMGSCDEINNWLDEWHRENPEGKLCSEYGAEGITKYHSDSPVQGDYSEDYQAIYHEHYIK
AINERDWLWGSYVNMFDGFSASRNEGGVGRNNKGLVTFDRKTKKDAFWLYKAYWSDKKFVHITGERYV
NRPAGKSTIKVYSNCPEVTLEVNGKTYTKKADRVFVDDIELKLGENVIKARGGKVRPSHTITVNGIEKA
TDEYKMPAEYSSFVRNWFVATDGE GPDNKL SVEDSMGEVLSNGEVNKIIKNAAGDFAAKAVSSPLLKPIY
PVKVKTMIKLAGKLGVDKMLSLAGGFLQTIDKN

>SRS015217.97161-NL *Roseburia hominis*

MRNTIVLEKDWTFYKNPQSESGEAVTLPHWTNAVDGQDGGNDYYRGTC KYVRHFAPKELEKGGRAYLEFN
GAAMTADV VVNGTKLFHHEGGFSTFRVDVTEQLTEDNLLLEVYVDNSDNTKVYPQKADFTFYGGLYRMVKL
VTVPKVHFMVDYAGGNMKTPEVTILDAAEQADADVTVELWMTGEATDVTVAVAGETQTPVVENGYAR
AVFALKNVHLWDGVDPPYLYTAKAELPGGDVVERTFGCRSFKVDSREGFFLNGRSYPLRGVSRHQDRAGA
GNALTYHEMHREDMAIVRELGANTIRLAHYQHAQEFYDLCDENGIIVWAEIPYITMHTMADGTENTLSQMKE
LIVQNYNHPSIVCWGLSNEITAASAVNEELLENHRLNDLCHELD KTRPTVMADVFMLETDS PMLEIPDM
NSYNLYFGWYIGELDQND SFFDEYHSTY PDRVIGLSEYGADANPAYHSANPERGDYTEEYQCVYHEHMAK
MIEERPYLWATHVWNLFDAADGRDEGGRHGENQKGLVTMDRRIKKDAFYVYKAYWSKAPFVHLCGSRYT
DRAEDVTEIKVYSNQKVSLEFVDGA EKETKEGARIFRFRVPITGHTTIRAVSGDCTDEITVRKVDEPNPD
YIFNKQGDVVNWF DKKEDFKADHYSISDTL GELAKNEMANAIVQSLMAQASASRGDVAESVKDNPALQRM
QRM TLASLLKQAGDAVSEEQMKALNDALQKIPKN

>SRS019161.7410-NL *Faecalibacterium prausnitzii* SL3/3

MSKTILLNHSNWHFTKENTGIPTAIMGEAITLPHWTNAVDGQDGGNDYYRGTCWYTLALEKPAIPAGGKA
VLQLDGAAMTAAVYLNGEKLA EHKGGYSTFRVDLTNHLKDENLLAISVDNRDNDTVYPQKADFTFYGGIY
RDVTLHIVPAEHFALPANGAPALKATPIVTDLAAKTVEVTLEAAVVGASVTFALEGQSIAAPVENGTA
AVFTLHDAHLWDGVEDPFLYTVSAKLGNS EASARFGCRKFEFD PQKGFI LNGRSYPLRGVSRHQDRKGM
GNALTKEEM EEDISIILEMGANTIRLAHYQHAQYFYDLCDEKGLVIWAEIPYITMHTMNGRANTLTQMEE
LIVQNYNHPCIAVWGLSNEITAASAVNEDLLENHRLNL ELAHLKLDPTRKTTMANVFMLETTSPILEIPDV
NSYNLYFGWYLGELEQND E FDKYHADFPDRCIGFSEYGADANPQYQSSHPEKGDYTESYQCVYHEHIAK
MIADRPWLWATHVWNMFDAADGRDEGGKHGENQKGLVTFDRKIKKDPFYLYKAYWSKEPFVHLCGSRYV
DRAEDVTEIKVYSNLPEVSLYKDGQLVETKQGDKVFTFQLPITGKHSIEARSGEHSSVILVNKVDAPNPD
YAMDNRKNVTNWF DGEDESCWSVKDNMAAAMADPKAGPILKQISDKAAAARGDVAAAVKDNPALVAMME
RAMQRM TIQSMMLQAGTSEEDIKQLNRVLQGIKE

>SRS050422.26053-NL *Faecalibacterium prausnitzii*

MSKTILLNHSNWHFTKENPGIPTAVMGEAITLPHWTNAVDGQDGGNDYYRGTCWYTLALAKPAIPAGGKA
ILQLDGAAMTAAVYLNGEKLA EHKGGYSTFRVDLTEHLKEENLLAISVDNSDNDTVYPQKADFTFYGGLY
RDVTLHIVPAEHFALPANGAPALKVTPIVTDLAAKTVEVTVEAAVVGASVTFALEGQALTAPVENGTA
AVFTLDHAHLWDGIDDPFLYTVSAKLDNGEEASARFGCRKFEIDPQKGFFLNGRSYPLRGVSRHQDRKDA
GIALTNEMMEEDIALILEIGANTIRLAHYQHAQH FYDLCDEKGLVIWAEIPYITMHTMNGRANTLTQMEE

LIRQNYNHPCIAVWGLSNEITAASAVNEDLLENHRLNLNELAHKLDPTRKTTMANVFMGLIDSPILEIPDV
NSYNLYFGWYLGELEQNDEFFDKYHADYPDRCIGLAEYGADANPQYQSSHPEKGDYTESYQCIYHEHIK
MIADRPWLWATHVWNMFDAADGRDEGGKHGENQKGLVTFDRKLKKDPFYLYKAYWSKEPFVHLCGSRV
DRAEDVTEIKVYSNLPEVSLYKDGQLVETRKGDKVFTFQLPITGKHSIEARSGEHSSVILVNKVDAPNPD
YAMDNRKNVTNWFDGELDESCWSVKDNMAAAMADPKAGPILKQISDKAAAARGDVAAAVKDNPALVAMME
RAMQRMPTIESMLMQAGASEEDIKQLNRVLQGISKE

>SRS024331.215647-NL *Roseburia inulinivorans*

MRKIIILLDKNWTIFYKNPDSTQGEVSLPHTWNAVDGQDGGNDYYRGTCCKYECFHSRPELETGKAYLEFD
GAAMTADVINGTKAVHHEGGYGAFRADITQYLKDENKLEVYVDNSDNTKVYPQKADFTFYGGLYRMVKL
VTVSKNHFELDYAGGTGIKVTPVTLTDAAAKKADAKVTVELWMVGDAEAVTVTIDGQQKETPVENGYAK
AVFDLANVHLWDGVDDPYLYTAKAELSDGDIVETTFGCRSFSIDPQKGFFLNGRSYPLRGVSRHQDRAGA
GNALTHERMHREDMEIIKELGANTIRLAHYQHAQEFYDLCDENGMIWVAEIPYITMHMSDGTENTLSQMRE
LVVQNYNHPSIVCWGLSNEITAASAVNEELLENHRKLNLDLCHALDMTRPTVMADVFMLETDSPMLEIPDM
NSYNLYFGWYLGELEQNDSFFDAYHTTYPDRVIGLSEYGADANPAYHSANPERGDYTEEYQCIYHEHMAK
MIEERPYLWATHVWNLFDAADGRDEGGKHGVNQKGLVTIDRKLKKDAFYLYKAYWSKTPFVHVCGRRYV
ERTEDVTEIKVYSNEQSVTLFVDGVERETSEGSKVFKVPITGTHKIRAVSGTCEDEITVKKVQKSNQD
YIFNKQGDVVNWFDRDDFKADHYSISDTLGELAQNEMANAIVQRLMAQASASRGDVAESVKDNPALQRM
QRM TLVSLKQAGDAVSEEQMKALNDALQKIPKNQA

>SRS016954.181615-NL *Roseburia inulinivorans*

MRQTTVLNWKWIFYKNPDSGEGENVTLPTHTWNAVDGQDGGNDYYRGTCCKYACTFEKPEMEQNGKAYLEFD
GAAMTADVFNGAELFHHEGGYSRFRVDVTEHLKEQNQLEVLVDNSDNTKVYPQKADFTFYGGLYRMVKL
VTVPENHFCMEYAGSNGIKVTPVTLTDAGAKKADAEVTVELWIEGKSADVMVTVGGEKKSVPEHGYAK
AVFDLTDVHLWDGVDDPYLYTAKAELSDGDSVETTFGCRSFSIDPQKGFFLNGRSYPLRGVSRHQDRAGA
GNALTNEMHREDMEIIKELGANTIRLAHYQHAQEFYDLCDENGMIWVAEIPYITMHMSDGTENTLSQMRE
LVVQNYNHPSIVCWGLSNEITAASAVNEELLENHRKLNLDLCHALDMTRPTVMADVFMLETDSPMLEIPDM
NSYNLYFGWYLGELEQNDSFFDAYHNTYPDRVIGLSEYGADANPAYHSANPERGDYTEEYQCIYHEHMAK
MIGERPYLWATHVWNLFDAADGRDEGGKHGVNQKGLVTIDRELKKDAFYLYKAYWSKTPFVHVCGRRYV
DRTEDVTEIKVYSNEQSVTLFVDGVERETAEGSKVFKFEVPITGTHKIRAVSGACEDEITVKKVEKSNPD
YIFNKQGDVVNWFDRDDFKADHYSISDTLGELAQNEMANAIVQRLMAQASASRGDVAESVKDNPALQRM
QRM TLVSLKQAGDAVSEEQMKALNDALQKIPKNQA

>SRS014923.307745-NL *Roseburia hominis*

MELKDNWFMVKEAENAEDTASKEGTAISLPHTWNAVDGQDGGNDYHRGTCWYVTKFQKPELEAGSQVYVE
FLGASVIADVNLNGEAVAHHEGGYSTFRVNLTDKLEENVLAVALLNADNNYVYPQKADFTFYGGLYRMV
NLIVVPESHFELDYAGGNGIAVTPTECEDENRVPTGKASVKVETWVTGNADSVVITITGEESESKTVSVV
DGHAEEATFELEHVHLWDGVDDPYLYHAKAELSSGDVTETTFGCRSFTYTDPEKGFVLNGRVYPLRGVSRHQ
DRTGAGNALS YEMHKEDMEIIKEIGANTIRLAHYQHAQEFYDLCD EYGMVVWAEIPYITMHMPNGRANTL
SQMEELVVQNYNHPSIVCWGLSNEITAATPVDEDLLENHRLNLNDLCHKLDKTRFTTMANVFMQETDSPLL
EIPDVNSYNLYFGWYLGELEQNDEFFDEYHAKYPDRCIGFSEYGADANPQYQSTNPTHGDYSETYQTVYH
EHMLKMIEERPYLWATHVWNMFDAADGRDEGGKHGVNQKGLVTMDRKLKKDAFYLYKAYWNKEDAFVHI
CGSRYVDRKEDTTEVKVYSNQTTVSLYVDGTLLETQDGSRIFRFAVPMTGEHEIVAKAGAEEDSIHIRKV
EEANPDYVMGEIKDVINWFDDEPYKPECYSIKDKMSEIKASPKAGAILAELMAQAPGAESRGDVAESVKD
NPNLVRMMGRMTLESMLGHMKDSITEEQVKGLNRMLQQIKKGE

>SRS015065.66511-NL *Faecalibacterium prausnitzii*

MREIISLNENWTLSPFKGERPTEQVTLPTHTWNAVDGMDGNGSYLRTTGVYTRTFTQPKQPLAGGRITYEV
LAAALAATVKVNGQVATTHEGGFSIFRADVTDLCHEGDNELTIEVSNEEDTPSMYPASADFTFYGGLYRGV
NLISVPGTHFDLDYFGGPGIKVTPKPTEDGGATFEIESWVTNGEEAYTMYSIEDCYGNEVASGVRPCDS
TKVTLYVPDAQLWSMDEPNLYTVTARLQKNNETFDEIYANVGVRSYTVTPNDGFSINGVPTPLRGVARHQ
DRLYKGNALTAEDHYEDAMLIKELGANTIRLAHYQHSQDFYDACDELGFVWAEIPFISVFKKDPSAHQN
CRHQMTIELIQNYNHPSIMFWGLSNEILIGGICQELVDNHHDLQKLVRELDPTRLTITIAHVSNTPVDGPM
HHITDVESYNHYFGWYGKMEQNGPWLDKFHAEHPDICIGISEYGTEGIINWHSNDPQCKDYTEEYQALY
HEHLAQVFEDRPVWATHCWNMFDFGCAARNEGGAAGRNNKGLMTIDRKTCKDSYFVYQAYWSKQPMVHI
AGRRAHQRAGETTEIKVYSNQDTVVLYVNGKEVGQQTARHVFKFNVALEEGFNTILAVAGDVKDSITLEK
VEKEPDCYTLPEFNERQEGVANWFKQVGSLLDKAPMEFPEGYYSIKDSMEDLSKNEEALALATRAVKLAT
NFDVKPGVGMWDMMKRMTPEAMAKMINMPDGFIESLNAQLIKK

>SRS015264.38260-NL *Faecalibacterium prausnitzii* L2-6

MREIISLNEGWTLSFPPKGERAAEAVTLPHWTNAVDGMDGNGSYLRTTGVYSRTFKKPVQPLAGGRVYVEV
LAAALDATVKVNGTVATTHEGGFSIFRADITDLCDGDNELTIEVSNEEDTPSMYPASADFTFYGGGLYRGV
NLISVPNAHFDDLYYGGPGIMVTPKPTADGGATFEIKSFVTNPDDSFVTMYSIEDPYGCEVASAVRPADS
TAVTIYVPDAELWSMDEPNLYTVVARLQRNNEAFDEIYANVGVRSYTVTPDGGFSINGEATPLRGVSRHQ
DKLYKGNALTVEDHYQDAQIIKELGANTIRLAHYQHSQDFYNACDELGFVWAEIPFISVFKSGKDAHEH
CRRELTELIVQNYNHPSILFWGISNEILIGGISQELVDCHHDLQKLCKELDPTRLTTIAHVSSTPTTGPM
HRITDVESYNHYFGWYGGKIEQNGPWLDKFHAENPDICLGISEYGCEGIINWHSNTPQCKDYSEYQALY
HEYMAQAFEDRPWIWASHVWNMFDFGCAARSEGGVKGRRNNKGLVTIDRKTRKDSFYVYQAYWAKDPMVHI
AGRRAHQRAGETTEVKVYSNQDTVTLTYCNGKEVGTQTAHRVFKFNVALDEGFNVLMVAVADTVKDSITLEK
VDAEPACYTLPEFNERQEGVANWFKQMGSMDLTAPMEFPEGYYSIKDSLEELAKNEEAFVAAKAVKLAS
NFDIKPGEGMWAMMKMTPEAMAGMISTMPDGFIESLNAKLKIKK

>SRS015663.18708-NL Faecalibacterium prausnitzii

MREIISLNNWTLSFPPKGDHATEQVNLPHWTNAVDGNDGNGSYLRTTGVYTRTFTAPKQPREGGRITYVEV
LAAALNSTVKVNGQVATTHEGGFSIFRADVTDLCHEGENELTIEVSNEEDTPSMYPSSADFTFYGGGLYRGV
NLISVPDAHFDLDLYYGGPGMMVTPVPTEDGGANFTIKSFVTNPADDLTMYSIEDCFGREVASAVRGSAD
TEVTIYVPDAQLWSMDEPNLYTVVATLQRNNEEVDEIAANVGVRSFKVTPDEGFSINGVPTPLRGVSRHQ
DRVFEGNALTAEHYDDAMLIELGANTIRLAHYQHSQDFYDACDEIGFVWAEIPFISVFKKGEGAHKH
VMEEMKELIIQNYNHPSIMFWGISNEILIGGISQELVDTHHDLKELCKELDPTRLTTIAHVSTTPVNGPM
HHITDLESYNHYFGWYGGKMEQNGPWLDQFHAHPDICIGISEYGCEGIINWHSNTPQCKDYTEYQALY
HEHMAQVFEDRPVWASHVWNMFDFGCAARNEGGVSGRRNNKGLITMDRKTCKDSFYVYQAYWTQTPMVHI
AGRRAHQRAGETTEIKVYSNQDTVVLVYNGKEVGQQTAAHRVFKFNAAALNEGFNTIVAVAGDVKDSITLEK
VAEPPGYTTLPEFNERQEGVANWFKQVGSMDLTAPMEFPEGYYSVKDTMEDLAKNEEALALAAKAVKLAT
NFDIKPGVGMWDMMKMTPEAMSNMISGMPEGFIESLNAQLIKVKK

>SRS017103.192921-NL Clostridiales bacterium KLE1615

MRKLISLDENWTLSFPPKGERLAEKVTLPHWTNAVDGMDGNGSYLRTTGVYRCTFLKPEQPLEGGRVYVEV
LAAALAAVVKVNGTVAVTHEGGFSAFRADITDLCKEGNNEMTIEVSNEEDTPSMYPASADFTFYGGGLYRGV
NLISVPHTHFDLDLYYGGSGIKVTPKPTEDGGATFEIESFVTNQEDCFTVMYSIEDACGCEVAAAVRPADD
TAVTIYVPDAKLWSMDEPNLYTVVARLQKNNETYDEIHVNTGVRYSVTPDGGFFINNKAIVSLRGVSRHQ
DRLYKGNALTAEHEVQDAKIIKELGANTIRLAHYQHSQDFYACDRLGFVWAEIPFISVFKSGKDAHEH
CRRELTELIVQNYNHPSIMFWGISNEILIGGISQELVDCHHDLQRLCKELDPTRLTTIAHVSFTPVSGPM
HHITDVESYNHYFGWYGGKIEDNAPWLDKFHADNPDICIGVSEYGCEGIINWHSNKPQCKDYTEYQALY
HEHMAQVFEDRPWIWASHVWNMFDFGCAARNEGGVGRNNKGLVTIDRKTRKDSFYVYQAYWSKEPMVHI
AGRRAHQRAGETTEVKIYSNQDSVTLYCNGKKVGTKEHRIHFHTVTLDIGINILLAVTDKVKDSITLEK
VEKEPDSYTLPEFNERQEGVANWFKQMGSLDLKAPMEFPEGFYSVRDTMEEISKNEEALAITSKALKLAT
NFDIKPGIGMWDMLKTMKPEAMTNMISGIQDGFIESLNARLIKIKK

>SRS023176.32079-NL Ruminococcus gnavus

MRRKIEINENWFFTKERPNTAPDFSAMQEITLPHWTNAEDGADGDDYYDRRECWYVRKLERPQGDEIYL
EVPAASQTAHVYLNAGAEIGHHIGGFSLFRIRLTERLEEKENILAIIVDNSESQDTYPQFADFTFFGGGLYR
GVNLLVPPERHFELEEYGGPGIYVTSHLNEDGKAEVLVQGVSGGQSEAYKVQIRNGEGEIILEANTEKP
EKVFFIDNPRVWNGKKDPYLYTAKMIMEGSGDEVSTRFGIREFYVDAETGFFLNQESYPLRGVSRHQDRQ
GKGWAISEEDQREDAMLIHEIGATAVRLAHYQHAQSFYDFCDELGMVWAEIPFISRFINTKEAKEDTVR
QMRELIMQNYNHPSICFWGISNEITMGGEDELVENQRILQKLCHKLDPSRKTVLNLTTVESQSEQNKI
TDL SAYNVYMGWYAGKVEDCGAWMDSLHKENS DMC MGISEYGADTNVQYHSDAPKRQDYTEYQSYHEK
MLQAIQERPYLWCSFVWNMFDAADKRKEGGTQGRNTKGLVTYDRKIKKDAFYLYKAYWTEKPFVHICSK
RFQKRPGDTTIIKVYATGIEKAEWLMDGKRIQEQKGTYCFLFEGIKLTQNHQITIIYGYCGDEKVCEDAA
FTHTDGLAEAYILES GENDGVN WFLDEYGEKKKLEATQGFLSVYDEIGTILDTEKGNEILNGLFISFGEG
GKALLTESVQNSIRSLTFVELAKMIGPAITPEMLNMLNEQLRHVKNS

>SRS016495.28558-NL Anaerotruncus sp. CAG:528

MRNIKNINDNWLFSKEAKSIPHVMPHDWEKVNLPHWTWNGTDGQDGGNDYYRGTCYYVKQLSLADFGDEDE
YYLQFDVAVNSSAKVYFNGELIAQHDGGYSAFRAKLGNIQESNLLVVAADNSPNDYVYPQTADFTFYGGIY
RSVKLIGVSKSHFDLDYFGAPGISVTPKVTGTGADVDVAVYVATDENASVRFEILDKDEVITSAEVSGDY
HANMHIDAVHLWNGVKDPYLYTLRAVLLVNGEIDCVSCRFGCRSFKIDPNKGFILNGREYPLRGVSRHQ
DRPQIGNALLPQHHKEDIDLICEMGANTIRLAHYQHAQEFYDLCDERGLVVWAEIPYISNHLPNARANTL
SQMKELIYQNYNHPSIVVWGLSNEITMGKKSTPDMISNHKELNELAHSIDSTRLTMTATVTMCQPDDEIN
AISDVLSYNHYFGWYGGNVHMYGPWFDFNHKKFPNKAIGISEYGCEALNWH TSEPEQGDYTEYQAYYHE

ELIKQISKRPYLWATHVWNMFDAADARAEGGENGMNHKGLVTFDRKYKKDSFYAYKAWLSDEPLVHICG
KRYVDRVEPVTKVTVYSNQPPQVELFVNGESLGKQTKGEYPPFFCFEVPNIGESKLIAGDCTDESVIKKV
EAFNEAYRMKEEGAVINWFEINTPPGYYSVNDTIGDIISTFKGKIFALKLVKILKKSLSASSGDKNGMSV
AGVKINKTMIEMAKGFTLKRALMMLGGKVSKEIDILKINDMLNKIKKKK

>SRS023176.83150-NL Lachnospiraceae bacterium 9_1_43BFAA

MKEILLNDNWLPHKENEVEGCVTLPHTYNAVDGQSGISMWRGKGFYKKEVLFSEEDLKEKIFLEIGASAM
KSTVYMNGKTIHNTCPYAMYRIALNEHIIVGKNLLEIVTDNTATDEVYPQMADFSFYGGVYRDVKLVFV
PQQLHFDYLDGSRDGVQVRTQKLDGQKWEVTVTGCVINEGKDCEGEVKVSVLDRQGVVFEKTLSMYFSEK
QAFTIRTEIDHPLHWMGVDPYLYQVQVEICTKDGVETRSIATGTFREITIDPEKGMFLNGKHIKIRGVA
RHQDFGGVGNVSKSEQMEEDLSLVKEVGANSVRLSHYQHDDYFYRRCDDEGILVWAEVFFISILSESEKA
DENIRQQMERLIKQCGNHISYCWGVQNEITIAGETAETHEKVKAMAAATKQLDPSRYTAEANIYSVANE
SPLNRMADLVGYNLYYGWYYDKMTGLQQRLEDFHMACPDVPLLVTEYGVDTNPGYHSYEPKTKDYSEEQ
LLFSENALKAFEERDYFVGSYVWNLCDFGSANRDEGGKQGGQNGKGLVTIDRKIKKDAYYLYKANWSKESF
VKLAGSRFVNRHQEENTITVLSNLKKLEVYVNGMLAGVKEQPEPMTNFTVVLPMGENKITVKGFDQVETV
YEDEMVLHRVTEPDKSYEYVQKDQGGNVINWFEKFDLEDAGTEEIALDPECYSSMDKLGEIMKHPKGEAI
ILKYFGEMVDHPKFAMMKVMTLDAMSKLKNLGIPKELILVINRELQQIKK

>SRS022071.228303-NL Firmicutes bacterium CAG:341

MREILNINKGWKFSKDVSSVPSLPDDWQDIDLPTYWNGDDGQDGGNDYYRGKGYFAKILTKDKMPQGEE
IYIQLDGVNSTAEVYFNGKKLCEHNGGYSAPRVKLDLIDENLLVVTADNSPNDYVYPQVADFTFYGGIY
REVKLIGVSKNHFDLDFYGSTSFVTPVISGNDAKITLDAYCNGEVEFEIDDGNEVVAKAKASGKNPSTV
LEIKDVHLWDGVDDPFLYNASAKLIVDGGVADEVKTRFGCRTFEIDAQKGFILNGRPYPLRGVSRHQDRP
HIGNALLPEHHKEDMDLICEMGANTIRLAHYQHAQYFYDLCDERGIIVWAEIPYISKHLEKGVENTISQM
KELIFQNYNHPSIVVWGLSNEISMGGATPAIVQNHKMLNDLVHSDKTRPTTEAVLTACNINEEYVHISD
VLSYNHYFGWYGGDVSMYGDWFDNFHKKYPNKAIGMSEYGCEALNWHTSNPEQGDYTEEYQAYYHEEVIK
QIAKRPMWATHVWNMFDAADARAEGGENGMNHKGLVTFDRKYKKDSFYAYQAWLSKKPMLHICGKRYV
DRVEDVTKITVYTNLDEVELFANGESVGKQKKGEFFFFFYFEVKNEGETTLVAKAGELTDESKIKKVDKFN
EDYRLKEEGAVINWFEIETPAGYYSVNDTLGDILSTFRGKICAVKLLMKMKKALTPKGPKKEGKKKQGA
IMGFKLNDINKTMIDMLKGFTVKRGLMMLGGKFTKEQILEINAMLNKVKKK

>SRS049712.200941-NL Eubacterium sp. CAG:251

MRRVINIDKDWKFSKEAFSVPTTLPDWSIDLPYTFNGDDGQDGGNDYYRGKGYFAKSLAKSDMPDGEE
VYLQLDGVNSTGTFVDFYNGKEICVHDGGYSTFRVKLDDIKDENLLVVVADNSPNDYVYPQVADFTFYGGIY
RDVTLIGVGNHFDLDFYGAAGIATPEINGDDAKITVDAYCDGEVEFEISDGNEVVARAKASGKNPSAK
LEIKDVHLWNGVDDPFLYTAVAKLIVDGEALDEVKTRFGCRTFEIDAQKGFILNGREYPLRGVSRHQDRP
HIGNALLPEHHKEDMDLICEMGANTIRLAHYQHAQYFYDLCDERGIIVWAEIPYISKHLEKGVENTKSQM
KELVYQNYNHPSIVVWGLSNEITMGGATSPIVENHKLMLNDLVHSLDKTRLTTEAVLTACNINEEYVHISD
VLSYNHYFGWYGGDVSMYGNWFDTFHKKYPNKAIGMSEYGCEALNWHTSHPEQGDYTEEYQAYYHEEVIK
QIAKRPYLWATHVWNMFDAADARAEGGENGMNHKGLVTFDRKYKKDSFYAYQAWLSKKPMLHICAKRYV
DRVEDVTKVTVYTNLDEVELFANGESVGKKKKGEFFFFHFVEVKNEGETTLVAKAGDLTDEAQIRKVDKFN
EDYRLKEEGAVINWFEIETPAGYYSVNDTLGDILSTFRGKICAVKLLLKMKKALTPDGPQKQKGGKGA
MGFKLSDINKTMIDMLKGFTVKRGLMMLGGKFTKEQILEINAMLNKVKKNRH

>SRS045004.190116-NL Eubacterium sp. CAG:180

MREVININKNWLFSKKEQVPVKTLPEDWESVNLPHWTWNGTDGQDGGNDYYRGKCCYVKLLKKADLGEKPV
HYIQDFGVNSSAEVWVWNGEKIGSHDGGYSAPRVRIPEISDENILTVYADNSPNDTVYPQVADFTFYGGIY
RDVTVIGVDESHFDLEFYGSSGIMITPKVSGSSAAVNISARVTNPQDCSVRFVVTADKKPVGEKNVDTS
DGKTVIEIENAHLWNGTDQPYLYSLTAELLKDGEKTDEISVRFGCRSFSIDPQKGFILNGKPCPLRGVSR
HQDRPGIGNALTEKEHREDMDLICELGANTIRLAHYQHSRVFYDLCDERGMVWAEIPYISRHMPGGREN
TVSQMKELIYQININHPSIIVWGLSNEITMNGASDSSLIENHRMLNDLVHKIDPTRPTTIAVLSMCDPGEE
YVRIPDVLSYNHYFGWYGGKTDMYGPWFDFKHKKYPDRAVGMSEYGCEALNWHTSDPQQGDYTEEYQAKY
HEDVIRQIAVRPWLWSTHVWNMFDAADARSEGGENGMNHKGLVTFDRKYKKDSFYAYKAWLSDEPFVHI
CGKRYIDRPESVTSVTVYTNPSVELFANGKSLGVQKRGEFFFFYFVSPNEGETVLTAKAGDCTDESRIR
KVDKANPDYVLQEEGAVINWFEIETPPGYMSVNDTIGDILATAKGLLALKILKMVRANMKKNKGKSTGG
MADMAKGMKINKSIIEMGKGFVSKRVCMAGGLFTKEQILEINASLNKIKKKQK

>SRS014923.1295-NL Clostridium sp. CAG:217

MRQIININKDWRFCCKPQSTPTALPTDWSVDLPHTWNGTDGQDGGNDYYRGKCAYAKALTADELGSAPV
HYLQDFGVNSSAAVYWNGQQIAKHGGYSTFRAKLPKILPENLLVYVADNSPNETVYPQMAFTFYGGIY

RDVTVLGVESHFDLDYYGAPGVQVPTMQGTDAATVAATAYVTAPAGCTVHFAITNMDGDPVAEADADAA
DAKTNIKIENAHLWHGTEDPYLYTLTVTLQNGKAVDEIATFRGCRSFAIDPQKGFILNGKPYPLRGVSR
HQDRPGIGNALTAKHTEDMDLICELGANTIRLAHYQHSQTFYDLCDERGMVWVAEIPYISRHPGGKAN
TISQMTLICQNSNHPSIVVWGLSNEITMNGASDPSLLENHRLNDLVHKMDPTRPTTIAVLSMCDPGEA
YVQIADVLSYNHYFGWYGGKTEMYGPWFDFHKKYPDRAVGMSEYGCEALNWHTSDPQQGDYTEEYQAKY
HEDVIRQIAARPWMWSTHVWNMFDAADARSEGGENGMNHKGLVTFDRQYKKDAFYAYKAWLSKEPFVHI
CGKRYVDRTEDVTKVTVYTNQPQVELFANGKSLGVQQKGEYPPFFYFDPVNSGETTLTAKAGDCTDESHLR
HVNEPNRDYVLQEEGAVINWFEIETPPGYMSINDTIGDILATTRGKLLALRIVQMVRANMKKNKGGSTGG
MADMAKGMKINKSLIDMGKFTVKRVCMMAGGLFTKEQILEINAKLNKIKKKQK

>SRS015578.91631-NL Eubacterium sp. CAG: 76

MATKNINASWQFTKDAITVAEGYSVLTQGNWEALDLPHTWNGKDQDGGNDYHRGTCYYVKNMKREEFGS
EPITYIEFNGANSSAWLYVNGKEAGHHDGGYSTWRVNVTDFLADENEIVVAVDNAPNDHVYPQMAFFTFFY
GGLYRDVNVISVPETHFDLDYYGTHGIAVTPIVEGANASVEVEVFVTDATEDTLEYVIKNGDEVIATKS
VSANETKVTFEIENVHLWNGRKDPHLYTAEVTLKRNDTVLDERSTRFGCRTFVIDPEKGFILNGERYPLH
GVSRHQDRPHIGNALTHKEHKEDIELILEMGANTIRLAHYQHDQYIYDLCDDETGLVLWAEIPYISKHLPN
GRENTISQMKELIIQNYNHPSIVVWGLSNEITMGGGEDDDLIENHNILNDLCHQMDPTRLTMAVVMCS
IDAKYIQIPDVISYNHYFGWYGGTTDMNGEFFDDFHKKYPNIPIGLSEYGCEALNWHTSNPVQGDYTEEY
QAYYHEELIKQLFTRDYIWATHVWNMYDFAADARAEGGENGMNHKGLVTFDRKYKKDAFYAYKAWLSDDP
FVHICGKRYVDRVEDTTKVTVYSNQPEVELFVNGESLGKQTSDVHFFYFDPVNAGESTLVAKAGDCSDS
KIRKVDTFNEEYRLKEEGEVLNWFIDIMPEGYFNINDKIGDIMKANEGKMFIDELMLKIMKGADGGAAEA
AEAAAGAGSEGLMKMLSGFTVKRMLNLMGTAGGGKQFTKEELLELSKLNKIKR

>SRS023176.73160-NL Lachnospiraceae bacterium 9_1_43BFAA

MREIININEGWYFSKEQKELPTACEESWEVVDLPHTWNQYDGDGSDYRGACWYVKELGPYTVKPHSR
IYLEFQAAANVCVAYVNGKKVCSHEGGFSIFRADITEQLDENGENLIAVCVDNAPKSNVYPQMAFFTFFY
GLYRDVNLVIVNESHFDLDYYGAPGIKVVSKIQGENALVDIHSYVTNPLEGDRVEFEIYDEDGETKVFV
RPATQETQVAVELLDAHLWQGVDPYLYTVSARLVHRNEVLDEIETRIGIREYTVDPQKGGFFLNGKSMPL
RGVSRHQDRYDLGNALTFFEEHYEDILLIKEVGANTIRLAHYQHSQDFYDLCDDEAGFIWAEIPFISAMNA
DPDAHENCERSQMRELIFQNFNHPSICFWGISNEITIGGEVPGLVENLKDHLALVHELDDTRMTTMAQVSN
LPMENEQNDITDVVSYNHYFGWYGGKLEDNEAWLDAFHEMHPERPIGISEYGCEGITTYHNDNPKGGDYS
EEFQAVYHEHMAKIIERPWLWATHVWNMFDFGCDARNEGGVKGRRNKGLMTIDRLIRKDAFYLYKAYWT
EEEFVHICSKRYAQRTTDTVTIKVYTNCPEVTLSIGRVIEWKQAGNKVFTFENVPLQEGFTTIVAKAGNC
TDTITLEKVAEANPSYIFEESEDDKVAGVTNWFNDVLSVVRPMEFKEGYFSLKDLVKDILKNTEACQVL
LNLGSSTMGINIKKSMLEVMGDRPAMDGMANSNGDEKKMAVMAYINTELQKIKKED

>SRS023176.7378-NL Ruminococcus gnavus

MRNVISLKNWAFIQENAGLPGEMPTDWHKVDLPHTWNAVDGHDGNGSYDRGTYWYAKTFETPKQPLGGG
KVYVEILAAGQQATVYVNGKEVITYHEGGYSTFRADITDVCHEKGDNLLVIACSNEMKDSVYPQSADFTFFY
GGLYRGVNLISVPDAHFDLEYYGGPGIQVTPKPCCEGGAAFEIVSYVKETDENFTVLYAICDAEGNEVAS
ACRPADETTVTYVVPDAKLWEIDAPYLYTVTAKLQRRNETYDEISARVGVRFSFCDPNKGFIIINGAETPL
RGVSRHQDMLYKGNALTKEHDYEDARLIKELGANTIRLAHYQHSQDFYDACDEMGFVWVAEIPFISVMNQ
DPDAHQNCIIQLKELIIQNYNHPSICFWGISNEILIGGISEQLVENHKELNALAKEMDPTRLTTIAHVSM
TPIEGPMHGITDVESYNHYFGWYGGKMEDNGPWMDNFHKIHPICLGLSEYGCEGIITYHGPNPACKDYS
EEYQALYHEHMAKMLEERPWIWSSHVWNMFDFGCAARNEGGVAGRNNKGLMTIDRKVKKDSYIYQAYWN
KKPMVHLCGKRYAQRAGEMTQIRVYSNQPSVILFLNGEKVEELSADKVVFVFTVALKDGFNIITAQAGEVQ
DTMTLEKVEKEPEIYVLPEVNERRAEGVANWFQTVGMDMLKAPMEFPEGYSIKDTMEELAKNPEAMEIAS
KAVKLMANMIVSPGEGMWDMMKSMSFERLSEMAGDLAPEGFVESVNAKLIQIKKA

>SRS023176.5275-NL Ruminococcus gnavus

MRNVINLNKDWAFIQQAGLPGEMPSDWEHVDLPHTWNAVDGHDGNGSYDRGTYWYAKTFETPKQPLGNG
KVYVEILAAGQQATVYVNGKEMTYHEGGYSTFRVDITDVCNEEGDNLLVIACSNELKDSVYPQSADFTFFY
GGLYRGVNLISVPDVHFDLEYYGGPGIQVTPRCPCEGGAAFEIVSYVKGADENFTVLYSVCDAGNEVAS
ACRPADETAVKVYVPDATLWEIDAPYLYTVTARLQRRNESYDEISARVGVRFSFCDPNKGFILNGTETPL
RGVSRHQDMLYKGNALTKEHDYEDARMIKELGANTIRLAHYQHSQDFYDACDEMGFVWVAEIPFISVMNQ
DPDAHQNCISQLKELIIQNYNHPSICFWGISNEILIGGISEKLVENHKELNALAKELDPSRLTTIAHVSM
TPIEGPMHGITDVESYNHYFGWYGGKMEDNGPWMDNFHKVHPEICLGLSEYGCEGIITYHGPNPACKDYS
EEYQALYHEHMAKMLEERPWIWSSHVWNMFDFGCAARDEGGVAGRNNKGLVTLDRKVKKDSYIYQAYWN
KQPMVHLCGKRYAQRAGETTEIRVYSNQPSVTLFLNGEKVEELSAEKVFVFTVALKDGFNILTAQAGEVK

DTMTLEKVEKEPEIYVLPEVNERAEGVANWFSTVGDMDLKAPMEFPEGMYSIKDSLEELAKCPEAIEIAA
KAVKLTMMNVSPGEGMWDMMKGMSLERLGEMAGSLAPEGFIESLNGKLIQIKKV

>SRS015217.229301-NL *Roseburia hominis*

MREVINFNKTKWAFTEATEVPKEMPEKWWVTLPKSHWNEIDGQDGGNDYYRGTCYYAKQLKKSELPEADC
YYLELRGANASADIYVNGKAVAHHDGGYSTWRVDITKELTEENLIVIAVENGVNDRVYPQNADFTFYGG
LYRDVNIIVAVNKSDFLDYYGGPGIKVTPEIKGADASVEVEVFLTNAAADQKLVTYTKDAEGKEVAKTET
AADETKAVLSIPAVHLWNGKKDPYLYTAEVALVSGEETVDAVSTRFGCRTFEIDPERGFILNGEYPLRG
VSRHQDRWGIGNALLPEHHREDIDLICELGATTIRLAHYQHDQYFYDLCDERGLVIWAEIPYISSHMPNG
RENTISQMKELVVQNYNHPSIVVWGLSNEITMAGSSEDELLNHRILNDMVHEMDHTRLTITIAVSMCDI
HDPYIQIPDVISYNHYFGWYGGDVSMNGPWWDNFHKFPNIPLMGSEYGCEALNWHTSDPKQGDYTEEYQ
AYYHEEMIKQLFTRKYIWATHVWNMFDFGADARNEGGENGQNHKGLVTFDRKYYKDSFYAYKAWLSDEPF
VHLCGKRYVDRVEDTTKVTVYSNLPEVELFVNGKSAGKLQAEDHFFHFVFPNVGESTLVAVAGEYKDESH
IRKVDTFNEEYSLKESGAILNWFDITEPEGYYSLNDRLSDIMKSEEGKALFMGLMSKVAAGMSQGNEKND
GNPAAGAMANPKMLEMLGGFTVIRMINLMGAAGPKVEWKKEDLLGLNAQLNKIKRVD

>SRS011239.2499-NL *Firmicutes bacterium* CAG:475

MRTTKIINDGWRFSGLTSEVPANLPSPDWQELNLPYTWNKGDKGQDGGNDYFRGAGCFVKELKKQDFPSGEE
VYLQIDGANSTAKVYFNGKHVCTHHGGYSTFRVKLDDVKENLLAIIVDNSPNDFVYPQVADFTFYGGIY
RDVSLIGVNRKHFDLDYFGAPGIKITPTVCGKDAKVEVQAFCEGEVEFSILADGKEIATKSVSGHNPIAV
LDIQDVRLWNGRKPPLYTMVAKLVCDGEVADEVSAFGRCFEIDPQRGFLLNGKEYSLRGVSRHQDRP
GIGNAISKEDHKEDVDLICEVGANTVRLAHYQHAQYFYDLCDERGLIVWAEIPYISRHMPLGKDNTFSQM
RELICQNYNHPCIVVWGLSNEISMGSAKGLVENHRELQKLVRS�DKTRPTAAVLTACDTGEEYVHVSD
VVSYNHYFGWYGGDVHEYGKWFDDFHKKYPNKAIGLSEYGCEALNWHASHPEQGDYTEEYQAYYHEELIK
QIAKRPYLWATHVWNMFDFGAADARAEGGESGMNHKGLVSFDRKYYKDAFYAYKAWLSSEFVLHICSKRYV
DRTEDTTLVKVYSNQPIVELYANGTLVERKEKEGFPPFEFQVKNEGETLLEARAGELRDEAKIRKVDTPN
MDYVMKDDHCVINWFEINTPVGYLSINDTIRDILKPFKGKIFAVKTLLSLKKAMSSGKKEKSDKKKKGK
ISVMGVNLRDVNKS MINMAMGFTLKRALMMLGGKFTKEQVLKINDDLNKIKKKKQNRKV

>SRS024132.99137-NL *Clostridium* sp. CAG:62

MRRILDFNENWWFIHADVDVSEARPEGELVSLPHTWNAALDGQKGDGTYDRGSYVYVKMLDKWEPSAKYS
RLFLEIPAAGQQAQKVFVNGEFVVSHEGGYSAFRADITEKCTEEFNITAIWCSNEKCDAIYPQSADFTFYG
GLYRGALHICVPETHFELLHYGDLGIHAATEPASWDKEDFMLHIRTICGVEKDNQVIYQIFDEGKEVA
SAARPADSPDTSIPLYKPHRWDIKNSYQYTLKALIMRNECLDEVELKFGVRTFCCDPGRGFLINGNCVP
LRGVSRHQCPLYQGNALTKEDHIRDAELIEEIGANTVRLAHYQHAQEFYNECDKRGFVVWAEIPFISVMN
EDPLAHENCITQLKELIYQNDHHPISICFWGISNEILIGGISEKLVENHRELNALAKKIDPYRLTTIAHVT
MTPKDGPMHEITDVESYNHYFGWYGGKTEDNGPWWDEYHEKYPNRCIGLSEYGCEGIITYQGNPKCKDY
SEAYQAGYHEYMAHMLERRPWIFCSYVWNMFDFGCAARSEGGVTGRNNKGLVTMDRKIKKDSFYLYKAYW
NPEPMVHICDKRYSLSRGENTQIRVYTNQERVTLFVNGEEMAVNEVKRHVASFPISLSGGKNAILVKAGD
VWDAVTIDRVEKEPESYVFPEASEREEGVANWFQEVGSLNLSEEMKYPEDRYHIRCTLEEISENDEAMEL
VTKAMKLITGMTLAKGEGMWDMMKMKLESMEKMLGAMAPDGFLSLNMKLNEIKKENLNG

>SRS012273.319820-NL *Subdoligranulum* sp. 4_3_54A2FAA

MKHILPINEGWAFVKGCEAPAAIPAGAARVDLPKHCWNAALDGQDGGSDYWRGPCCYLKTIEKPAGKNVYV
EVEAASSVGKIIVNGVEKASHKGGYSLFRANVTDALHDGENLLAIWVDNTQRDDVYPQTADFTFYGGLYR
GVNLITVEDSRFDLDYCGGKGFVSAAVEDGAAVVS�DAFITNPVEGQSVEFIVRDAAGVIVAGAVRPAA
ERVHAEVRIENAHLWQGVKDPYLYAVEALLTVHNETIDNVSANLGVRTYTVDPQKGFFLNGVLTPLRGVS
RHQDRLGQGNALTKEQHEEDVAFIREVGANTVRLAHYQHSPYFYDACDRAGLVVWAEIPFISVMNPAPGA
HENCRSQMKELIVQNYNHPSICFWGISNEITIGGERPGLLDNLRDLNALAHEMDKTRMTTIAHVSMPME
NDMHITDVLSYNHYFGWYGGELENNEQWLDKFHALHPDRPLGISEYGAEGIVSYHSDTPKCKDYTEEYQ
AVYHEHLAKVIDERPLWATHVWNMFDFGCDARDEGGVKGRNNKGLMTLDRKIRKDAFYLYKAYWSDEEF
VHLCSTRRYAQRTEETTQVYVSNLKPVALYVDGRLFAEQEGSKVFVFEHVPMSDGFTQISARAGACADAM
SIEKVAEPNQAYIYVDPDDTGDDAGANWFNVDDYKDVDTIVVREGYFSVKDKIGDIMKNEEAGNVLMQF
MQATVKMKLKKSMGLMVKDMTLEGMAGMASSMGIGGKAGADPEQGRMLITLNEMLNKIKK

>SRS055982.32560-NL *Bacteroides salyersiae*

TKVSITDNWKFTPSDSENFASSGLDDSKWQTVSIPHTWNDKDIIDEQRGYRRAASWYRKHLNIPLEMKGR
KIVLLFEGVGSKADVYLNGEHLKTHMGAYTAFTVDITGHCKFGEKNILAVRADNSGSLSEVLPPVSGDFS
MLGGIYRRVFLQTYHPVHFDPEPYASTPVLVQTPVVSEESAQVKLSCRKNDADKNSKLSLVFALLDKEG
KVVSRKKMKVKVDVGQSLPVDLEMASVNNPHLWSPDSPYLYTLKATIVDDKESTLLQEVDSPLGFRWFSV

DKTGFYLNGEYLKLRGAARHQDYWGLGAAIPSEINYQDMKLLKEMGANFIRISHYPQDPEIYRACDELGL
VAWSEICIVNEVKKNETFALNSAEMLKEMIYQNYNHPSVVMWGAMNELWDYHDEAIRLAKELERIKKELD
PYRLSCVAFHAFTWEKPYKQNSKEMFNISDINAVNVYESWYHGNFSTITPMFDEFNRNYSNKPRFLSEFG
AGSDERVHTYSPRTFDFSPFEQLAFNREYINQMESRPDYVGYSIWNLIDFQVDGRGDSKPNLNQKGM LTS
DRKKKEIYYYYQARWSKEPMIHFAGDWTERVMVCDGEVSRPVTVFSNQKEVELFHNGKSLGSHPVVNG
EAEFDVFFVDGDNRLKARCGELEDILNISMVLLPSKLADNKRLSEGLYINMGQDHCYFTDPLIRKTWLPD
QPYRLGSWGYVDGKPFNSWPGSSHDGVRNGIGTDIKGTGLEPLYQTFHMGATAYRLDVPDGHYEVTFCFA
EPFNDREKDKGHTGVSENGERIFDVEVNGEMVAQRLNMAEEYGVQTAFTKTILITVSGGEGLDIRFHSY
EGQSVVNLKVLKLC

>SRS011529.32451-NL *Bacteroides dorei*

SEISITDSWKYKAENDERFSSMDWNDSWVTVDLPHWTNAGDVIDEQRGYRRGISWYRKKLFIPSEARDK
KITLRFDGVASKADVYLNGLKLLKTHLGAYTAFGVDITDICEVGKENLLAVKVDNSSSLGEILPPVSGDFS
IFGGIYRRVFLQWTEKVHFVTEPYAAVPVRIQTPEVSVSEASMQIVAFLKNDFTDTKHVHVNVFLCDEM N
RIVKEKQLKLLKLIPIGRKYPISTSVGRIENPHLWSPELPYLYTVKVQVCDANGEMYQEVISPVGFRWFSV
DKTGFYLNKGKYLKLRGAARHQDYAGLGTAIPVEMNRRDMRLKEMGANFVRISHYPQDPEIYRACDELGL
IVWSEICVVNEVRKNAFAHNCKEMLKEMILQNYNHPSVVLWGAMNELWDYHKQAIALARELEALKKELD
PYRLSCVAFHAFTWEKPYTQSSKEMFISISDVNGVNVYESWYQGDSATIAPMFDKFCSSYSTAKPRFLSEFG
AGSDERIHSYTPRTFDFTPFEQLDFNRRYINEMEKRPDIYGYSIWNLVDFQVDGRGDSKPNLNQKGM L TE
DRKKKEIYYCQARWSDIPMIHIAGADWTKRVEICDD SINVRKISVFSNQKTVELIHNGKSLGVREVVNG
EAVFAVPFINGENLLDARSGALSDRLKIQMKLLSSRLTDSVDVLLDGLCINLGQEHCFIDPQLQEIWIPD
KPYTKGSWGYMDGKPFNSWPGSSHDGVRVGVGADIKNTFLEPLFQTFLLGTTCYRLDVPDGVYEIGFYFT
EPFSKDERKNIVHTGVSAEGQRFVDVSVNGEKLIESNLADSYGEQTAVVKTLLVNVNRNHEGLEILLSPQ
KGQGVISGLKVKKIR

>SRS024331.217018-NL *Coprobacter secundus*

LEVSITDNWRYFTGDSIQYSENGYCDKKWDKITLPHTWNAADLIDESRGYRRAVSWYRKTLLQIPEAEKGK
KLILRFDGVAIKAEVYLNCELLKTHVGAYTAFVTVDITDKYRFGLDNILAVKVDNSSSLAEIVPPVSGDFS
FGGGIYRRVFLQTLNPNVHFDPEPYASEDLLVQTPNVSESSASVRILGTVKNESMHTYKIRVRHIIYDKNR
NKVSEKLDITYKIKPDAALSIKRDLPVNTNCLWSPENPYLYTLCVIEDISSGKVLQQTESVFGFRWFSV
DNTGFYLNKKIKLRGAARHQDYAGLGIAIPSEINRNDMEALKDMGANFVRISHYPQDKEIYRACDELGL
IAWSEISIVNEVKKNREFARNCKEMLKEMIYQNYNHPSVVMWGAMNELWDYHPEALALAHAELEQIKKSID
PYRLSCVAFHAFMWEKPYKQDTKEMFQISDVNGVNVYEAWYHGDFSTIVPTFDKLRYSYPDRPMFLSEFG
AGSDRVHSYSPQCDFSTEYQLAFNREYINEMEKRDYFIGYSIWNFIDFQVDGRGDSKPNLNQKGM LTS
ERRKKDIYYYSKPNMIHISGADWTERMEVCNTEVNTRPVTVYSNQEEVELFHNRRSLGKKKVSAG
EVTDFVPPISGKNQLEAVAGIEKDVLDIDMTFVSPVMKENVKLLSEGLYINLGQEHCFIDPQLISKIWIIP
DQPYRQGGWGYVDGKPFDSWPGNPTHGVRKGVGANIKDTDLEPLYQTFLLGTTRYRLDVPPEGKYEVLH
FTEPFNGNERKDKGRSGTDGNGDRVFDVSVNGTVCISNLNLAREYGEQRAVSKTIIVDTNNGLEVELVPV
KGKPVISGLSIKKLL

>SRS018656.100250-NL *Bacteroides nordii*

NEIGRQKININASWKFSKILSNNPAFVDFDDKSWSIINIPHTWNNTDAQDGGNNYLRTVGWYRKSLPWN
DSYNGKKLYIEFLGACLQAE CFVNGKSAGIHKGGYTAFRFDITSLMKPGENIIAVKVDNRHSEEIMPLGG
DFSVLGGIYRNVSLIVVDPVHVDLMDNGSSGLYLPVNVSESSADLEIKALIFNHSSSESKTVIINGKL RN
PDSFESIADVSKPTFDVTAMCPGGKVIKSLNELVTIEPGKSYRFEKKITIERPHLWNGKKDPYRYLVDLE
IEENGVIDMVSEYVGFYYSVTKEGFFLNGLYPLRGVGKHQDRYNMGYAVTDKEQNEDFGMIYDMGAN
AIRLAHYPHDPYIYDLCDRYGLAVWTEIPLVDRPGTAESFNEVTKDQLRELIRQRYNCPSVFFWGLQNEI
REQHDTHMRVLMKELNELAHAEDPSRLTVQATNHHTAKNWDSDVFAWNYYPGWYVQNPLFSGKLDSFKDS
ESRPTALSEYGAGGSIYHHEINPAKPVNTGGDFHPPEYQNKVHEQAIIDISTRDFVWGTFLLWNIFDFASD
ARKEGDQHGVDNKLITYDRKVKKDSYAYKANWSNEPFVYITSRRYTNRNEEVTPISIYSNCESVELFV
NGESKGKIVSKDVRCGFFKWDDIILLANTENKIKVVGTEQEQKEYTDEVVWTKIAGTTTTLQSDKLVDND
NKRIIFKENISVEQIYSYLQALDGATFIVVEADGKTPVAKGMVSLGMKIVTSEDSQNKATYEFIAGHIA
LKKNITTDSEQTNEADNLACYAVDGDYNTRWASKDKNPHWLEIDLKEYVLNKITIHWFNPNGSRITYQYHI
KTAGNEKEFEIEVDRSGNTQPNVSDNLQNSRFRVREVLGGNASTAYPSLYEIEVLGWTIESEYDIDF
ENRKIKILNTKENVQNLTTTEKFLSNISFEGNLSYVLNSMSYFIQENDELVITDYSGNKMKFKCVFTNNLT
SVKNTDRKTFQITSQNNLTTLVRLTDFIEPVTLEIRDLLGHTIVTEEISNSYTCRLDNGFYLVSLSNPQLR
SETVKCLIK

>SRS050925.151496-NL *Porphyromonadaceae bacterium GM3*

NEIRWRINLNTNWKFKILTDAPASVDYADETWTTVNI PHTWNSTDAQDGGNNYLRTIGWYRKSLPWND
SFQGGKLYIEFLGACLQAEFCVNGKSVGNHKGGYTAFRFDIT AQMQGNNIIAVKVDNRLSEEIIPLGGD
FSVLGGLYRDVSLIVVDPVHIDLMDCGSSGLYLTPTNVSEASADLEIRAKILNNSSETKTITINGILRNP
DNFTTIEDVPNPAPFDVNAMECPGGKEIKKVNTVAIEPGSSYQFKEKVTVENPHLWNGKKDPYRYLVDLEI
TEGENIIDAVSEYVGFYYSVTKDGFLLNGNLYPLRGVGKHQDRYNMGYAVTSKEQNEFDGMIYDMGANA
IRLAHYPHDPYIYDLCDRYGLAVWTEIPLVDRPGTAATFNDITKTQLRELIRQHYNRPSVFFWGLQNEIR
EQHDTHMRGLMKELNELAQIEDPTRLTQVQATNHYTARNWDSDFAWNYYPGWYVQNPLFSSKLDDEFKNS
SRPTALSEYGAGGSYHHEINPSKPVTNGGKFHPPEEYQNKVHEQAIVDISTRNFVWGTFLLWNMFDFASDG
RNEGDQPGTNDKGLVTYDRKVKKDSYAYKANWSDDPFVYITSRRYTEREETITPVNIYSNCESVELFVN
GASKGKKLSKDVQCGFFKWDNIELLSGSENKIKVIGIKNKEEYSDEVIWTKIAGTTTNLQSDKLVDIDN
KRIVFEENIPVKNIIDTYLQALDGATFIIVESDGETLVTEGFISIGMKVIVTSEDKKNKAIYEFVSGHIAL
NKPITTDSEQINENNLARYAVDGDYTRWASKDKNAHWLEIDLKEYILNKITIHWNPNNGTRTYQYYVK
TADNRKNFVTEVDRSTNNQSNTISDNLKSKSRFVKVEVIGGNASTAYPSLYEIEVFGWIIIESDEYVIDFE
NKQIKILNSEENVQNLTTTEKFLGNISFEGNMSYSLNSMSYFIQENDELVITDHSGNVTKFKCVFSDDLTS
INDKKKESPFQITKQGHELTSIDLVDSVKLEIRNLLGHTIVSEQFSDSYTCQLGSGLYLISLSTPQLN
YKTVKYLIK

>SRS011586.205727-NL Firmicutes uncultured bacterium fragment

RAALVRGGEEDAVETRVGFRTISFDHEKGCFLNGKHIKLGVSRRHQDRYHMGNALTEKEHREDLALIRE
VGANSIRLAHYQQDQRFYDACDEMGLLVWAEVPVISHFVDAKFENAKQQLTELITQNKNHPCICYCWGVQN
EITMSGKTKSLEQGIRTLNDLAHRLDPTRPSTSAQVMCGISSPMNRITDLQGYNLYFGWYVQTYHAIDQ
WLDQFHKAYPEIKLCLSEYGAEGIIKYQAEQGVQGDYSESYQARFHEHYARAIAREWLWGSYVWNMFDF
GAANRDEGGVKGRNNKGLVTIDRKTRKDSFYVYKAFWSDEPFVHIGGERFVDRVIGETTVPVYSNLPEVT
LTVNGEPITKPCDRVVYFEKIPVTAGETVLTAEAGGCTHSIHIRGVEEENPAYKMQGDGSGSFVNWFVNA
DGKRNPDYFSIDDRVGALLKSKDVQSLIRMGLGSRKVPVSVLFTIAKPFVRVTLKLKLDGDMKEIADQY
LQTIHK

>SRS024435.43879-NL Bacteroidetes

HDVIDFCGGWTFKKAPSTTETMQAAAANWRWDSVSPHTWNATDMLTRVNNFYAGVAYYRKHSVPSSW
KDKVIYLRFEFVASCAEVYVNGKLVATHKGGYSAFACNISGDVRHGQSNEIVVKADNSARPDPVIVNHNH
FGIYGGMYRPVSLIVAEPCHIAVTDACAGPGVYVRQQQVSSKSATVCVTTKTRNATRQAPATIETTITDD
KGNCVARVRKLTFSPOGIIQAHEATMQLSKPHLWQGREDPYLYKVTVRILQNGKVTDESVPGLGLRNIEI
KPGKGVWLNGKRYPMYGVTRHQDRWGKGSALSADHEDLAMIMDVGATTIRLAHYQQSDYFYSRCDSLG
LLVWAEIPFVNRVTGQEWDNAHTQLRELIRQSYNHPSIYIWGLHNEVYTPHAYTADLTSSLHELAKQEDP
DRLTASVNGYGHADHPVNMNADIQGMNRYFGWYERKIKDINGWVEGLEKNYPSMPLMLTEYGADGNIRQQ
TEYLGESLWGGPFYPETFETKVHEYQWSVIAKHPIYILASYLWNMFDFGCPMWERGGVPARNMKGLVTID
RKVKKDAYYWKANWSKSPVLYLTQRRNTDRERRVTSITVYSNIGQPKVFLNGQELEGIHKGFTDVHYVI
DNVTCLKDGENIVTTTATDKQGNKLSDEIRWQYTGEKRRSARQGNNAEHAGF

>SRS023583.12646-NL Bacteroidetes

REVISFNDGWLFFKKGPATDMMKAAAQWEGKWTEVTIPHTWNAKDMQVKTNSFYEGAAYYRKTYLIPETM
KKGRLFLRFEGVGTCTEVYVNGYLVGTHKGGYSAFVCEIGEQVKYGESNEIVVKADNTARPDPVIVNHS
FGVYGGIYRPVSLVVTPECHIAVEDYASPGVYITQKNVSKSAAEVNVKVKLDNASLQPEALELENAIYDA
EGKLVKKHVRRLELTPQGMQSYTASFQKLRPHLWQGRKDPYLYKVVSRLKQKGRVIDEVVQPLGIRKYEI
VAGKGFLLNGEKYPMYGVTRHQDWWGKGSALTNQEHADLAMIMDIGATTIRLAHYQQAEYMYAKCDSL
LIVWAEIPFVNRVTGQEWDNAHQMMRELVRQNFNHPSIYIWGIHNEVYHHPHYTAALTRSLHDVCKTDDP
DRYTAAVNGYGHAEHPVNQNAADVQGMNRYFGWYERKIQDIKPWVEGLEKNYPWQKMLMTEYGADANLEHQ
TEYLGDAxxxxxNMFDFAVPMWERGGVPARNLKGMTFDRKIKKDAYYWKANWSKEPVLYLTQRRNVDR
ERKETSVTVYSNIGTPVVYLNKGPLEGIRKGYTEVHYVFDKVTLAQGKNILKAVVSYQGKEYVDEIEWNY
TGEKNRGTVIHVNKKEHAGW

>SRS021948.213386-NL Bacteroidetes fragment

KVDNSMRLDVIPVNHNLFGVYGGIYRPVSLIVTERYNIAVTDHASPGVYITQKNISQNSADICVQVKLDN
VTFQSVPPVLETAIYDQVGKQVGVQQHAFELSPQGIQKRECLFRIKPHLWHGRKNPYLYRVVCRLYRDN
ELVDEVIQPLGIRKYEIVAGKGFYLNGEKYPYGVTRHQDWWGVGSALTNVQHDIDLDMIMDIGATTVRF
AHYQQSDYVYSRCDSLGLIIWAEIPFVNRVTGMEWENAHSQLRELIRQSFNHPSIYVWGLHNEVYHPSY
TSSLTRSLHDLAKTEDPDRFTVSVNGYGYMEHPVNMMNADIQGMNRYFGWYEEKIEDIDSWIEGLEKEYPT
QRLMLTEYGADANLKHQTEFVGESLNWTKEFYETFQTKTHEYQWGVIAKHPIIIASYLWNMFDFACPMW
ERGGVAARNMKGLVTFDRTIKKDAYFWYKANWTDSPVVYLTQRRNFDRESRETKVTVYSNVGIPSVYLN

KKLEHVRMGYTDVHYIFEEVILVKGENILYA EVMGVDGQKYSDEIKWYNGEKRRIVDNKENKTEHFGF
>SRS011452.30249-NL Bacteroidetes
MLLGQFFSLAAREVTSFNEGWLFRGPFSEEPVKVAAQWNGKWETVNLPHWTWNAKDMQVKAASFYEGVGY
YKKKQFFGEGLGKGRVFLRFEGVGANTEVYVNGKLVGTHKGGYSAFAFEIGTALKLGAENEIIVKADNVS
RADVIPVNHNLFGVYGGIYRPVWLIITEQNNITVTDCASPGVYITQKNVSRRSADIYVVKLDNGSLTPA
SLVLENIITYTREGKKVDSHRLLELTPQGTQTYVSSFELNRPHLWQGRKDPYLYKVVSRLMADGKVIDEV
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>SRS021948.204621-NL Bacteroidetes
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>SRS011734.333180-NL Bacteroidetes
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>SRS017701.130646-NL Bacteroidetes fragment
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>SRS056519.10028-NL Bacteroidetes
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>SRS017307.212630-NL Bacteroidetes

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>SRS045713.77206-NL Bacteroidetes

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>SRS045713.77020-NL Bacteroidetes

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>SRS016095.87929-NL Bacteroidetes fragment

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>SRS015663.4145-NL Bacteroidetes

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>SRS015782.13092-NL Bacteroidetes

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>SRS018656.108594-NL Bacteroidetes

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>SRS020233.379797-NL Bacteroidetes

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>SRS015133.10516-NL Bacteroidetes fragment

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>SRS020869.216543-NL Firmicutes fragment

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>SRS013951.108115-NL Firmicutes

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>SRS017307.218563-NL Firmicutes

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SEEFKIVDVKEVKPEPFDCRGKNVRKTKVKEEALDCGEKNAGETEVKAEVFLNKVRLWDGLKDPYLYILR
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>SRS024132.104031-NL Firmicutes

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>SRS019601.107807-NL Firmicutes

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>SRS064276.161266-NL Firmicutes

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>SRS015217.11114-NL Firmicutes

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>SRS018351.57975-NL Firmicutes
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>SRS013476.215840-NL Firmicutes fragment
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>SRS020328.32922-NL Firmicutes
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>SRS045645.174240-NL Firmicutes fragment
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>SRS024435.253488-NL Firmicutes fragment
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>SRS011061.156348-NL Firmicutes fragment
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>SRS051031.146690-NL Firmicutes

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DSKNWM

>SRS023914.87176-NL Firmicutes

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>SRS014287.153454-NL Firmicutes fragment

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RGDHTEEYQAVYHEYMLKCFERNPYLWATHVWNMFDAADARDQGGEPMNHKGLITFDRRTKKDSFYLY
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>SRS015190.73379-NL Firmicutes

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NDRVYPQKADFTFYGGIYRDVSLMVVSKNHFALDYFGGPGIRITPTVQGADASVQVTTWHDGEGEV SIEL
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YISEHMPNGRENTISQMKEIIQYNHPCIVCWGVSNEITISTKDKKDMLDNHRQLNDLICHEMDKTRLTTL
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HRGDHTEEYQAKYHEYMLRCFKRHPMMWATHVWNMFDAADARDQGGEPMNHKGLVTFDRKTKKDSFYLY
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>SRS017521.23841-NL Firmicutes

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SWLWSTHVWNMYDFAADARDQGGEPMNHKGLVTFDRKTKKDSFYIYKAWWSDEPFVHICSKRFADRTQS
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>SRS014287.8565-NL Firmicutes

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VVRNHIALDYLGGSGVQITPTVNGANADIEVKTWMEGDGEVEFSIYDAAGAEVLTGKGRDITVTLLEPH
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>SRS056259.8724-NL Firmicutes

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>SRS077730.60690-NL Firmicutes fragment

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WSKDPFVHICSKRYVERCEKTTTVKVYSTEKKIKLLVNGKPFQAEQTRDDHVFTFRVPL

>SRS024435.88193-NL Firmicutes

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EPMVHVCGRSRYAQRAGETTQIRVYSNQPSVALYLNDELVQELSADKVVFVFNVALKDGFNTVLAVAGDVKD
SITLEKVEKEPSIYVLPEVNERAEGVKNWFEAVGDMDLKAPMEFPEGMYSIKDTMEEISHNKDAFAIVQK
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>SRS063040.264469-NL Firmicutes fragment

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SITLEKVEKEPSIYVLPEVNERAEGVKNWFEAVGDMDLKAPMEFPEGMYSIKDTMEEISHNKDAFAIVQK
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>SRS017521.296625-NL Firmicutes fragment

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>SRS015578.76196-NL Firmicutes

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NGRRYAQRAGETTEVKVYSNQDCVTLYLNGKEVGTQQAHRVFHFTVALAEDFNTLLAVAGSAKDSITLEK
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>SRS019601.65570-NL Firmicutes

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>SRS023914.205103-NL Firmicutes

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DIVRKRRYIWGTYYWLLADCGSDGRNQGGDAGKNHKGGLVTFDRKIRKDAYYLYKAFWLDPEEQKFVHICG
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>SRS020233.378352-NL Firmicutes

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LILQNYNHPSICFWGVANEIGIGGESEAMFHVHRLHELCKQLDPSRLTGIANVGMTPTSSQLFRLTDVT
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LQERPWIWATWLNMFDFAVDVRNEGGVQGRNNKGLVTFDRKQKKQAFYFYKACWSKEPFVYICGERYK
HTAAPMTVKVYSNAAQVTLNGLHKLGTVQGGPVFLFPNVLDPRVNMELMAVTDTDCHSLIWECEVAAEP
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DRMIQGGALSFSIWNHINKYLPDEAYTLINARLSRIKKTQ

>SRS011586.67984-NL Firmicutes

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YPLHGVSRHQDRLDKGWAIGEREHEEDMKLIQEVGANTIRLAHYQHDAFYDLCDEAGMVVWAEIPFISM
FLNTPEARENTLQQMHELVVQNYNHPSICFWGISNEISIGGESEELLSNLHALNDLVYELDQTRLTTMAN
LSMVENDSPLNHITDVISYNHYFGWYLGKVEDNAPWLDTFHAENPEICLGISEYGCEGITTLSASPKVR
DYSEEQAYYHEKMLETFAQRPYLWSTHVWNMFDFASDMRDEGGVQGRNNKGLVTFDRQTRKDSFYIYKA
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FLGQQEVCCESIPLVRVTEPNASYMLAEAAEKAGQNVENWFAASGGEGEEPLQFPEGYFSIKDKVGALLK
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>SRS017433.39923-NL Firmicutes

MREIFNLNEKWAFSKEAIKPPVAMPQNWYVWNLPHWTWNAIDGQDGGNDYYRGTCFYAKEVQKEELPEGEQ
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MYRDVNI IAVNKS HFDLDYYGGNGLKVTPEVADKNAKIAVEVFLSGEKAGQQLVYQITDAEGVLAAETKT
GISDKQVNLEITDVHLWNGRKDPYLYTATVRLMEDGVCIDSVSTRFGCRTFTIDPDKGFFLNNGNSYPLRG
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RENTISQMKELIVQNYNHPSIVVWGLSNEITMSGAEDEDLNENHILNDLAHEMDPTRLTMAVVSMDM
HNPYIQIPDVVSYNHYFGWYGGDTSMNGPWWDEFHKEFPKIPIGMSEYGCEALNWHTSDPKQGDYTEEYQ
AHYHEELIKQLYTRPYI WATHVWNMFDFAADARNEGGENGMMNHKGLVTFDRKYKKDAFYAYKAWLSDEPF
VHICGKRYVDRVENVTKVTVYSNQKTVELFANGESLGKKEASDHFFYFEVFNHGETKLLAVAGECRDES
F IKKVEKFNEAYRLTEKGAVLNWFDITAPDGGFFSLNDTVGDIMSNDGKNVMQELFQSFQQAQAGGGMNS
GMMKMLSGFTVLRMINMMGLTSQMTSQPGAEKTEVSKEQLLAINDRLNQIRK

>SRS015578.244827-NL Firmicutes

MRKVYNFNTKWAFSKEATEAPKVMPEKWWVSLPHWTWNDIDGQDGGNDLYRGTAAYAKELEKIDLPAKADK
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YRDVNILAVNNSHFDLEYYGGPGIKVTPEVVENNANVEVEVFTTNTLDTQKLNYILKDREGNIIAQKETS
SSETKVTFEIENVHLWHGKKDPYLYSAEVYKQDDEVLDNVSTRFGCRTFKIDPELGFILNGEYPLRGV
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RKVEVFNEDYRLKEKGAILNWFDIDAPEGYYSLNKIKISEIMESEEATKIFNEFINPLMSGMMGADQVGK
KKEETENNLSSMMTMLGSFTILRLTSLLGAGHVEVTKEQLLNLNEKLNKVECSVKTLA

>SRS017307.210466-NL Firmicutes

VEREVTMRTIVKLNKKWAFSKEADSAPQKMPQNWYVWNLPHWTWNAIDGQDGGNDYYRGTCYYAREIEKAS
LPKADRYYLELKGANASADVIWNGEVLAHHDGGYSARVVDVTEAIKDQNL LVI AVDNSANESVYPQMADF
TFYGGLYREVELIAVNNSHFALEHYGDQGIKVTATVEGKITDRAIEKGNRSDQTNAAEKNTAKEINA EK
TKKVQTEGNRIEKVIKTAPARLDIALNLADIREGQKVRVKVFDGEGKLAVKEHSISAAFLEKVEEEKKK
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WGLSNEITMSGASDEDLLENHRLNDMVHAMDPGRKTVEACISMCDIHDPYVQIPDVVSYNHYFGWYGGD
TSMNGPWLDFHFEFPDTPIGVSEYGCEGLNWHSSSPTQGDYTEEYQAHYHEELIKQLYTRPYLWATHVW
NMFDFGADGRNEGGENGQNHKGLVTFDRKYKKDAFYAYKAWLSCEPFVHICGKRYEKRVEPVTKVTVYSN
QPKVELFVNGKSLGVKESPEHFFYFDVPNEGESVLTAVAGITVSNTPDAFIPVCRDESRLIKVDTFPE
EYRLRENGALLNWFDITEKEGYLSLSNKIKEIAVTEEGSSILKELMGQMAFKNARQKDEKKKAEKGEQ
SGLDMMKMLEGFTLLRLVNMSGVTGELLSKEKLLALNEKLNIPKTGRLA

>SRS016095.241000-NL Firmicutes fragment

LYRDVNI IAVEDTHFDLMYRGSSGIMVTPEIKGDDADVKVEVFLTDPKEGQKLRYVIKDQEGTVVTEQVQ
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VSRHQDRWGIGNALLKEHHREDMDLICELGATTVRLAHYQHDAFYDLCDEKGMVVWAEIPYISAHMKNG

RENTVSQMKEILILQNYNHPSIVVWGLSNEITMAQGKSDMLENHQLLNELAHKLDPTRLTITIAAVTMCD
IHDAIYQIPDVVSYNHYFGWYGGDTSMNGPWFDNFHKEFPDIPIGVSEYGCEALNWHITSEPKQGDYTEEY
QAHYHEELIKQLFTREYLWATYVWNMFDFGADARNEGGENGQNHKGLVTFDRKYKKDAFYAYKAWLSEKP
FVHICGKRYVNRMEPVTKVTVYSNQPDVELFVNGKSLGKQESCEHFFYFDVPNEGTSILKAVAGACTDTG
TICKVDQFDESRLKEKGAVLNWFDITEKEGYSLNDTIEDLMKAEAKTLFLGLMQSQDHFSSSENTGAL
MKMMGSFTMIRLLNTFGAIGSAMTKEEFLALNHKLNLIKPPD

>SRS014235.308239-NL Firmicutes fragment

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ADA EVMQAFADGEVEFEIFDGENVISSTTAQGNPTSSVTIKNVHLWNGTKDPFLYTLTARLLVNGEAV
DEIKTRFGCRSYEINPQKGFI LNGQEYPLRGVSRHQDRPDIGNALLPCHHKEDIDLICEMGANTVRLAHY
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IDNHKMLNDLVHSLDKTRPTTEAVLTACNIDE EYVHISDVLSYNHYFGWYGGDV KAYKPWFDFKHKKYPN
KAIGMSEYGCEALNWHITSYPEQGDYTEEYQAYYHEELIKQIMACPYLWATHVWNMFDFAADARAEGGENG
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NTDTPFYHFTVPNTAEETI AVKANDIDGIEDRMTIRHVDELNEDYIMKDEHGVINWFEIDTPTGYLSVN
DTLGDILKPLKGKIFAIRFLKLKVLAPPKEPGQKKKKGASVMGDFDKNIDKSMIDMAKGFTLKRATM
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>SRS049959.116680-NL unclear classification

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YFDLTDLLKHENRILIQVDSTRNNQVPALD TDWFN YGGIYRDI ELIRVPKVHVKEFKIALEPDSNFRKI
RVFVRMSEEITGTAILTIRELGIEKEIPVLGGRGELLFEAEPEVWSPENPKLYDVELSCEGDEVDRVGF
REIRVKGMDIVLNGKLVFLHGISCHEDSVPNGKGLTDEER IENIRIAKELGCNFMRV AHYPHHERMAQLA
DELGILLWEEVPVYWN IHFDS EDTYQDAENQLKELIHRDYNRASVI IWSVGNENEDTDLRLQFMGNLAKC
AHELDHTRAVSAACLVNFKKNAIEDRLEQYLDI IGLNEYCGWYTADLKTLPALFANSKPQKPVIVTEFGA
DAYAGLHGTITDKGTEECQAFVYEKQIETIRNIPYIKGMT PWILYDFRCPRRTSVNQKYNTKGLLSADK
KYRKQAFYVLQKFYRELQ

>SRS017307.204178-NL Firmicutes

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VYIERVHFTPSFADGVWSAKLEVTVSNAGELPAEFVLGALTEQDGEEVEKLRLLPGRVEAGESVTLEST
QIYPGVKAWCYEEP NLYLLKVRLYMDQNEESDGESVFG RKEAGEVKKEGIPDKEENKDKKEAMEWEPAEL
ADDFIERVGFVRVSIQGRKLQINGKDVFLKGFN RHEDHGMAGCAIPLQLMVQDMELMEDMGANAVRTCHY
PNDERFLDLCDERGIMVWEENHARGFTLDKMMNPNFDRQCEDCIREMILNHYNHPGIVIWGLLNECASET
KEGREKYAKQYAQVRSMDQSRPVTSATCRHFNDICLDLPDVVSYNMYSGWYVDVDVKERHEQLMDWIEGA
GGAGKPVIVSEFGAAAMYGYRDRGRCKWSEERQADIIRANL DAYKDDSRLTGVFIWQFADCRVTEEEGWF
ASRARCHNNKGVD EYRRPKMAYDVVRELYPKMNGVTESDKDKA

>SRS022609.316191-NL Firmicutes fragment

RMQNP NFDRQCHDCIKEMIEQHFNHPGIVIWGILNECASETEEGREKYAMQYEQIKSLDTSRPTTSATCR
HFKDICLDLPDVVSFNMYSGWYQDVSVKERHEQEMEWIDSVAGGIKPVIVSEFGAAAMYGFRDRGRCKWS
EERQADILRENLEVYAKDDRLTGAFIWQFADCKVTEEEWFPTVRCHNNKGVD EYRRPKMAYDVVKEYF
PKLGQESR

>SRS055982.56993-NL Firmicutes

MVRTFCTNQIRNTEELSGSVWSFVPESGEHAGTSYSVVVPTVWENLPEFASYRGRGCFRRKFYGEGNVRL
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GIVLEHLKKAYIEYLVHVEVGKAGKNGWKVLLTTKINGISDSSERYTLRLQADPLFTDEIAIDLETNESRI
VQKEIYLENVSEWTPENPVLYKVRAILSDTEPMDDLIERFGFREIKVSGKYILLNGRP IRIKGFNRHEDH
PMFGCAMP LQAIEQDIQIAMDLGANAI RTSHYPNDELFLDLCD EQILVWEESHARGLKEYHMQNPYFQK
QSEDCIREMVMNHKNHPCIFVWGILNECASETLYGRDCYTKQYRQLKELDRTRPRTSASNKHKKDICLDI
PEIVSFNLYPEWYSQNETAVDCIQDMMQWIRNNPVTRNKPPIVSEIGAGALYGYHTRTKDKWSE EYQCI
ALEHQLRAVLESAECCGVFIWQLTDIRISRERFENRPRMTMNNKGIVDEYRRPKMSYEIVKDI FTSYDDYF
ETNL

>SRS021948.194172-NL Firmicutes fragment

VDNSFSEASALHVENDYQSYGGISRGVMLEQLPKHYIKQLHFTPVCKDDTWYAQLELVIASLDGKPWKGK

VSVSLKGCDDVNNVLYTQSCSTGAGELVLKSDMIACKNAKPWAPEHPTLYMLSAQLEDEDGKVIDDLTD
RVGFREVRTQGKDILLNGKKLQIKGFCRHEDHPQFGCALPLSAIMQDLQLARDMGANSIRTSHYPNDELFL
LDLCDELGILVWEENHARGLSEEQMKNPNERQCEACIREMIAAHYNHPSIYIWGILNECASDTAYGREC
YKTQLELIKSLDVTNRPSFSSCRFKTDICFDLVDVVSYNIYPKWYHNTPVAMYLDDLYKWVQTTGGAGKP
FLITEVGAGAIYGYRTPAKVKWSEYQVLALEEQLGAILS YKDCSGVYIWQFCDVRVTNDWWNTRPRTMN
NKGIVDEYRRPKLSYETVKRIFGSVD TYRK

>SRS017103.219270-NL Firmicutes fragment

EKLPHKHYIRQMHFTPIYKDGGWYARIKLVITSLTEDRWDTVNVTLKNGETEAFLCSETCSMKENEDELT
IETDEIACKAVSAWAPEHPTLYQVTARLQDESGCVIDDLTDRVGFREVKTQGKDILLNGKKLRIKGFCRH
EDHPQFGCALPLSAIMQDLQLARDMGANSIRTSHYPNDELFLDLCDLGLVWEENHARGLSEEQMRNP
FERQCETCIREMIAAHYNHPSIYIWGILNECASDTEYKSCYKAQFELIESLDKTRPSFSSCRFKTDIC
FDLVDVVSYNIYPKWYHNTPVAEYLLDDLYKWVQTTPEGGKPFLLITEVGAGAIYGYRTSAKVWSEYQVQ
ALEEQLNAILS YDDCSGVYIWQFCDVRVTNDWWNTRPRTMNNKGIVDEYRRPKLAYETVKKIFSSVD TYR
E

>SRS064276.279741-NL Firmicutes fragment

RSFEAGGDMRIVFKGVSH TAKVFVDDKFVCEHYNAFTQFEAVVKNLPSGVHKL SVLVDNSFSEASALHVE
NDYQSYGGISRGVVLEKLPKH YIRQMHFTTIYREGSWYAKIKLVIASLTRETWDGT VNVTLKNGETEAFL
CSETCSMKENEDELT IETDEIACKAVSAWAPEHPTLYQVTAQLQDESGCVIDDLTDRVGFREVRTQGKDI
LLNGKKLQIKGFCRHEDHPQFGCALPLSAIMQDLQLARDMGANSIRTSHYPNDELFLDLCDLGLVWEE
NHARGLSEEQMRNPNERQCECIREMIAAHYNHPSIYIWGILNECASDTAYGRECYKTQLELIKSLDVT
RPSFSSCRFKTDICFDLVDVVSYNIYPKWYHNTPVAMYLDDLYKWIQTTS GEGKPFLLITEVGAGAIYGY
RTPAKVKWSEYQVLALEEQLGAILS YKDCSGVYIWQFCDVRVTNDWWNTRPRTMNNKGIVDEYRRPKLS
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>SRS049995.245815-NL unclear classification

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SVFYDFYGYGGIYRSLELFTLPQGS RFDRVRVTVDPAARRVRLEGRMAGGSTPLRLGFDGAEPQPV
WHGDRFELELAVPGGQLWTPAAPKLHTVRLESTDAIVERFGIRTVTAARGQILLNGEPLRLYGYNRHDA
HPQFGPAMPVELQIADLQLMRDLCNFVRGCHYPQSQRFLDLCDQLGMLVWEESLWGDRLEVQQDPKFR
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AYDVISFNTYPGWYDDPDGAPRPLNR IAEELDLILSKLATPDCIDKPVILSEIGAGAIYGWRDPLNSFWS
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TSWN

>SRS017701.154589-NL unclear classification

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FYGYGGLYRSVELHQLPEYSVDRAEVRIVNAEEGRVRVRLRVSGTLPSPLSATFCFDGTQTVEQKLDASP
EAEFVLDDKALWSPESP NLHTVRVQVGESVVTERTFGLRTVKTEGRKLLNGREIRLKGICRHESHVLFV
SVPHTLALEDVRLKDLGCNFVRCVHPQDSVFLDLCDLGLVWEESLWGNRAEQFLIPEFAEAQKEQ
TRLLTEKSINHP SIIIWGFLNEGRSNAPESQELYRD LAAILRSAEDGRLVSYASMFPDDH SFALADIIS
INRYPGWYASDQELVRPLSEIEKEMKRLETMLDERGFGDKPLLI SEIGAGAI PGWHD TFRVHWSEYQSD
YVQEACRAILGNPRYLGIAVWHFADCR TYASSYALGRPRSLNNKGVLD EYRRPKLAYGTLRQAFRG

>SRS015663.266489-NL Firmicutes

MRKQILLNDNWYLLRDGREKAVSLPHCFNENDGQGEGKMFRGELLYKKIMKLDENDLKNEIYLQLDGASN
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SGPHFDLLDGSRDGFDANTYLS DNKGVEINARIKMLDES NLTLATIMDKDGKEITKKKLKASYNNSLS
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TYLISQVRNHTCVYCYGVQNEVCMVT KNEYSFKLVRELASLAKELDPSRFTAQANEYTTENDCIINSYTD
IIGYNLYYGYGNMEELGPRLD AIHEANPSKLLILTEYGVDTNPQIHSSMPKRNDYSEYQLLYSKNAL
DTIKSRPWLGGSYAWVMFDFGSSMRDEGGKRGONQKGLVTIDRKIFKDAYYLYQACWSKEKFIHIAGKRY
INRAEELTNIYILSNLKNIVLNLNGLSYKAFSDAGIATFRVPLRVGENKITAYAFENPEFKDTILINRVL
KVDKSYVIEKKEERSTAVNWFENV DVENIVGMNKPLRPEGYTIDDRICDIFSSKKAKEIFMKYFAFLTES
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>SRS057717.12790-NL Firmicutes

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NIYSVANESPINSLTELVGYNLYYGWYYDTMTGLQERLDSFHAARPDIPLMVTEYGVDTNPQYHSYEPKT
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KGYSKHGTVYHDMVTLYRVKDKDKNYEYIKKEDGNSVINWFEKFDLTQINKIKLDPNCFSSADKIGDIMK
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>SRS052027.152578-NL Firmicutes

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LWQGENPYMYRCIAELRIENEIYDSVELPFGFRNIEMKPESGFYLNGEHIKLGHVAKHQDFEGKACGTG
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IYFGWYYGKLT DYEDFLNKFHEENPNVSLGISEYGVDCNTKFHSEEPCKDYSEEFQCVYHEAAYGAFAEK
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DTTITITVYSNEEQVELYLNGEFFSRKSGKRKF EFTEVPIDKETKVTAKAGDCVDETSICKVDS PKSYTY
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>SRS017701.165653-NL unclear classification

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LIRGVELIKKNKLHITAEDYGSNGVYVRQDKVSDKEAMLSFTAKVRNTEKKAQKAVISFLIKERDGRVLK
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>SRS013476.22122-NL unclear classification

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FFFNGKPYRLAGVNRHQDRPGKWAI SQADHEEDMALIREIGANTIRLAHYPHAEQFYNLCD DNGMLVWA
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>SRS019397.48796-NL Bacteroidetes

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EAVSQEADVFLNGQYLG NHKGAFNAFCFEITPYLKKKNNELVVKASNVLPDIAPLSGDFTVFGGIYRPV
SLILPRTCITPLDYASSGIYIKQTKIQDHKATIDITTKIDGDIYGNKKLRLRTSVFDPTGKLVITSENE
EIKASENTAAIDQRIYIDNPILWNGKAEPNQYRFCELLQGKKVIDT LSVYTG LRYFKVDPKKGF FLNGK
SYQIRGVNRHQDRPGKWAI SEEDHQEDMQLIKEIGANGVRLAHYPHSDYFYSLCDKEGLLVWAEIPFIG
NGTESIAFEENLKQQLTELIRQNFNHPSIFCWSL FNLVKG NPEKLV AELNDLAHKEDPTRFTVAAANIE
GRPENTITDIMAFNTYPGWYWAEPETMGPSIDWKKDEKRGVCISEYGAGASIKHHQQRMKKAPKCEGDFH
PEEWQAI VHEGNYKEIDKRDFVWGSFVWNMFDFASGGRKEGDTYGMNDKGLV TYDRKTPKDAFYFYKANW
SEKPV LHITSKRHIERTEPETEIKIYSNCS DITLHVNGIKYPSSAGDHNI FTWKNVRLQPGKNKISVSGT

FDNKTLTDQCEWILEN

>SRS024435.60746-NL Bacteroidetes

QLRQKHLLDQEWEFKSDKDGNNLAWEKVTIPHTWNAATDGLTPDYIRGVGEYRYHLNLLPEMLKRRRAFLRF
EAVSQVADVYVNGIPVGTHTKGSFNAFCFEVTSLLKKGDNLIEVKVSNRLDNDIAPLVGDFTVFGGIYRPV
SLLLLPKTCITPLDHASSGIYIHQDKINSDAQELSIITKVNSNRAEKDLAVRTSIFDAQGKLVSNHITRK
STGKEETVEFINKQSIIEHPVLWDGRQSPHLYRVLVEVLKGNMVDVSLSQHTGLRYTVDADKGFFLNGKP
YKIKGVNRHQDRADKGWAITNKEHNEDMELIKEIGANGIRLAHYPHSDFFYSLCDREGMLVWAEIPLVGK
ASPTPAFTENIKLQLTELIRQNFNHPISFCWSLNFELSEGDVKSILQELNELAHKEDPGRLTVAAANVEN
RPNVITDIMAYNTYPGWYWAQPSMTKSSVAHWNRLVGNKGICVSEYGAGANIWHHQQDVQSAPKTDGIF
HPEEWQAIVHEQNYRGISSDFVWGSFVWNMFDFASASRNEGDI LGLNDKGLVTYDRKVKKDAFYFYQSA
WSESPVLHITSKRDIARREPATDIKIYSNCDHIHLKVNQGDCGHPDREDNILIWNVSLKKGENRIEASG
KKGTVDLTDQCKWVLLDKKK

>SRS057717.67703-NL Bacteroidetes

IVTGMLCCFSFFHINAQLRQQHILEEGWEFKLDKTDVWEKVSIPHTWNAHDGTTSEYFRGMGEYRYQLY
VSSAMLEKRIFLRF EAVSQVADVFNKGNI GTHRGGFNAFCFELSHLLKKGNLIEVKVSNSENNYIAPL
AGDFTVFGGIYRPVSLLLLSEVCITPLDFASSGIYIRQKTNSEQSWLSIQTKVDNSRKNASRLTVRTTLF
DTKGELIDCNSTQEAIEITGNDLSFINTFHVKDPVLWDGRKSPHLYRIFVEIMKGSALVDTLSQYVGLRYC
SIDDQKGFLNGLRYKIKGVNRHQDREDKGWAINHKDQLQDMEI I KEIGATGIRLAHYPHSDFFYSLCDK
EGMLVWAEIPLVGKACSSNDFTENIKLQLTELIRQNFNHPISFCWSLNFELSEGDVKGVLTELNDLAHAE
DPDRFTVAAANIENRPNKVTDIMAYNTYPGWYWGEP SIMKSSLAHWNRLVGEKGI AVSEYGAGANVWHH
WQSDRKVPKADGIFHPEEWQSI VHEQNYQAIEDSPFVWGSFVWNMFDFASAFRNEGSMQGMNDKGLVTYD
RKIKKDAFYFYKSLWSEEPVVHITSKRDIARNDSLTRIKVYSNCDHIVLVKNGKEYGNPLKNDNIYIWEN
VTLRKGENQINVS GDRNQIKLTDDCKWILL

>SRS050422.25351-mL1mL2 Bacteroides ovatus

ENHPRTSILLDKGWGYKPLTSNKKDASLTPTVTLPHWTNANYLEGTTTYNREMMVYKRDLEITPEMKDKRL
FLYFEGVNSAADIFINRYRTVGQHLGGYTAFCFEITDFVQNGNNDLEVWVSNAAYRTDILPISGDFNIYGGI
HRPCHLIITEKNCISPVFYASPGVFIHQNRITKQQAIEICVESILSLKEVKSGLKLKTTITDTNGKEVTSN
ITAVSEETVKQPLTITNPILWNGKENPYLYNVSVELYDGDKVIDKVTQKTGFRYFHADPEKGFFLNGKYL
NLYGFCRHEDEVKKGSAQQEDYDKDISLIKEVGATAMRLAHYPHAEPMYDLCDQNGIILWTEIPMCGPG
GYAYTGYLHNKG FEDNARQSLKELVYQKFNHPISICFWGIFNELLITDNKKFQEYDNPVPFVREINELYKK
LDPSRLTAFATCVDQNTNYLGCSDLIAWNKYFGWKTSQQSAEKFFDDVKNSNNQPVGVSEYGAGASIHQH
ACPLDHNTQRPKREHPPEEYQAICHEGYWTAFTERPYLWAKFIWQFADAQSSIRQEGD TDGINDKGTITYD
RKVKKDVFFYFKANWNSEPMPLYICSRRTERTNTPLVKVYSNLNEATLYLNDKKIKGQKKDKVNRIIWD
NITLSPGENVIRVEGKSGKEILNDTCIWTLK

>SRS016018.40585-mL1mL2 Bacteroides sp. CAG:754

QETSPRKVFTLDEGWECRPITTVNRKAPFTPVTIPHTWNTSYIEGTTLYERKTVVYQRPLEVTEEMDGKR
LFLYFEGVNSAAQVFLNRRTIGEHLGGYTAFCLEITDQVKKGENLLEVWASNAAYRTDVLPISGDFNVNGG
IHRPCHLIITGQDCITPLFYASPGVFIHQENISRETAGINVETHLSLKNKKQGLKLKTTIADASNKIVAS
NETGISGDIVRQPIQVERPVLWEGKKNPYLYTVTVELYDGNQLVDRMVQRTGFRYFSVDHEKGFFLNGNY
LNLYGFCRHEDAVGRASALLPEDYKMDMELIKESGATAMRLAHYPHAEPMYDLS DENG IILWTEIPMCGP
GGQAFTGLVDTEGFKVNARLAVKELVYQKFNHPISICFWGICNEILVSDGKR FVEYDNPIPFIKELNGLYK
SIDTSRLTALATCVDQSYLLGCSDLIAWNKYFGWKDAAPSASKFFDDCKNSSQGIPVGVSEYGGGASIN
HHQWPLSMEDRADSHFHPEEAQTFCHEGNWESFMKRPYLWAKFIWVFADFPSYMRQEGEEDGYNDKGLIT
QDRKVKKDAFYFYKANWNPEPMVYITSRRFVKRNNPQTDIKVFTNLKEVTLYINGKKIGTMKPDEMHRVI
WKDIKLDEGRNTVRVEGKNGKENLSDSCEWYIYK

>SRS019601.35894-mL1mL2 Bacteroides ovatus

QETSPRTIFSLNEGWECPITTVNRKAPFTPVTIPHTWNTSYIEGTTLYERKMMVYQRPVLVVTAMKNKR
LFLYFEGVNSAAQVFMNRRTVGEHLGGYTAFCIEITDEVKEGENLLEVWASNAAYRTDILPVSGDFNVNGG
IHRPCHLIVTGQDCISPLFYASPGVFIHQENISKTVADNVETHLSLKNKKQGLRLKTTVADADNKTVAS
NEVEVSDVIVKQPMKIHRPILWDGKKNPYLYTVTVELYDGNLLKDRMVQRTGFRYFSVDHEKGFFLNGEY
LNLYGFCRHEDAVGRASALLPEDYRMDMELIKESGATAMRLAHYPHAEPMYDLS DENG IILWTEIPMCGP
GGQAFTGFVDTEGYKDARLAVKELVYQKFNHPISICFWGICNEILVSDGKR FVEYDNPIPFIKELNGIYK
SIDSSRLTALATCVDQSYLLGCSDLIAWNKYFGWKDAAPSASKFFDDCRDSSKGI PVGVSEYGGGASIN
HHQWPLAMEDRSDSHFHPEEAQTFCHEGNWESFAKRPYLWAKFIWVFADFPSYMRQEGEKDGYNDKGLVT
HDRKTKKDAFYFYKANWNPEPMIYITSRRFTKRDNPKTDIKVFTNLKEATLYINNRKIGTMKPDEMNRVI

WKDIRLNDGRNIICVEGKNGKGLLSDTCEWYCIK

>SRS049900.8240-mL1mL2 Bacteroides sp. CAG:754

QETSPRKVFTLDEGWECRPITTVNRKAPFTPVTIPHTWNTSYIEGTTLYERKTVVYQRPLEVTKEMDGKRL
FLFYFEGVNSAAQVFLNRRITIGEHLGGYTAFCTEITDQVKKGENLLEVWASNAYRTDVLPISGDFNVNGG
IHRPCHLIITGQDCITPLFYASPGVFIHQENISRETAGINVETHLSLKNKKQGLKLKTTIADASNKIVAS
NETGISGDIVRQPIQVERPVLWEGKKNPYLYTVTVELYDGNQLVDRMVQRTGFRYFSVDHEKGFFLNNGY
LNLYGFCRHEDAVGRASALLPEDYKMDMELIKESGATAMRLAHYPHAEPMYNLSDENGIILWTEIPMCGP
GGQAFRTGFVDTEGYKDNARLAVKELVYQKFNHPSICFWGICNEILVSDGKRFEYDNPPIFIKELNGLYK
SIDTSRLTALATCVDQSYLLGCSDLIANWKYFGWKDAAPSASKFFDDCKKSSQRIIPVGVSEYGGGASIN
HHQWPLSMEDRADSHFHPEEAQTFCHEGNWESFMKRPYLWAKFIWVFADFPSYMRQEGEKDGYNDKGLVT
QDRKTKKDAFYFYKANWNTEPMAYITSRRFIHRMSDKTDIKIYTNLKEVTLYLNGTKIGVMKPDELKRVI
WKNVQLCKGRNTVRVEGKNGKELIVDNCVWYCLGD

>SRS022137.24481-mL1mL2 Bacteroides multispecies

SISPRLFIPLNKEWGYKSITSATKKAPLTPVTIPHTWNAEYLPNQISYNREMMIYKRTLTISPDMKGKRL
FLFFEGVNSVADVFNQYQTVAAHLLGGYTAFCTEITDQVEDGENDLEVWVSNAAYRTDVLPISGDFNVYGGI
HRPCHLIITENNCISPLFFASSGVFVHQKNISEKKAEFEIESRSLSLKGMKSNLKIIRTITILDEAGKIVTSQ
EKKVINERVVQAMAIQNPVLWDKHNPNLYTVNVELYEDNRLVDKVEQTGFRYFSVDPDNGFFLNKPY
DIHGFCRHEDVKGKGSALLPDDYAKDMELIKEVGATGMRLTHYPHAKKMHDLCDRNGIILWAEIPLVGP
GYAYTGYVSNQGLHDNARQTLKELVFQNYNHPSICFWGIFNELLITDGKTFQAYDNPVSFVKELDSLFR
LDPSRLTTFATCVDQTNLYGCADLMANWKYFNWNDGKQIGSFMDKVKANSQNPVGVSEYGFGGSIYQHR
CPVNESEILLGRLHPEEFQTSHEKNWEALAERSYLWAKFIWVFADFQSVIRKEGDRDGINDKGLVTYDR
QIKKDAFYFYKANWNTEPMLHLCDSRFTERFYSITSIKAYTNLEAAILYVNGKKIGKMKRDKYNRVIWNN
ILLQPGENIIRIDAKNGKKIISESCVWTLKL

>SRS056519.9840-mL1mL2 Bacteroidetes

MEKEHSPRIIKLLNEKWEYRPISTVGKKALFTPVAIPHTWNAAPVEGTTLYERKTMVYQRPLEIAEAKEG
KRFFLYFEGVNSVAQVFLNRRITVGEHLGGYTAFCTEITDLVKQGENLLEVWVSNAAYRTDVLPISGDFNIN
GGIHRPCYLIITGEDCINPAFYASPGVFIHQDNISDKQADIRVKTLLSVKSKQQLDLKLTITDAENKVI
ASNEVPVTDSEIEQPIRIEHPHLWNGKKNPYLYKVTVELYNGSDLEDVVQRTGFRYFKVDSQDGFFLN
EYLNLYGFCRHEDVAGRGSAALLLEDYQLDMELIKEAGATAMRLAHYPHAEPMYDLADENGIVLWTEIPMC
GPGGQAYTGYVGAEGYKNNARQVLKELVYQKFNHPSICFWGICNEILVNDGKKFIEYDNPPIFIKELNNM
YKSLDSSRLTALATCVEQSYIIGTSDLIANWKYFGWKDAAPSVSKFFDDAYKAAKGTVPVGVSEYAGAGS
INQHQRPLRIEDRADSRFHPPEEAQTYCHEGNWEAFAKRPFLLWSKFIWVFADFPSYMRMEGEVEGYNDKGL
ITRDRRIKKDAFYFYKANWNTEPMAYITSRRFTQRTNNLTDIKVYTNLKEVTLYLNGVKIGTMAPDEIKR
VVWNNVRLAEGRNVRRIEGKSGKMLVTDCCVWYFTEK

>SRS024132.318813-mL1 Faecalibacterium sp. CAG:74

MLAPQSNASRTILDFSGVWDFRLKPDAPWQSIAPVASYNDQSADPEFRRHVGLAWYRRSITVPSMLKGQR
LMLRFDAVTHNARVFLDGQLLCTHRRGGFLPFVADVTNLLYPGQTALLEVEVDNTISHHTLPVGNEGGTAF
FGSDNPGIPSVVAGKQHQHAQGINLPNFDFFNYAGLNRPVRLYTTPADYIADVTLPVSMGTVRYTVATH
GEGDVTLDVLDADGQVVASGTGASGEVHVANPHLWEPAPGVPLYTARVRFAQDEYSQPFQIREIKVDGC
RFLINGKPFHFHGPCKHEDSFFHGRGLDQCLNVTDINLFHWLHANAFRTSHYPYSEEMLSLCDREGIVVI
AETPAVGIGEGGKDPYRWAPADYHAQVLTDMIARDKNHPCIVLWSLGNPNTPDAHPQSAIDYWHPLYLLA
HQLDPQNRPVTLVGCQNDYTRDITIPSMVVCINRYYGWYNLSGDLEAAAFAMRMEMDYWATVNKPTILS
EYGADTIAGMHGTEMFTEEFQVDYKTIINACLDERPFFVGETPWNFADFQGTQGGPMRAGGNRKGLFTRDR
KPKMAAHYFRQRWENFKK

>SRS011239.11144-mL1 Faecalibacterium prausnitzii

MLYPEQNEARLKLSLDGTWAFALGSCAETQFDPKPLPDAQPIAPVASYNDQNDQTTALRRHYGWVYQQR
KVTLPAPFCAGQVRVLRFGSVTHAKVWLNQGLIAQHKGGFTEFVADVTALLQPGETALLTVACDNRVNS
TLPVGNEDGQLAFFGSDNAGIPSVVAAKRAAAPQNRPNFDFFNYAGIHRPVVLYTTPKEYIEDVTIVPAV
DGTVQYAVKTTGSAPVRVTVLADGNAVASAEAEGTITIPVHLWEPRPGTPYLYTLHATCGADVYDQT
FGVRSIEVRGTQVLLNGKPLYFKGFCKHEDFTAHRGFDVPLNVKDVNLIHWANANAVRTSHYPYAEFEY
DLCDREGILVMDETPAVGIGGGAANPYKEYPLAEHHRQVLAEMIHRDKNHPCVVLWSLGNPNLEHFPQ
DAYDYWHPLYELAHQLDPQDRPVTLVCCQNDYTKDITTRTMDIVCINRYYGWYNLSGDMDAACYGLNQEL
DFWAEQHKPVMSEYGADTVAGLHTAGAEMFSEEFQVEFYRRLDAEFDKRPWFVGEFVWNFADYDVTVQGP
MRVDGNKKGLFTRDRRPKLGMHFLRQRWSEIPTFGFKE

>SRS015217.92661-mL1 Roseburia hominis

MLYPRENVRVGVIDLGGVWNFKLGDEKEPGDSFAMGEDTELMAVPASYNQKDDPKYRNHYGWAYYERKI
TVPSYLKGQRLVLRFDVTHAAKVYLNGRFLTAHKGGFLPFEVEVTDQLVPGETAELVVAADNRVNHSTL
PVGNEGSTAFFGSDNPGVPSVEAAKVWRKPQNLPNFDFFNAGINRPVRIYTTTAAAYIEDITLVPDIDGD
TGVVNYEIKTAGSVCEDEVIRIRILDADGNEVAAADGAGGTVRIPDARLWWPWPGEPLYTASVTYGEDLY
EETFGIRTVAVEGTQFLINGKPFYFKGFGKHEDAAFRGRGLDVCLDVKDINLIHNLNANSFRTSHYPYAE
EMYQLCDREGIVVIDETPAVGIGAGDKIDPYKTFHIREHHEQVLRREMIARDKNHPCVVMWSMGNEPDTEH
FPESAYDYWHSYELTHSLDPADRPVTFVCCQNNYERDLVTRTMDVVCVNRYYGWYNLSGDLDAACYALN
MELDFWEKQNKPMITEYGADAVAGIHESVPEMFSEEFQMEYYRRQNAEFDKRSFFIGEQQVWNFADFATI
QGCMRADGNKKGLFTRDRRPKMAAHYFKERWQIPNFEYK

>SRS054956.59831-mL1 Roseburia intestinalis

MLYPQENKIRGIIDLNGVWDFALGTTEEPKDTVTLPDQMEPIAVPASYNQKDDIAYRNHYGYAYYRRNI
TVPSYYKGQRLVLRFDVTHFAKIYLNGLVLTQHKGGLPFEVDITDKLCAGESAELVVAVDNRINHSTL
PVGNEEGTSFMGADNAGVPGVEAAKRWKRPQNLPNFDFFNAGINRPVRIYTTTPKAYIKDVTLVTDIRGT
DGIVNYQVKTSDDTDGQEVVLQILDANGNEVAQAKGTSGEIVIPDAKLWEPPGTPYLYTAAVTFGDDYYE
EPFGVRTVRVEGTSFLINGKPFYFKGFGKHEDSAFHGRGMDVCLDVKDINLIHNLHANSFRTSHYPYAAE
MYRLCDREGIVIIIDEVPAVGIGAGAGINPYETFPPIREHHEQVIKDMIARDKNHPCVVMWSLGNPDTEH
PESAYEYWHSYELAHETDPSDRPVTLVCCQNNYEKDIVTRSMVDVVCINRYYGWYNLSGDLDAACYGLNL
ELDFWESQNKPMITEYGADAVAGIHECVPEMFSEEFQVEFYKRQNAQFDRKFFIGEHHVWNFADFATVQ
GCMRVDGNKKGLFTRERRPKMAAHYFKERWGEIPNFEYK

>SRS015854.61990-mL1 Clostridium sp. CAG:91

MLYPIMNRSRSLIDLGGVWQFQLGDETEPEDLWSGPQEMDLIAPASYNQKDDPAYRNHYGWAYYERTV
VLPALLCGNNGQRQVLRFDVTHVAKVYINGKLIMTHKGGFLPFEVEITDLVPEDGHILTIGVAVDNRINH
STLPIGNEGNIAFFGSDNPGIPSVEAAKIWRKKQNLPNFDFFNAGINRPVRIYSTPASYIRDITLVPSM
EGDTGVLNIEILTDGADSDVPVILKIIDAECSCKAEGTGRKGELRIPQVHLWQPPGTPYLYKAEVHYGD
DYYTEEFGRFTVEVKGTQFLINGEPFYFKGFSKHEDSFFCGRGLDLCLDVKDINLLHWHANSFRTSHYP
YAEIYRLCDREGIVIIIDETPAVGIGAGDSCDPYKTFPIREHHEQVLRDLIARDKNHPCVVMWSMGNEPD
TEHFPESAYDYWHSYELFTHSLDPQNRPVTFVCCQNNYEKDIVTRTMDVVCVNRYYGWYNLSGDLAASY
AWNMEELDFWEKIGKPVMMTEYGADTLAGLHNCAPEMWSEEFQAALYKMYSKAFDGRDFFIGEQAOWNFADY
ATLQGCMRADGNRKGIPTDRRPKLAHYFRGRWEHIPNFHYKDKNNS

>SRS011405.139207-mL1 Clostridium sp. CAG:91

MLYPIMNRSRSLIDLRGVWQFQLGDEKEPGDTWIEPQAMEMICVPASYNQKDDPAYRNHYGWAYYERTV
ELPALLCGSNGQRQVLRFDVTHVAKVYINGKLIMTHKGGFLPFEVEITDLVPEDGHILTICVAVDNRISH
STLPIGNEGNIAFFGSDNPGIPSVEAAKIWRKKQNLPNFDFFNAGINRPVRIYSTPASYIRDITLVPSI
EGDTGVLNIEILTDGMDSDAPVTLKIIDAESNCKAEATGRKGELRIPQVHLWQPPGTPYLYKAEVHYGE
DCYTEEFGRFTVEVKGTQFLINGEPFYFKGFSKHEDSFFRGRGLDLCLDVKDINLLHWHANSFRTSHYP
YAEIYRLCDREGIVIIIDETPAVGIGAGESCDPYKTFPIREHHEQVLRDLIVRDKNHPCIVMWSMGNEPD
TEHFPESAYDYWHSYELFTHSLDPQNRPVTFVCCQNNYEKDIVTRTMDVVCVNRYYGWYNLSGDLAASY
AWNMEELDFWEKIGKPVMMTEYGADTLAGLHNCAPEMWSEEFQVALYKMYSKTFDGRDFFIGEQAOWNFADY
ATLQGCMRADGNRKGIPTDRRPKLIHYFRDRWEHIPNFHYKNKNNS

>SRS019582.80136-mL1 Bacteroides massiliensis

QRENILINQDWNFRFSHQVDKNSSRRVDLPHTWNAQDALSGKPDYKRGIGNYDKKLFIRSEWKGRFLR
FEGANCVSNVFINGKQIGEHRRGGYGAFFIFEITDKVNYGKDNTVLIRVNNGEQLDVMPVGVDFNFYGGIYR
DVHLLVTEDICISPLDYASPGVYLFQQHVGEKQAAVLARINLSNGTEHPRQATLRLQVKEGDKVVYQADK
KVTVPAPHTSVQPEEMSFTLLNPRLWNGREDPFMYQTVITLVKDGKEIDKVEQPLGLRYTTDADRGFFLN
GKHLPLHGVCRHQEWAIEVGNALRPMHHEEDTRLMLEMGVNAILAHYPQATYMYDLMDRNGIVTWAEIPF
VGPGGYADKGFVDQPSFRENGKEQLKEMIRQHFNHPSICFWGLFNEKENGDNPLEYIKELNVLAHQEDP
TRPTTSASNQGGAINFITDNIAWNRYDGWYGATPATLASWLDKTHQAHPEIKIAISEYGAGASIYHQQDS
LVQTSPPGSWWHPENWQTEYHIQNWKIIISERPYPVWGSFVWNMFDFGAHRTEGDRPGINDKGLVTHDRKVK
KDAFYFYKANWNPEPMVYIAGRRSVNRVKPVTEVQIFSNCAEVTCLKVNGQIIKKMQPDGVKVCIFKDIRL
KKGENNIEVSAKNGKRVITDACCWIL

>SRS019582.46068-mL1 Bacteroides multispecies

ERQDILLNNQWKFRFSHQVQKGTEVRVDLPHTWNAQDALSGKIDYKRGIGNYEKKLFVQSQWKGRFLR
FEGVNSIANVFINRKHIGEHRRGGYGAFFIFEITDRVEYGKDNSILVRVNNGEQLDIMPLVGVDFNFYGGIYR
DVHLLVTDEICISPLDYASPGVRLIQDSVSHEYACVRALIDLSNGSDKLELELNIILSNAGKVVEKNE
KVTLYPDSVVSQKEMRIELNNPHLWNGRQDPFLYQVEVSLHNGEILDQVIQPLGLRFYRIDPDEGFFLNG

EHLPLRGVCRHQDRSEIGNALHPEHHEEDAALMLEMGVNAVRLAHYPQATYFYDLMDKLGIVVWAEIPFI
GPGGYEDKGFVDSFSPFRANGKEQLKELIRQHYNHPSICVWGLFNELAEYGDNPVEYINELNVLAHQEDTT
RPTTSASNQSGKINFVTDLIAWNRYDGWYGGTPQDLGKWLDMHKEYPEIRIAISEYGAGASIYHQQDSL
VKTNPSTSWHPENWQTYHYHINWKAITERPFVWGSFVWNMFDFGAAHRTGDRPGVNDKGLVTFDRKVRK
DAFYFYKANWNKEDPMIYILIGKRNIIRVQRQQTIIAFSNQPEAELFVNGKSCGKVETDRYSILEWKHVIL
TPGKNKIEVKTTNKKQTLTDMYYCIL

>SRS045713.38032-mL1 *Bacteroides multispecies*

QRQDILLNDNWKFRRFSHQVQKGSEIRVDLPHTWNAQDALSGKIDYKRGIGNYKELFIRPEWQGKRLFLR
FEGVNSIADVFINRRHIGEHRRGGYGAFVFEITGEVNYGKENSILVRVNNGEQLDVMPLVGDFNFYGGIYR
DVHLLITDEACISPLNYASPGVRLIQDSVSHQYAKVRAVVDLANGNNAGQEVELGVRLLDGQKVVAQQKQ
TLTLAGNAALQQELTFEINNPHLWNGRQDPFLYQAEVSLSRGGQLVDCVTQPLGLRYYRIDPDKGFFLNG
KHLPLHGVCRHQDRSEVGNALRPQHHEEDVALMLEMGVNAVRLAHYPQATYFYDLMDKNGIIVWAEIPFI
GPGGYDDKGFVNLPSPFRANGKEQLKELIRQHFHNSPSICVWGLFNELLEFGDNPVEYIKELNELAHQEDTT
RPTTSASNQMGLNFITDAIAWNRYDGWYGGTPADLGRWLDGMHRDHPEIRIAISEYGAGASIYHQQDSL
VKSVPNSWWHPENWQTYHYHINWKAI SARPYIWGSFVWNMFDFGAAHRTGDRPGINDKGLVTFDRKVRK
DAFYFYKANWNKEEPVLYLAGKRNTVRSRRLQTITAF TNQPGAELFVNGKSYGKAMPDQYAILWKNVEL
QPGENEIKVVSINKKVKLMDSFRCCL

>SRS015782.59238-mL1 *Bacteroides vulgatus*

QREDKLINQDWSFRFSHQVNASAAARRVDLPHTWNAQDALGGKHIDYKRGIGNYTKKIFIRPEWQGKRLFLR
FEGANCVSNVFNKGKHIGEHRRGGYGAFVFEITDKVEYKENTLLVRVNNGEQLDVMPLVGDFNFYGGIYR
DVHLLLTDLNLCISPLDYASSGVYLIQQQVTDKQATICARVNLSNGTGELRKVVLRQLQVNDGKKTVEYETEK
EVSMIPHTDVQVENIEFILKNPRLWNGMQDPFMYQAVLTLIKDGKELDKVEQPLGLRYYVTDPDKGFFLN
GKHLPLHGVCRHQERAEVGNALCPVHHEEDTRIMLDMGVNAVRLAHYPQATYMYDLMDKHGIVTWAEIPF
VGPGGYADKGFVDQPSFRENGKEQLKEMIRQHYNHPSICFWGLFNELKEQGDNPVEYIKELNAIAHQEDP
TRPTTSASNQEGALNFITDHI AWNRYDGWYGATPATLATWLDATHKNHPEMKIAISEYGAGASIYHQQDS
LVQTVPGSWWHPENWQTEYHIQNWKI INERPYVWASFVWNMFDFGAAHRTGDRPGINDKGLVTHDRKIK
KDAYFYFRANWNPEPMIYIAGRNVNRVKPLVDVQVFSNVEEVILIVNDCQCGMKPDSLKVCLFKDVPL
RKGRNEIEVRANDSKKQLIDRCTWILQ

>SRS050925.51312-mL1 *Bacteroides dorei*

QREDKLINQDWSFRFSHQVNANAAARRVDLPHTWNAQDALGGKYDYKRGIGNYTKKIFIRPEWQSKRLFLR
FEGANCVSNVFNKGKHIGEHRRGGYGAFVFEITDKVEYKENTLLVRVNNGEQLDVMPLVGDFNFYGGIYR
DVHLLLTDLNLCISPLDYASSGVYLIQQQITDKQAAICARINLSNGTGELRKAVLRQLQVNDGKKTVEYETEK
EVSMIPHTDVQVENIEFILKNPRLWNGTQDPFMYQTVVTLIKDGKELDKVEQPLGVRYITDPDKGFFLN
GKHLPLHGVCRHQERAEVGNALYPVHHEEDTRIMLDMGVNAVRLAHYPQATYMYDLMDKYGIVTWAEIPF
VGPGGYADKGFVDQPSFRENGKEQLKEMIRQHYNHPSICFWGLFNELKEQGDNPVEYIKELNAMAHREDP
TRPTTSASNQDGALNFITDHI AWNRYDGWYGATPATLATWLDATHKNHPEIKIAISEYGAGASIYHQQDS
LVQTVPGSWWHPENWQTEYHIQNWKI INERPYVWASFVWNMFDFGAAHRMEGDRSGINDKGLVTHDRKIK
KDAYFYFRANWNPEPMIYIAGRNVNRVKPLVDVQVFSNVEEVILIVNDCQCRMKPDSLKVCLFKEVPL
RKGRNEIEVRASDSKKQLIDRCTWILQ

>SRS013687.231718-mL1 *Bacteroides multispecies*

QREDKLINQDWSFRFSHHVNANAAHRVDLPHTWNAQDALGGKQDYKRGIGNYTKKIFIRPDWKGKRLFLR
FEGANCVSNVFNKGKHIGEHRRGGYGAFVFEITDKVEYKENTLLVRVNNGEQLDVMPLVGDFNFYGGIYR
DVHLLLTDLNLCISPLDYASSGVYLIQQQVTDKQAAVCARVNLSNGTGELRKAVVRLQINDGEKTVEAEK
EVSMTPRTDVQAEDIEFILTNPRLWNGTRDPFMYQAVVTLIKDGQELDKVEQPLGLRYYTTDPDKGFFLN
GKHLPLHGVCRHQERAEVGNALHPEHHEEDTRIMLDMGVNAIRLAHYPQATYMYDLMDKNGIVTWAEIPF
VGPGGYADKGFVDQPSFRENGKEQLKEMIRQHYNHPSICFWGLFNELKEQGDNPVEYIKELNTMAHREDP
TRPTTSASNQGGALNFITDHI AWNRYDGWYGATPATLATWLDATHKSYPGIKIAISEYGAGASIYHQQDS
LVQTVPGSWWHPENWQTEYHIQNWKI I SERPYVWGSFVWNMFDFGAAHRTGDRPGVNDKGLVTHDRKIK
KDTYYFYFRANWNPEPMVYIAGRNVDRVKPLVDVQVFSNVEEVRLIVNGRLCGMKPDKGLKVCLFKDVPL
RKGSNEIEVRAGDGKKQLIDRCTWILQ

>SRS011405.39318-mL1 *Bacteroides thetaiotaomicron*

DDNPRTFILLDKGWGRPVSDTGLKSSMKQVTPHTWNANYIPGTRS YNREMMVYRRDLEITPDMKDKRL
FLYFEGVNSSATVLVNNSVSGSHKGGYTAFCMEVTDYAKQGTNKLEVWVTNAYNPEILPISGDFNIYGGI
HRPCHLLVTEQDCISPLFYASPGVFIHQDKVSEKQAQITVETMLSLRGKKQGLKVRTTIEDAKGNIISQN
IEQNVTTIENVKQPFVIEHPVLWNAKQNPPLYKVIVELLELDNGKVIDRVEQRTGLRYFSVDADKGFFLNGKY

LDLYGFCLHEEVEGKGSALSADHERDMELVKESGATSLRLVHYPHSESIYHLSDENGIVLWTEIPMVGP
GGYDFCGFINTDGLKEHARQVLKELVYQKYNHPSICFWGIFNEIRTNYDNAEPFARELHELYKEIDPSRL
TTLASCDPKFYQNCSDLMAWNKYIGWYGSRNAPETAGNFFDKAKAASNGKPVAISEYGGGANVEHHFSM
KENDVKPSGQFHPPEGQTYIHEGNWSAFAQRPYMWAKYIWVFADFQSAIRNEGGKPGINDKGLVTYDRKI
KKDAFYFYKANWSTEPMIYITSRRFTKRPEASVQVKVYSNLRENTLYVNGKKIGKQKSDSLNRVMWQNV
LSKGENRIRVEGSKAGVIEDTCVWYLK

>SRS045004.32036-mL1 Parabacteroides merdae

TRKEILLNENWMFRFSHQVQFKSEERVNLPHTWNSGDALSGKQDYRRGMGNYSKTLFVKPEWKGKRLFLR
FEGVNTIANVFINDVHIGEHRRGGYGAFVFEITDKVKYDADNRILVKVNNALQQDVMPLVGDNFYGGIYR
DVNLIITDPVNISLTDYASPGVYLIQNKVTKEQADVAKVLVSNNGSAKVQPVKVDIKIWEDDKLVQQQDI
KVTVDANDWATTGMDLTIRNPRLWNGRKPFFMYKAEISLLSEGOIIDRIVQPLGLRYYHVDAAEKGFFLNG
EHLKLKGVCRRHQDRAERGNALYKMHHEDEAAIMVEMGANAVRLAHYPQATYFYDLMDKYGIVTWAEIPQI
GPGGYQDQGYINQPSFRENGKEQLKELIRQHYNHPSICFWGLFNLKTIGDNPTEYIEELNDLAHQEDPT
RLTTSASFLSYDDDINKVTDVIAWNQYFGWYGGSPSDMGKWL DANHKAHPEYKIAISEYGAGASIYHQQD
SVKRGIASGWWHPENYQTYHYHGNWKALAEPRFVWGSFIWNLFDFGAAHRVEGDRPGINDKGLVTFDRKV
KKDAFYFYKANWNTEEPFVYITNRRHRDRSLAVTDIMIFSNQPEVELFVNGKSLGRQKPDEYATFEWKGV
ALQDGENTIEARSTQKKNPVNDKVVWTVK

>SRS016056.36511-mL1 Bacteroides intestinihominis

QRSYLSLNDNWQFRFSHQVGKNSTQRIPLPHTWNAPDALSGKLDYKRGIGNYEKDLYISPEWKGKRLFLY
FEGVNSVADLFINGKYRGEHRRGGYSAFTFEITDDVEYIGIDNKILVRVNNAEQFDIMPLVGDNFYGGIYR
DVSLIITEPTCISLLDYASPGVYLTQKSVTEKQAHIVATVKLSNAATTKQVNVKVTVKDHDKTIQEIKRH
VTLNAHSEGETIIPFTIQNPRLWNGREDPFMYRAEISLWENDKLLDKVIQPLGIRYYRVDPDNGFFLNGK
HLPLYGVCRRHQDRAEIGNALRSMHHEEDELMLMGVNAVRLAHYQQAEMYDLMDQSGIITWAEIPFVG
PGGYADKGFIDSEAFKKNKEEQQLKELIRQYNHPSICFWGLFNLTERGDNPTVYITELNELAHSEDPTR
LTTASASNQNGGLNFITDCIAWNRYDGWYGGMPDALAKFLDQTHNQYPLRIASEYGAGASIYHQSDSLK
RPNPAGWWHPENWQTFYHIENWKIHERPFVWGSFIWNMFDFGAAHRREGDRPGINDKGLVTFDRKYKKD
AFYFYKANWNKTDPMVHLAEKRCDDRPTAQONFMAFTNLPEVELFVNGKSMGKQKPDYAIKWEQIELP
PGKNTILVQSTTKDKVSDTMTLWITDSAKNRKE

>SRS024132.66972-mL1 Odoribacter laneus

REVISLNRSWKFTPGYETAKNTYTEVNLPHPTWNQDALAAKVNYERGLGNYEKDIDIPAAWQGRLYLRFK
GANTINTNVFINGKHAGEHRRGGYTAFAMDITPYIHWGTKNNIWVRVNNSPQLDIMPLLGDFNMYGGIYRDV
ELIITDPFHISLSDYSGSYGLYFTQTDVSREKARILAKILVKGETGCSSVIRVKIEDPQGNIVSAEQEIT
FDEQGEAVGELIILNISKPHLWNGRKDPFLYKTITELLRKQGITDRVEEKIGIRYFRVDPDKGFFLNDEPL
QLRGACRHQDRAEIGNALFPSSHREDEIMKEMGVNAVRLAHYPQDRYVYDLCDEYGFIVWAEIPFVGPG
GYRDKGFVDQPSFRENGKIQLKELIRQYNHPSICFWGLFNLKQSGDDPLNYVEELNTLAHEEDPYRLT
TAASNQNGALNEVTDLIAWNQYWGYYGGEPAIGKWADAVHKKYPEKPVGISSEYSGASILHQQEELKKP
VANSYWHPENWQAFFHEEHWKAIDQRPYLWGTFFVWNMFDFGAAHRQEGEMAGKNDKGLVTFDRKIKKDAF
YFYKANWNLEEPFVYITGKRFEKRSEMSQQIKIYSNQPEVELILNGKSLGVKKGVYGIFTWTEIQWKKGE
NILTARSKNGITDEVKILF

>SRS050925.174250-mL1 Coprobacter secundus

VRRDVLINDNWMFRFAHQVSHKSERRVDLPHTWNAKDALAGIPDYRGTANYTRVLKIPEGWKGRFLRFLR
FEGANTVTDVFINQRHIGEHRRGGYGAFVFEITDWIEYKDNQLQVSVSNALHLDVMPLVGDNFYGGIYR
DVHLLVTEETCISPLDYASPGVYLIQESVSPQRADV KARIAVSGLAVDNVNLNVRVYDGEKVVKAEAVQPV
EVKPSKVMYVDVPITIGTPHLWNGRQDPFMYRVEVTLSKGDRILDKVEQPLGVRFFELDADRGFSLNGQP
LRLHGVCRHQDRIEVEGNALRPEHHIEDAALIAEVGANAIRLAHYPAATMIYDLMDKYGLVWAEIPFVG
GGYQDQGFVNTSFRQNGKEQLIELIRQHYNHPSICFWGLFNLKTPADNPLEYVRELNELAHKEDPTRI
ITSASNLNDNDELNRVTDACWNKYFGWYGGNPDDLGAWADKVHAAYPQLKIGISEYGAGASIYHQQDSL
KKVVPTGWWHPENWQTYHYHMKSWGQISERPFIWGSFIWAFFDFGAAHRTEGDRPGINDKGLVTFDRKVKK
DAFYFYKANWDKDNKFVYIAERRCKNRVNPQVDIMVFTNLPEASLWVNGQYMGKAKADKWATVRWENMSL
IPGANRIEVKSVDKKKEYSDTVEWILQ

>SRS011239.197778-mL1 Parabacteroides goldsteinii

SRKEILLNNDWMFRFSHQVQKKSEQRVNLPHTWNSDALSGKQDYRRGMGNYSKTLFVKPEWKGKRLFLR
FEGVNTIATVFVNSVCLGEHRRGGYGAFVFEITDKVQYGAENKILVKVNNALQLDIMPLVGDNFYGGIYR
DVNLIITDPVNISLTDYASPGVYLIQNNVTKEQADVAKVLVSNNGSSTSKPVQVNVKIWDGNKIVQQQDI
NLTVDANQIH TAGLEFSIKNPHLWNGRKPFFMYRTEVTLLADGQEI DKVTQPLGLRFYHVDAAEKGFFLNG

EHLKLHGVCRHQDRAERGNALYKEHHEEDAIIAEMGTNAIRLAHYPQATYFYDLMDKYGIVAWAEIPFI
GPGGYQDRGFNDLPSFRENGKEQLKELIRQHYNHPSICFWGLFNLKTHGDNPIEYIKELNKL AHQEDPT
RPTTAASFLSYSDISKITDVI AWNQYFGWYGGSPSDMGKWL DANHKAHPDYKIGISEYGAGASIYHQOD
SIKPGVASGWWHPENFQTFYHMGNWEAIAERPFVWGSFIWNLFDFGAAHRTEGDRPGINDKGLVTFDRKV
KKDAFYYYKANWNKDDKFVYIANRRHRDRDTKVTDIMVFSNLPQVELFVNGKSIGVQGP DNYATFVWNGV
NLKD GENTIEVRGIQKKNITDTVNWTK

>SRS015190.51659-mL1 *Bacteroides ovatus*

QDILLNNDWNFRFSHQVQKGTEVRVDLPHTWNAQDALSGKIDYKRGIGNYEKNLFI RPEWKGRFLFIRFE
GVNNIADVFINRRHIGEHRRGGYGAFIFEITGKVEYKENSILVRVNNGEQLDIMPLVGDFNFYGGIYRDV
HLLITDETCISPLDYASPGVRLIQDSVSHRYAKVRAIVDLSNGSSGNQVELVNRLLDGQRVVKEGTKNV
NLSGNEVMQOQELTFEIDQPHLWNGRQDPFLYQAEVTLFRNGQMVD RVTQPLGLRFYRIDPKGFFLNGKH
LPLQGVCRRHQDRSEVGNALRPHHEEDVALMLEMGVNAVRLAHYPQATYFYDLMDKNGIIVWAEIPFVGP
GGYNDKGFVDLPAFRANGKEQLKELIRQHYNHLSICVWGLFNLTELGDNPVEYIKELNVLAHQEDTTRP
TTSASNQMGDLNFITDAIAWNRYDGWYGGTPADLGKWLDRMHKDHPEICIAISEYGAGASIYHQQDSL VK
TVPTSWWHPENWQTYHYHIENWKTISSRPYVWGSFVWNMFDFGAAHRTEGDRPGINDKGLVTFDRKVRKDA
FYFYKANWNREEPMLYLTGKRNTVRTQRLQTITAFTNLSGAELFVNGKSYGKAIPDSYAIL EWK NVELEP
GENEIKVVSTNKKLPLSDSFHCRL

>SRS016335.280078-mL1 *Eubacterium* sp. CAG:115

MRTYEYINSGWRFSKVGGLPEGRQTEPKSDDNGWELVELPHTWNAADCTGVSGRASEYSEGYR GAGCYR
RKVKLPHEQYHGKHVFIEFAGANTTTVLFVNGKKAGVHVGGYSAFRFDITKL VKPGADNQFTVYVSNATS
EFVAPIGDCGDFTKMGGLYRAVRLIAVDKLRIALRDNSSGVYITAKPEADGGA EVRVLVKLEND SAE EK
NAVVRVTISGGAANIQREAKLTIAPEGEASAEFSLKLPGAHLWDGVRDPFLYSAKIAVLSGGTALDGVTEQ
FGVREFSIDREKGFFLNGKHTPLHGVNYHQDSPRAGWAMTNKQRRERDYAVMRDMGCNAVRMAHYQHASEE
YALCDKLGMCVWTEIGIIGKLCPEGPAEPQLSREFADSAMQQLTELITQTYNHPSVMFYGMSNEIYQMSD
SIHSLYEQMYDYISENGGGRAAVYADAQFWGKFCTLPADTVGYNRYFGWYKEAGSVEQFREWFDKAHEEQ
TRPLCISEYGGGGAISQHKDNINWLT DIDPNGERHYENYQSRLHEQIWAQIGSLDY LWAKLVWCMFDFPA
AGRREGDTAGINDKGLCTRERVTKDAYWFYRSVWSGEKTCRLTEKRFTERPANVPEVRAYANAERAELL L
NGNSLGFGKVPDNR LPTVFVWKDVKL RPGKNDLLLR AFYFDGTVLEDGASWETSC

>SRS011134.351238-mL1 *Firmicutes bacterium* CAG:95

MRTIREFNFWKFTKLQDEIGISKDAISPEYADTSWQFVVLPH TYNSEDGAGRTMDMSENRRGGY YHGSV
CYRNSLVLSSEECGKEVYLEFEGANTVAEVYINGRFVKGHSGGYSAFRFNLT DYICADKDNLI AVIVSN
APTNYIAPIADQGDFTKMGGGLYRGVKLIIVDPVHIALNDSGSCGVYIMPHNISESAANINMLVKLDKAEI
AEAKAIYDPPQKLVLEELFGRADKDRIMLSGEINCPKLWNGVNDPALYRADITLFYDGKPVDAVEQKFGI
RSYQIDADKGFFLNGKSYPLHGVNYHQDSYESGWAMSDAQRERDYHIIRDMGCTAVRMAHYQHCDHEYS L
CDRLGLCVWTEIGIVNKMSDDSDTHALSNGFADNAKQQRLRELIRQYNHPSVIVWGLSNE LHQMTDEIF
ELYNELYEIIACEEDDTRLKTFADNQFYGRFLELPADVVGYNRYFGWYKEAGSAEEFGKWL DLYHNEKEKR
PVCLSEYGGGGAISQHKDNVDWKS DIDPVGVRHYENYQSQLHEILWKQFSVRKYLWAQFIWCMFDFAS YE
RVEGDTKSQNDKGLCTRERVPKDAYFFYRSVWSSEKTVYITERRHEYRACDVFPVKVYSNADTVELCIND
VSLGRISRCECPNNESTVFVWENVKIKPGTRNKIYAKAYFCDGTTKTDYAFWIGE

>SRS017433.51215-mL1 *Bacteroides fragilis*

QRQDILLNNWNFRFSHQVQGDTRRVDLPHTWNAQDALAGKIDYKRGIGNYEKALYIRPEWKGRFLFLRF
DGVNSIADVFINRKHIGEHRRGGYGAFIFEITDLVKYGEKNSVLVRANNGEQLDIMPLVGDFNFYGGIYRD
VHLLITDETCISPLDYASPGVYL VQEVVSPQEAKVCAKVNLSNRAADGTAELQVLVTDG TKVICKESRNV
SLKQGADIQEQPLPLLIQKPRLWNGCEDPFMYQVVISLHKDGKQIDSVTQPLGLRYYHTDPDKGFFLNGKH
LPLHGVCRHQDRAE VGNALRPHHEEDVALMREMGNVNAIRLAHYPQATYMYDLMDKHGIVTWAEIPFVGP
GGYADKGFVDQASFRENGKQQLIELIRQHYNHPSICFWGLFNLKEVGDNPVEYVKELNVLAHQEDPTRP
TTSASNQDGNLNFITENIAWNRYDGWYGSTPKTLATFLDRTHKKHPELRIGISEYGAGASIYHQQDSL KQ
PSASGWWHPENWQTYHYHMENWKIIAERPFVWGTFVWNMFDFGAAHRTEGDRPGINDKGLVTFDRKVRKDA
FYFYKANWNKQEPMIYLAEKRCRLRYQPEQTFMAFTTAPEAELFVNGISCGKQKADTYSTV VWNV KLTS
GENIIRVTTPGKKPLTDEVTVEYKEDRPL

>SRS011586.37825-mL1 *Bacteroides fragilis*

QRQDILLNNWNFRFSHQVQGDTRRVDLPHTWNAQDALAGKTDYKRGIGNYEKALYVRPEWKGRFLFLRF
EGVNSVANVFINRKHIGEHRRGGYGAFIFEITDLVKYGEKNSILVRANNGEQLDIMPLVGDFNFYGGIYRD
VHLLITDETCISPLDYASPGVYL VQEA VSAQEAKVCAKVNLSNRATDGAAELRIQVTDGSKVICKENRNI
SLKQGTDIQEQLPLLIQKPRLWNGCEDPFMYQVTVSLHKNGKQIDSVTQPLGLRYYHTDPAQGGFFLNGKH

LPLHGVCRHQDRAEVLGNALRPQHHEEDVALMREMGMVNAIRLAHYQATYMYDLMDRHGIVTWAEIPFVGP
GGYADKGFVNQLSFRENGKQQLIELIRQHYNHPSICFWGLFNLKEVGDNPVEYIKELNVLAKQEDPTRP
TTSASNQNGDLNFITDNIAWNRYDGWYGSTPKTLATFLDRTHEKYPEIRIGISEYGAGASIYHQQDSLRO
PSASGWWHPENWQTYHYHMENWKIIAERPFVWGTFVWNMFDFGAHRTEGDRPGINDKGLVTFDRKVKKDS
YYFYKANWNKQEPMIYLAEKRCRLRYQPEQTFMAFTTAPEAELFVNGVSCGKQKADTYSTVVWKGVKLV
GENTIRVTPGKKPLTDVVVVVEYKEDGSLQ

>SRS017821.165554-mL1 unclear fragment

AFGIRKVSYDEKGLYINDRPVYLKGFGMHEDFFVLGKGASSAVNVKSFELLKWINANSVRTSHYPYVEEI
YDLADRYGILVIDEVPVGMNHWEPFAEGRADEQTLALHKKTLMSLIERDKNHPCVVMISVANEAAATYE
DGAREYFKKIVNHTRSLDLPITSPEETKFAEDNKCADLDFVSLNRYYGWYDEIGRIEAIPLYKNDLC
NYYEKFRKPIIVAEFGADTVEGVHSLQSYVFSEYQRDYITECCRVFDELFPVVGEHVWNFADFRTKEGT
IRMRGNRKGVFTK

>SRS024075.141365-mL1 unclear

MLKIQTNACRTAVNLDGIWKFKTVGEEFFPAEPLKNFILMPVPASMNDIVTDIDLREYVGKVVEYETESF
PLENGREYRVRIGAASHKCQVFLNGDFIGGSDSAFLPVDLKLPLKEVNRRLTVIVDNRLDYHTLPSGRLV
NGNDLFVYDQLGKYIVPGDKKQFQKQKQIINHDFYNYTGIHRSVYVYSLPEKHVEDIVIRTVADGDYRKV
EVKVTGDTMGISYTIADNRNGDMVAESKTGEFLIENPRLWQVKDSYLYTLTVRTATDLREKFGIRKISFD
DKRLYVNDKPVYLKGFKGKHEDFPVLGKASNSAVNVRDFELLKWTNANCFRTSHYPYSEEMMDLADEYGLL
VVDEVPVAVGCNNWPEYTYGENRLDDITLSLHKNSLKLMYERDKNHPCLAMISVANEAAATYEDKARSYFSE
VVGFRVSLDLPITIVELTRACEGNKVADLVDFIGLNRYFGWYDEIGNLKAETLLKRDNLAYYMQFKKP
IIMFEFGAEAIIEGLHSLPVVAFSEYQSEFLKEYCRIFDELPHYVQGELVWNFADFMTKQGTIRVNGNRKG
VFTRDRQPKTAAFYLKERWANEIKK

>SRS011134.278894-mL1 unclear fragment

KLFGKEKQIINHDFYNYTGIHRDVYIYSLPQKHIDDI IKTVVEDDYHAVSVELKGDKTAVYSVTDENG
NTVATSETGELYIENPHLWQIRNAYLYTLTVRTATDCYHEKFGIRKISFDDKCLYVNDSPVYLKGFGMHE
DFPIIGKGNNSAVNIRDFELLKWINANCFRTSHYPYAEIIMDLADEYGLLVIDEVPVAVGCNNWPDYTYGE
NRLDDITLALHKESSLRLMLERDKNHPCLAMVSVANEAAATYEDTGREYFSKVIEYVRSLLDLPVTIAELTR
ACEGNKVGDLDVFIGLNRYFGWYDEIGNIKAAEP LLKRDLEAYYEQFKKPVIMFEFGAEAIIEGMHSLPSV
AFSEYQTDFLREYCRIFDELTYVQGELVWNFADFMTKQGTIRVNGNRKGVFTTRERQPKSAAFYLKERWQ
NKTEK

>SRS016267.108985-mL1 Firmicutes

MLYPEMNASRALIDLSGIWEFQLAQAEGFEEEWTKVPFEHPEHIAVPASYNDQKEGTQFRDHYGWAVYQK
KLEISTMLKKERVVLRFGSVTQAARVYLNGLTICHTKGGFLPFEVEINPYLEENLLCVAVDNRVNYGTL
PIGNEVGAAFFGSDLPDIESVRNMKARPQNLPNFDFFNYAGIHRPVKLYTTPWNYIKDITIVPDIEGTDG
IVQYQIDCEGTGDLHVEIYDEAGQKVCEGFEPGTLRIPDAHLWEPGNAYLYQACVTFGEDSYTQSFGIR
TVRVEGIRFLINNKPFFYFKGFKGHEDSAVHGRGLDEALNVKDLSLMNWLHANSFRTSHYPYSEEMMNLCD
REGIVVIDETPAVGLNFGWDDTEGFAKFRTEKHEDVIRDMIARDKNHPCVVMWSIANEPDTEKQPQASY
DYFKPLYDLAHACDPQDRPVTLVICQNDYIRDLVAPAMDVICFNRYYGWYVYGGNLEISKEAFEYEMKYW
AELGKPAMITEYGADTVAGLHSATPNMFTEEYQMEYYRAMNAVLDQCPFIIGEQMWNFADFATIQQVMRV
DGNKKGLFTRDRRPKLAAHYFRQRWEEIPDFGYKK

>SRS015578.88564-mL1 Firmicutes

MYPQSNSARSVRDLSGVWDFRNVDEPWQPIAVPASYNQSPDPEFRRHGMAYYRTTFTVPEGARRVLR
FDAVTHNAVVRNLNGQEIASHRGGFLPFEADITALAQPGETVTLDEVDNRIGHSSLPVGNEGGTAFFGSD
NAGIPAVEAGKARQQAQGINLPNFDFFNYAGITRPVHLYTTPAAYIRDVTVAADMTGTLRYTVDTVGEYS
VKVEILDAEDRVVACADGAAGTVQVENARLWQPRPGTPYLYTALVRFGKDEYRQKFGFRSVEVSGHKFLI
NGAPFFYFKGPKHEDSAFRGRGYDACVTVDLKLKLYEWNANCLRMSHYPYAEVYDLCDRLGIVVIDETP
AVGIGAGAACDPYQTFPLKTYHSAVLRAMIERDKNHPCVVLWSLGNEDTEHFPESARDYWRPLYEQAHT
QDPQDRPVTMVCCQNDYTKDITTRTMDVVCINRYYGWYNLSGDLNAAAFKKELDFWETIDKPLILSEY
GADTVAGMHGFAPEMFTEEFQVEYYRTINGCLDERRFVVGEPWNFADFSTQQGPMRVGSCNRKGLFTRE
RTPKLAAHYFRDRWGKKEPNDR

>SRS012902.38584-mL1 Firmicutes

MLYPKMNRARTVMDLSGLWEFKLGDYETPGDNCELAGGERIAVPASYNDQKDDPAYKNHYGWAWYEREV
TVPAFFFKGQRLFLRFDVTHRARIYINGTFVMEHKGGLPFELELTELVPGETLLLTVAADNRISHSTL
PVGNEGGTAFFGSDNPGVPSVEAAKVWRKQGNLPNFDFFNYAGINRPVRLYTTVPVSRICGLTLIPDIDDT
DGILQYEAETEGEGAVHVEIFDVEGNKVAEAEAGASGTITIPDARLWWPWPGETPYLYRAVFTFGEDIYEET

FGIRTVKVEGTQFLINGKPFYFKGFGKHEDSPFHGRGLDCLDVKDVNLIHWHANSFRTSHYPYAEEMY
QLCDRGGIIVIIDETPAVGIGAGEHCDPYQTFFIREHHEAVLRDMIERDKNHPCVVMWSMGNEPDTEHFPE
SAYEYWHSLYEYTHTLDPQDRPVTFCVCCQNNYEKDLVTRTMDVVCINRYYGWYNLSGDLDAACYALNMEL
DFWEKIGKPVMITTEYGADAVAGIHQCVPEMFSEYQMEYYRRQNEEFDKRSFFMGEQVWNFADFADFQGC
MRVDGNKKGLFTRERRRPKLAHYFRKRWENIPNFYKAGGNTDEK

>SRS020869.14863-mL1 Bacteroidetes

MKNGMKIRYKATDFFLSVTHKKPYLDKKVPFLRQKVPFFLKGsfYLCYRNKEINNMKHILIGLLLVLCST
LTYASRKEILLNNDWMFRFSHQVQKNSAERVNLPHTWNSGDALSGKQDYRGMGNYEKKLFVKPEWKGKR
LFLRFEGVNTVATVFNVSCLGEHRGGYGAFVFEITDKVKYGAENEILVKVNNALQLDIMPLVGDFNFYGG
GIYRDVNLIIITDPVNISLTDYASPGVYLIQNKVTKEQADVAKVVLVSNSSENVPVQVNVKIWDGNKLVQ
QQDLNLTVDAGQVHTAGLEFSIKNPHLWNGRKDPFIYRTEVTLLGDGKEIDRVVQPLGLRFYHVDAAEQGF
FLNGEHLKLRGVCRHQDRAERGNALYKEHHEEDAAIIAEMGTNAIRLAHYPQATYFYNLMDKYGIVAWAE
IPFIGPGGYLDRGFNDLPSFRENGKEQLKELIRQHYNHPSICFWGLFNEKTHGDNPIEYIKELNELAHQ
EDPTRPTTAASFLSYSDISKITDVIWNNQYFGWYGGSPSDMGKWL DANHKAHPDYKIAISEYGAGASIY
HQQDSIKPGVASGWWHPENFQTFYHMGWNEAIAERPFWGFSFIWNLDFDGAHRTEGDRPGINDKGLVTF
DRKVKKDAFYFYKANWNKDDKFVYIANRRHRDRTTQVADVMVFSNLPEVELFVNGKSMGKQTPDNYATFV
WKGINLADGENTIEARAVQKKNSITDSVNWTK

>SRS020869.254180-mL1 Bacteroidetes

SRKEILINENWKFRFWHQVQFNSEHRVDLPHTWNAQDALSGKQDYFRGVGNYTEKELFINPEWQGKRLFLR
FEGVNTVANVFINEVHIGEHRRGGYGAFVFEITDKVKYGETNKKLLVKVNNALQLDIMPLVGDFNFYGGIYR
DVKLIVTEPVNISLTDYASPGVYLIQKVTKEQADVAKVVLVSNGSEIAQSTRVRVRIWEGRQLAGEQDL
PITIDAKDYATTGLEFHIKNPHLWNGRKDPFMYRAEITLLSGGKETDRVEQPLGLRFYHVDAAENGFFLNG
EHLKLKGVCRRHQERAIEVGNALRKEHHEEDAAIMVEMGANAVRLAHYPQATCFYDLMDKYGLVTWAEIPFI
GPGGYQDRGYINQPSFRANGKEQLKELIRQHYNHPSICFWGLFNEKTHGDNPIEYIKELNELAHQ
RPTTSASFLPYDNEINGITDVIWNNQYFGWYGGSPSDMGWTLDAHKSHPYKIGISEYGAGASIYHQQD
SMKRGEASGWWHPENYQTYHMGWNEAIAERPFWGFSFIWNLDFDGAHRTEGDRPGINDKGLVTHDRKV
KKDAFYFYKANWNQEDKFVYISNRNRDRVNPVTDIMVFSNLPEVELFVNGKSLGKQRPDELATFIWKS
VLQKGVNKIEACSVQKKEKIQDKVEFFLNAE

>SRS016585.25350-mL1 Bacteroidetes

MNLRHIIILVILSYWALCGYSQRICKSLDDNWMFRFSHQVEKGTGVRITLPHWTNAQDALSGKLDYKRGIG
NYERTLYIDSQWKGKRLFLRFEGVNSIANIFINGKCLGEHKGGSYSAFVFEITDKVDYGKNNKLLVRVNSA
ERLDVMLVGDVDFNMGYIRGVKLVITESACISLQDYASPGVYLTQQQVSPDEAEISATVLLSNRLAKRD
VVVRVEVKGQGRVISRHEQQMTLEKEKDEKVVPFIKIESPHLWNGKRDPFMYQTETTLWSDGRLLDKVVQ
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>SRS049959.63877-mL1 Bacteroidetes

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>SRS024625.141426-mL1 Bacteroidetes

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>SRS013476.24854-mL1 unclear fragment

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>SRS014923.117525-mL1 Firmicutes

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>SRS063985.92081-mL1 Firmicutes

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>SRS011239.205837-mL1 unclear

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>SRS013800.181271-mL1 Bacteroidetes

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>SRS064276.138185-mL1 unclear

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>SRS011586.85243-mL1 unclear

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>SRS015217.3050-mL2 Akkermansia sp. CAG:344

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>SRS011586.70098-mL2 Akkermansia sp. CAG:344

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>SRS018656.109612-mL2 Akkermansia sp. KLE1797

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>SRS024132.332613-mL2 Bacteroides cellulosilyticus

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>SRS020328.271517-mL2 Parabacteroides goldsteinii

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>SRS024009.5309-mL2 Bacteroides multispecies

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>SRS011586.60073-mL2 Parabacteroides merdae

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DRKEILVYSNCEVELFVNGVS QGVKRRNSQDYP AAGLRWNCVYREGMNEIRAVGVKKKEKKEVSDVIRQ
EYQTAKWDKEAACQVSL LSEEGDTALVQVQLIDKNGSRCLSSKKQITFEIAGNGSLICNLGTSTGSRKVQ
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>SRS017821.49250-mL2 Bacteroidetes

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DIREATMEII IKDPAGKIVLEEEKKV VPLGKAPLLTTTVQK PRLWSDKHPLLYTCEITLKHNGQE QKAVE
HFGFRHFEFIEKGP FMLNGSRLLLRGTHR HEDHAGVGAAMTKEQMRAEMI LMKQMGVNFIRLGHYQQSSI
ILSLCDELGILVWEEIPWCR GGLGGKRYQNQAKRMLRNMIH THYNHPAVIIWGLGNENDWPND FPEFDKG
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EWGGDCHARRHAEDPFVNL DKIEAGKGADERAGDASLYGGNARASKDGDWSESYMVRLIDWHLKEQETMD
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TRKWEAPARLQTQITSCG KDSLLEVVQIVDIRGVPCLDARNYIEFGLAGDGK LLENLGTSTGSRKVQAYN
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>SRS016335.53806-mL2 Bacteroidetes

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KIRTFMKELHDLAHLQ LDDNRVTAIRRCD FCKDIVDVYSPSIWAGWYRGIFTDYKNASYHEM QQVKHFFHA
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>SRS020869.55801-mL2 Bacteroidetes

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FDKGAIIRAFMKEQHDLAHLDDTRMTAIRRCEFCSDIVDVYSPSIWAGWYRGVFTDYKSVSEEEMKKVKH
FLHVEWGGDSHARRHSENVFANLKGIESGKGADERDGDASLYGGAARASKDGDWSESYIVRLIDWHLKEQ
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VQEYQTAKWGKEAVATLSVVSREGD TVLLEVQLLDKEGVRCLDSKKYIEFEIAGDGS LIQNQGTSTGSRK
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>SRS019601.24033-mL2 Akkermansia muciniphila

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>SRS019582.130094-L1 Clostridium sp. CAG:307

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>SRS017191.56930-L1 Escherichia coli

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DVSVELRDADQQVATGQGTSGTLQVNVPHLWQPGEGYLYELCVTAKSQTECDFYPLRVGIRSVAVKGEQ
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LRVGGNKKGIFTRDRPKSAAFLLQKRWTKGMNFGKEKPQQGGKQ

>SRS023176.24667-L1 Ruminococcus gnavus

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>SRS018351.29729-L1 Faecalibacterium prausnitzii

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>SRS022071.224144-L1 Faecalibacterium prausnitzii

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 >SRS077730.142013-L1 Firmicutes uncultured bacterium fragment
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 KGE
 >SRS014979.79627-L1 Firmicutes

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>SRS022609.165334-L1 Firmicutes

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>SRS011405.145603-L1 Firmicutes

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>SRS019161.247428-L1 unclear fragment

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>SRS014923.300030-L1 unclear

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>SRS011586.88897-L2 Bacteroidetes unconfirmed sequence

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ASPGVYIDQYKVTSHVAEIAVR TKLENRYKEPRSVMDYLLFEADGTLKEKKSHEITLSPQGRQTDIQPF
TIRRPHLWQGLDDPYLYRIVVRVKSGGHIVDEITQPLGLRH FELRAGDGMYLNGRRIPMYGVCRHQDWWQ

SGSALTPQQHDTDLSSI IRELGATTIRLAHYQQSEYVYSRCD SIGFLVWAEIPFVNRVSTHEAENAKEQLR
ELIRQNYNHPSIYVWGVHNEVYGYGEDDGLYTDNTRFSPVVALTTDLHDLAKTEDPARLTVSVNGHAAPG
HGVNMNTDIQGINRYFGWYERRIQDLEPWIESMERDYPGYRIMLTEYGAEANPEHQEELAGDVGDCCGFD
RRYTEVFATRSHEIQWGIISRHPYLLASYVWNTFDFATPASSQGGVEARNMKGLVTFDRQCRKDPFYWK
ANWSKDPVLYITQRRVVERRQKHTPVTVYSNVGIPRLFVNEQEVTVIKGTTSVHYIFPDVELKEGDNQL
SVRVVEGEKAIEDTIYWHYSGILSPDVPVYFREKEGEHVGL

>SRS011405.14712-L2 *Bacteroides intestinalis* CAG:564

RMMKTINDGWDFRKDGETRWQPINLPHTFNLDAYSQRNYYQGKGEYRKKLSLSEIAPTKRYYLKIDAASK
AADVKVNGQVAGSHAGGYSAFILDVTGLIRENNEIEITVDNARRDITPLWADFTFWGGIYRDVWLITPPE
QHFNMANYGSTGVFVSTPVANEREGVVQVKA EVTNDADSRIRLKMQNRIYDAQGKLLQTNAQPFSLKAGE
TATVEYKSDVIDNPQLWTPPEMPTLYRVVTAIVNAKTGEVLDEISNKIGFRWFSFDADKGFSLNGKPKYKLR
GVNRHQDQAPVGVAIDDEVNRRDIRQIKEIGCNFIRISHYPQDDALLDACDELGLLAWEEIPIIDIVPDT
PGYDDHCEMNLVEMIRQHYNHPCIMAWGYMNEILLIAPTPDKPEWPAVRDRTVKLAQRLEKRLKEEDSSR
VSVMAFHGSDLYNTIGLALGDVAGWNLYQGWFYFGE LPHFEQWLEDQHARYPHRPIIVSEWGAGSDKRIHS
VKGRPFDFSM EYQQEYIEHYLPYIENNDYIAGCAYWGYVDFNVAARQESMPRVNNKGLFYNDRTPKDIAY
YFKAMWRKDV PVLHIASRDWALRTGKAGEEHPKI KIYTNLDEVELILNGASC GKQCTS NYNTIFRVKLPEG
NSSLVARGYRNGVLVEDVMMMTLALFPDLHQGDELAVNVGSNCYFISDVSHLTWLPDRPYTAGEWGYVNG
KGRSTTSEIYNTVDGPVYQ TWMEDITEYKIDAPAGTYEV ELLMADVSRPARQQANLLGKGDERASAVSKR
FDITICGEVVEQNFSPADNNRYLNACRRRYIVNNNDGCIDIRFTPLQGKPVLSGLKVRRL

>SRS019787.27227-L2 *Bacteroides intestinalis*

RMMKTINDGWDFRKDGEAQWRSVNLPHTFNLDAYSQRNYYQGKGEYRKKLSLAEIDPAKRYYLKIDAASK
AANVKVNGQVAGNHAGGYSAFILDVTGLIRKENEIEIIVDNARRDITPLWADFTFWGGIYRDVWLITPPE
QHFNMANYGSTGVFLSTPVANDREGVVQIKAEVTNDASSKVRLKMRNCIYDAQGKLLQTKIQPFLLKAGE
TATVECRSDVIANPQLWTPPEMPALYCVTTTIVNAKNGDILDEVSNKVGFRWFSFDSGKGFSLNGKPKYKLR
GVNRHQDQAPVGVAIDDEVNRRDIRQMKEIGCNFIRISHYPQDDALLDACDELGLLAWEEIPIVNMVPDT
PGYEDNCETNLVEMIRQHYNHPCIIAWGYMNEILLIAPTPDKPEWPTVRDRTVQLAQRLEKRLKEEDPSR
VSVMAFHGSDLYNTIGLALGDVAGWNLYQGWFYFGE LPHFEQWLEDQHTRYPHRPIIVSEWGAGSDKRIHS
IKSRPFDFSI EYQQQYVEHYLPYIENNDYIAGCAYWGYVDFNVAARQESMPRVNNKGLFYNDRTPKDVAY
YFKAMWRNDI PVLHIASRDWASRIGKAE EEQFVKIYTNLDEVELILNDTSCGKQRP SNYNTTFRVKLPKG
TSSLIARGSKNGTFVEDVMMITFNLLPDL SHGDELAINIGSNQCFTSNVSQLTWLPDRSYTVGGWGFVNG
NSRSTTSEIYNTIDGPIYQ TWMEDITEYKVDAPAGTYEV ELLMADISRPSRQQANLLGKGDERASSASNR
FDITICGKVEQNFSPADNNRYLNACRRRYIVYNADGCIDIRFTPLQGKPILSGLKIRKL

>SRS011529.64867-L2 *Paraprevotella clara*

QRDTKTINDGWFFRKASTENWQPVNVPHTYNLDAYSQRNYYMGKAFYQKSFVLPEVPQ GKRCFLKIDAAS
KAADLMLNGQKIGSHAGGYS AFTFDITPYLQKNNTIEITVDNSRKDITPISADFTFWGGIYRDVWLITTS
EQHFNLLNHGSDGIFISTPHVDEKEATIEIRSEITNDATEKATLEVIHQIYDPQGKLQQT LKQRLSVKAG
ETQTVRKTS AIGQPQLWSPESPRLYEIVTTLRNPKTGMELDRSTHKTGFRWFSFDGEKGFSLNGKPKYKLR
GVNRHQDQAPVGVALDDEAHRRDIRLIKEMGCNFIRIAHYPQDDALVEACDEMGLLAWEEIPIINIVPDT
PGYDDNCETNLTEMIRQHYNHPSIIAWGYMNEILLTAPGPDRPEWPACRERTVKLAQRLEKTLKQEDPYR
ASVMAFNMTNLYNEIGLDLVDVSGWNLYHGWYVGR LNDFNRMEDQHARYPKRPTII SEWGAGSDRRLHS
RHPKAFDFSI EYQQTYVEHYLPFIEGTEYISGCTYWNFIDFNVAARQESMPRVNNKGLFYNDRTPKDVSY
YFKAMWRKDT PVLHIASRDWPVRTGDKNEKHPVKIYTNLDEVELIINGKSIGKQRPANCHTVFNVTLPYG
TSTLKAQGDKNQMVEDVMFLRLDAMPDLAKGDELAINVGS DCAFTSDVSGLTWLPDQPYFKGSWGYIEG
KARSTTSEI ENT TDGP LYQTWREN LRAYQIDAPSGTYEV ELLMADVSRSRPQLANLLGRGDDGQAIADSR
FNITICGRMETDFSPADGGHYRQAFRRRYIIQNKENKIDVLFETLKGKCHLAGIKIRKL

>SRS019601.88766-L2 *Bacteroides vulgatus*

QREMKTINDNWEFRKSIDESWESVNLPHTFNIDAYQQRNYYQGKG FYRRTLVLPEIVAERRYYMKIDAAS
KAANIRVNGKEVGS HVGGYTACIVDITEYIRKENLIEITVDNGRKDITPISADFTFWGGIYRDVWLITP
KQHFNMNSMGS DGI FISTPVVNEKRGVLVKVCEVTNDSHES SILEVRS AIYSPQGKLLQTIKQKQKLKSG
ETYIFENTSGAIESPD LWPETPSLYLVKTTLVDPKSGKLLDEKNHKVGFRWFTFDGSKGFFLNGKSYKL
RGLNRHQDQAPAGVALDDEAHRRDIFLMKELGCNFIRISHFPQDDAILEMCDELGLLAWEEIPIINIVPN
TPGYDDNCEYNLREMIRQHYNHSSVITWGYMNEILLTAPS IGKPEWLACKERTVNLAQRLEAVLKEEDPG
RASVMAFNMTNLYNEIGLNLVDVVGWNLYHGWYVDKLKDFNAWCEDQHRRYPDQPMI ISEWGAGSDKRLH
STQGRAFD FSI EYQQTYIEHYLPFI ENT EWISGCAYWNFIDFNVAARQESMPRVNNKGLAYNNRIWKDVA
YYFKAMWRKDIPVIRIASRDWEMRTGEINKPQSIKIYSNMPEVELFINGQSIGCQKIANCHTVFNVILPE

GISVLMAQGIKNGKTVQDAMTIQFKSLPNIAEGDELAINVGSNCYFTSDISNLTWLPDQPYTAGGWGYIG
 GKAHSTTSEIYNTLDGPIYQTWREGNWSYKIDAPIGEYEIELLIADVTKPAAQLANLLNKNKEEKHSGET
 RFHISICGKQVETNFSPIDGGKHRTAFKRRYIIRNEHDHIVISAGAVKGEPFLAGIKVRKL
 >SRS018656.37495-L2 Candidatus bacteroides timonensis
 QREMKTINDNWEFRKPMDEKWESVNLPHFTFNQDAYKQRNYYQGEGLYRRTLVLPLDIVKGRRYYLKVDAAS
 KAATVSVNGNKAGCHAGGYNAFTMDITNYIQKENTIEITVDNSRKDITPISADFTFWGGIYRDVWLSTP
 EQHFNMNTMGS DGIFINASITNKKSGVLKVRSEVSNDALKITTLLEVKNAIYSPEGELLQ TINRKLKLKAG
 ETRTVECISDDIQNP SLWTPETPSLYLIKTTLTDLKTGKLLDEKIHKTGFRWFAFDGNKGFSLNGKPYKL
 RGFNRHQDQAPVGVALDDEAHR RDIALMKELGCNFIRISHYPQDDTVLEM CDELGLLAWEEIPIINIVPD
 TPGYDDNCEQNLREMIRQHYNHPSVIAWGYMNEILLTTPAVDKPEWPACKERTVALAQRLETVLKEEDPS
 RVSVMAFNMTNLYNEIGLNLLEDVVGWNLYHG WYVGELKDFNAWCEDQHRRYPNQPII ISEWGAGSDKRLH
 TTNGRAFD FSIEYQQTYIEHYLPFIEETEWISGCAYWNFIDFNVAARQESMPRVNNKG VAYNNRTLKDVA
 YYFKALWRKDVPI LR IASQDWDTRVGETNKSQAIKIYSNMPEVELFVNGQSVGSQKIFNCHTIFNVLLPK
 GESTILVQGVNDGKTVQDAMKICFPFPNIAAGDELAINVGSNCYFTSDISSLTWLPDQPYTSGN WGYID
 GEEHSTTSEIHNTVDGPLYQTWREGNFAYKIDAPQGDYEIELLMADVTKPATQLANLLGKSSGEKHSGET
 YFNISICGKQVETNFS PVDYGHYRTAFKRRYIVQNNDNCITINLNTIKGAFLSGIKVRRL
 >SRS049959.38447-L2 Prevotella copri
 QRKAATINDNWEFKLPTSQKWT SVNIPHTYTLDAYQGRNYYKGKAEYRRILTLPEINPDRRYFLKIDAAN
 KAAEVKVNGKEVGCHAGGYSSFTFDITDFLNTNPARQGEKSENTIEITVDNSRPDVTPI MADFTFWGGIY
 RDVWL VSTPDIHFNMLNMGSDGIFVSTPIVNEKQSAVKVSEVSNDGQKASMVELRNEIFSPDGKLLQTI
 KK RITLKAGETQRTALQSKPIANPLLWTPERTLYKVKTSIIDTKSGKVIDEKNHKVGFRWFSFDGEKGF
 CLNGKTYKLRGFNRHQDQAPVGVALPDEAHR RDIKLMKELGSNYIRISHYPQDDALLDACDELGLLAWEE
 IPIIDLV PDPHYADNCERNLREMIRQHYNHPSIINWGYMNEILLCTPWPGTKEWPAFKERTLALAHRL
 KVLKDEDPTRKSVMAFNMTNTYNEIGLNLVDVVGWNLYHG WYQGELNGFNHWCEDQHORYPKKPMI ISEW
 GAGSDQRLRSNSAHAFDFSIEYQQTYIEHYLPFIEEKPWISGCTYWNFIDFNVAERQESMPRVNNKGIAY
 NDRTLKDVA YYYFKSMWRKDI PVVHIASRDWSIRTGHINEPQRIKVYSNMPEVELIVNGRSYGKKS VQNCF
 AVFDVVL PFGSSTLEAKGFNEVLTDYKNKVDGNTGDMKIQYNPLPNLAKGEELAINVGSNCYFISSLSQ
 LTWLPDQAYKPGAWGYVGAESKSTTSEIENTIDGPIYQTWREGDLEYRIDAPCGEYEV ELLMADVTKPAV
 QLPNLLAKSNTESSSKDVRFDVSINDERKESDFTPTDGRHYRTAFKRKYIVENNRGSINIQLKSLQGKAF
 LNGIKIRKLN
 >SRS045713.14514-L2 Bacteroides uniformis
 QRQTQTINDSWKFLKGECTAAADSAFDDSKWTSIHLPH TWNTDAYTEKDYYRGTGWYRRQLTLPQGWEK
 QIILRLDAASKAATIIYINGKNVGEHAGGYTACSFNITPFLSFDTPNTLAICVDNARQDIAPISGDTFFG
 GIYRDVWLTA VPNQHFNLTNHSSDGLFISTPQVSEERGIISIRGEVKNDAKEKATLELHTTIYRPDGTLL
 QTLKKNIQLKAGETYAFSNEATPVLKPELWTPETPRLYRVETTLRN RKTLLDQSNHYTAFRWF RF DGD
 EGFFLNGKPYKLRGICRHQDQKPIGPALTD EMHRRDFLLMKEMGANFIRISHYPQDDALLEMC DKLGLMA
 WEEIPIIDIVPNTPGYGDNCERNLREMIRQHYNHPSIITWGYMNEILLVTQRKYKTEAELKPV LERTLAL
 ANRLERVLKEEDSTRISTMAFHGSNSYNETGLSKITDIVGWNLYQG WYGGDLTGFEKFLAQQHQNHPHP
 MIVSEYGAGSDKRLHSLHPRAFD FSIEYQQKYLEHYLPVLEDTPYICGGTHWNFIDFSSALRDESMPRIN
 NKGLVYADRTPKDVYHYYYQAAWRKDI PVLHIASRDWTD RAGVQQGNAPVYLPVKIYTNLSEVELFIDGIS
 LGKQKTENYTATFEVPFSNRNPFLFAQGNYQ GKTVQDGLRINF TPIPACLDANNLKGLELAVNVGSQCFF
 TSDESQLTWLPDQPYAAGSWGYIGGKEGTAQTEIQNTADG PLFQTLRNEIEGYRFDAPQGVYEIELLFTD
 IFRNAGIAYQLDRNGQ QENRESTFGISINGEVVEESLSPCKESGYFRALRKKYYITNDKEYIDIRFHST
 SGTCFLNGIKLRNIY
 >SRS019161.95752-L2 Alistipes timonensis
 RVKYTVNDAWRFTKGSPYNAFLPAADDSAWEVVNIPHTWNAEDAVDDVPGFYRGPAWYRKRIVLPADAAS
 KQVYLCFEGANQVVRLYVNGHYAGKHTGGYTRFVFDITKYIVPGGENLFAAEVDNAHDPDIPPLSADFTF
 FGGIYRDVSLMTDKVHVAPTDFASPGVYLSTPEVTAERATVEVRTLVANDGPETARVRIEHRIVAPDGR
 EAARTEDTAEIAAGSVAEQLRGITI ASPRLWDIDEPNLYRVLTRIVAEDGRVLDEVSNPLGLRWFS FDP
 DKGFSLNGRPRKLVGTCRHQDYLGRGWALPDEMHERDVRLIKEMGGNFLRVSHYPQDPVVMEMCDRLGIV
 TSVEIPV VNAVTESDGFLENSVRMVREMIRQDFNHPSVMIWAYMNETLLRPPYTDEARTKEYYKAI EHVA
 RTLEETVRS EDP SRYTMIAFH NAPERYAAANLTQIPMIQGWNL YQG WYEK DITGFERILDR LHAQYPDKV
 LIVTEYGADVD PRLHSFSPEQFDFSQEYGLVYNRHYLSEMQRSF IAGSSLWNLNAFYSEPRADVVP HVN
 NKGVTGLDRELKDTYLFYKTVLNRT PQLVIGNRGWQNRSGADDGTGRCTQPV PVFTNQSSVTLT LNGRKV
 GTRPEGGVAMFDPVFADGENRLVARAGALTDVLT VGVFRLAQASPERFTGLNVMLGSPRYFDDREAGLAW

IPERPYTPGSWGYVGGEALRRSGAGWLGSASDILGTKNDPVFQTQRVGIESFRADVPDGTYSVYLYWAE
DAARQREALAYNLGADGAQQQYTGRRFDVSVNGRKVLENFSIAAEVGFSSRAMVRKVDVIVRDGKGLRVDF
GRIEGEPLNARIYRNH

>SRS017307.71236-L2 *Alistipes senagalensis*

RVRYTVNDAWRFTKGSFYNAFQPAADDSAWEVVNIPHTWNAEDAADDVPGFYRGPWYRKRIVLPA
KQAYLCFEGANQVRLYVNGHYAGKHTGGYTRFVFDITKYIVPGGENLFAVEVDNAHDPDIPPLSADFTF
FGGIYRDVSLLLTDKVHVAPTDFGSPGVYLSSTPEVTAERAABEVRTLIVANSVTEPVRVRVEHRIVAPDGR
EAARAEDMAEIAAGSVAEQLRDITIASPRLWDIDEPNLYRVLTRIVAEDGRVLDEVSNPLGLRWFSFDP
DKGFSNLGRPRKLVGTCTRHQDYLGRGWALPDEMHERDVRLIKEMGGNFLRVSHYPQDPVVMEMCDRLGIV
ASVEIPVNAVTESLEFLENSVRMVREMRQDFNHPSVMIWAYMNETLLHPPYTDETRTKEYKAIHVA
RTLEETIRREDPARYTMIAFHNAPEYAAARLTQIPMIQGWNLQGWYKEDITGFERILDRHAQYPDKV
LIVTEYGADVDPRLHSFSPEQDFDSQEYGLVYNRHYLEQMRKRPFIAAGSSLWNLNAFYSEPRADVPHVN
NKGVTGLDRELKDTYLFYKTVLSRTPQLVIGNREWRNRSGADDGSGRCTQPVVPFTNQPSVTTLTVNGRKV
GTRKEGGVAMFDPFADGENRLVARAGALTDVLTGFRLAQASPERFTELVNMLGSPRYFDDREAGLAW
IPEQPYTPGSWGYVGGEALRRSGTGWLGSASDILGTKNDPVFQTQRVGIESFRADVPDGTYSVYLYWAE
DAARQREALAYNLGADGAQQQYTGRRFDVSVNGRKVLENFSIAAEVGFSSRAMVRKVEVIARDGQGLRVDF
GRIKGEPLNARIYRNH

>SRS020869.18805-L2 *Bacteroides intestinalis*

QRESKTINDSWKFLQGECPAIDSAFNDANWTSVHLPHTWNTDAYTEKDYRGTGWYRRQISIPQSWKEK
QIFLKLDAASKAATIIYINGKKVGEHTGGYTACTLDITPFCSLNTPNLSLAVCVDNSRQDIPPISGDFTFFG
GIYRDVWLTAWSKQHFHLTNYGSEGIFISTPKVSEEKGTILIRGDIKNATQKALLELEHIIYHPNGSIA
QSLKQSIQIKAGEQYSFRSETSPILSPQLWTPETPHLYRVESILRDKKTKTVLDQSNHYTGFRWFSFDGE
RGFSLNGKPYKLRGICRHQDQKPIGVALTDEMHRDRIKLMKEMGANFIRISHYPQDDALLEMCCKLGLMA
WEEIPIIDIVPDTPGYTENCERNLREMIRQHYNHPSIITWGYMNEILLVTQRLYKKEELKPILERTLAL
ANRLEVVLKEEDTTRVSTMAFHGSNSYNEVGLGSITDIVGWNLYQGWYGGNLTGFEQYLEEQHKYPSHP
MIVSEYGAGSDKRLHSLQPHAFDFSIEYQQKYLEHYLPVLEETSYICGGTHWNFIDFSSALRDESMPRIN
NKGLVYSNRTPKDIYYYYKAAWRQDIPVLHIASRDWTHRSIGHQGKEPVILPVKIYTNLPEVELFIDGKS
LGKQKTKNFAVTFEVPFCQKEHFIAAKAENRKAVQDISFVEDGIRLNFPTIPANLNGTNLRDLELAINVG
SYCFYTSDESNTLWLPDQPYTENGGWYIGGESKSSQIQIKNTNDGPLFQTLRNGIEGYRFDVPNGVYEIE
FLFTDIFRENATTVYQLGRNSDVENRENRFDIINDQTIEESFSSCRESDYFHALRKRYNITNFRNKIEI
RFSVRNGKSFLNGIKLRKLY

>SRS015190.42469-L2 *Candidates bactericides timonensis*

QHESKTINDGWKFHKGEECAAASPAYDDSAWNSIHLPHWNTDAYTEKKYYQGTGWYRRTLMLPQHWQGK
QVFLKLDAASKAATIIYINGKEVGEHNGGYTACTFDITSLCSFNPNLSLAIRVDNARQDIPPIADFTFFG
GIYRDVWLTAWSKQHFHLTNHGSIDGIFISTPQVSEEKGTILIRGEIKNDAEQKASLELEHIVNPDGSIA
QTQKQSIQIKGGELYSFHTETTPILSPRLWTPETPHLYRVESILRDKKTKAILDRSNHHTGFRWFGFDGK
TGFSNLNGKPYKLRGICRHQDQKPIGVALTDEMHRDMMMLMKEMGANFIRISHYPQDDALLEMCCKLGLMA
WEEIPIIDIVPDTPGYAENCENNLREMIRQHYNHPSIITWGYMNEILLVTQRRYKKEELKPVLERTLAL
ANRLEKVLKEEDSTRASTMAFHGSNSYNEVGLGNITDIVGWNLYQGWYGGDLTGFERFLAEQHKLPVSH
MIVSEYGAGSDKRIHSLQPNAFDFSIEYQQKYLEHYLPVLEETPYVCGGTHWNFIDFSSALRDESMPRIN
NKGLVYSNRTPKDVYYYYKAAWRQDIPVLHIASRDWTHRSVGHGETPVLLPVKVYTNLPEVELFIDSKS
LGKQKTKENYTVIFEVPFSRKEHFISAKAESKEAARNLHTIEDGIHLSFTPAPAKLNETNLRLNLEAVNVG
SNCFYTSDESRLTWLPDQPYTENSWSGYIGGKSKSSQTQVENTNDGPLFQTLRNDIEGYCFDVPNGVYEIE
FLFTDIFRENAATAYQLGRNSDVESRENCFSIIVNDETVEEDFSPCRESGYFHALRKRYNITNFRNKIEI
RFPVRNGKSFLNGIKLRKLY

>SRS019601.55249-L2 *Bacteroides uniformis*

GRVVQSFNDGWYYVAYNDNRQKVRCHEEGWQVHLPHWNTDAYSTRNYRRASSWYKKEFRTDAAQMRD
KQLYLKFDVNSLADVNLGELLTHKGGYSAFTVDITDKVRTDAPNLLMVHVNQNDRIPLSGDFTIF
GGIYRNVWLISAQKHISLTDYASTGIYVDLPSVSEDRAEISVRGTLVNRDVRRAVLKLDVEVLDTDGKT
VAQSLRSVRIDADGAFAFEERLQLEKPQLWSPDSPYLYSVKVSLLKDVGRGVKLDKETRVPLGVRWFSVDAQ
EGFKLNGEPLKLMGACRHQDQMPMGIALSDEMHRDMMQLLKDMGVNFVRLAHYPQDDAVLRACDELGLMV
WEEIPVVDLIALGDEFANATSALREMIRQHYNHPSVIMWGYMNEAVIQLQYRVKKQRLNGFYENTLALA
RHLEETLKREDPYRLSTMAYHGNQIYNKIGLSNITDISGWNLYQGWYENDFKSFDRFVDEEHRKYPHRPL
IISFEGAGSDPRLQSLEPQIFDFSMQWQQLYLEYYLPAIMRRPFIVGATEWNFIDFSSANRQEATPHINN
KGLMYNDRRPKDVFFYFQAFRLRKDIPVLHIAVDDWKHRTTVSDGEAVEHPVKVYSNLDKVELSVNGKKLS

VQDIENCHAGWRVPLVVGRNTLVASGIYQGKKVEQVSDIFVKMQPRYIAAAGSGQLELAVNVGSNCFFTD
DKSDLCLWLPDQAYTPGSWGYIGGEIFRRSPGRIGTTAEVKDTRNVPLLQTKRKGIKAYRFDLPDGDYEVE
LLFADLNARSERVTYDLGAVATLDNADFRGSVFNVSVNGRPWLSHFSPAIEVGGNRCISKKLRIAVTGGN
LTVDFEAVKGMTFLNGIKIFRIH

>SRS048870.28850-L2 *Bacteroidetes*

GRVVQSFNDGWYYVAYNDNRFQKVRCHEEGWEQVHLPHTWNTDAYSTRNYRRASSWYKKEFRLEAAEMWD
KQLYLKFDVNSLADVYLNGELLTTHKGGYSAFIVDITDKVRNDVSNLLMVHVNNQNNQIPPLSGDFTIF
GGIYRDVWLISVGKQHISLTDYASPGVYVDLPFVSEEKAEISVRGTLVNRDVQRAALKLNVEVLADGKS
VVQSLRNIRLDAEGTFAVKERLQLENPRLWSPESPYLVSVKVSLKDERGKVLKDEMRVPLGVRWFMSDAQ
EGFKLNGKPMKLIGACRHQDQMPMGIALSDEMHRDMQLLKDMGANFVRLAHYPQDDAVLRACDELGMLV
WEEIPVVDLIALGDEFRTNATSALREMIRQHYNHPSVIMWGYMNEAVIQQLQYRVKKQRLNGFYENTLALA
RHLEETLKMEDPYRLSTMAYHGNQIYNQIGLSNITDVSGWNLYQGWEYENDFKSFDRFVDEEHRKYPYRPL
IISEFGAGSDSRLQSLEPQIFDFSMQWQQLYLEYYLPAIMKRPFIVGATEWNFIDFSSASRQETTPHINN
KGLMYNDRRPKDVFFYYFQAFLRKDIPVLHIAVDDWKYRTVVSDGEAVEHPVKVYSNLDRLVELSVNGKKLS
VQGIENCHAGWRVPLVVGRNTLVASGIYQGKKVEQVSDVFKIQPRHIAANIRQLELAVNVGSNCFFMD
DKSDLCLWLPDQVYTPGSWGYVGGEVYFVRSPGRIGTTAEVKNTRNVPLLQTKRKDIKAYRFDLSDGEYEVE
LLFADLNAPSERVTYDLGGTAVLSNADFRGSVFNVSVNGRLWLNHFSPEMEVGGNRCLSKKLRVIVTDGN
LTVGFEAVKGMTFLNGIKLFRIG

>SRS016203.125956-L2 *Bacteroides ovatus* ATCC 8483

QRVTQTINDGWKFSLFEGDASTADFDVSGWTDVSIPTWNAKDAEDEIPGYFRGKGWYRKAVTVEELIVG
QRVYLCFEGANQETNVFVNGKLVGNHKGGSYSAFTFDVTDYVHTGRNLVAVSVDNSYNPDIAPLSADFTFF
GGLYRDVYLVYTSPVQLSTTHYASSGVYLKTVGITDAQAEVCAKTFLSNALKSNQALILETEILDTDGNR
VALSAKKVNVKAGEKNVAFALMTIAQPKRWDVDSPLYKVSRLKNKKGEVLDCVVNPLGIREYHFDAE
KGFFLNGKYRKLIGTSRHQDYKGMGNALRDEMHIRDIQLSKDMGSNFLRVAHYPDQPVVMQMCCKLGLLT
SVEIPVNAITQSRAFMDNCVEQATEMVYQNHNYPSVIIWAYMNEVLLRPPFNPNDKKERAEMYKSLHQI
ASAVEAQIRSLNSERYTMLPCHSASQIYQEAGITELPMLLGFNLNGWYGGNLDGFEEKLEELHKEFPKH
PLLITEYGADVDTRIHSFSPVRFDSCFEFGSVYHEHYLPEILKRDYIVGAMVWNLNDFYSEARNAMPHV
NNKGLVSTDREKRDGYFLYQAYLKESPVHLIASKSWKNRAGASRDGKSCTQPLKVYTNADKVEVFLNGKS
LGVYPVADKVVSDIPFVNGKNVDAVIEKEGREYRDQYVCDFKCVNVKNGFTEINVLLGARRYFEDRIA
EMCWIPEQAYAEKGSWGYIGGEVAPNKTRYGSLPASDKDILGTDQDPIFQTQRVGIEAFKADVPDGVYAVY
LYWTELTSENKREALVYNLGNDDVREDYINRVFSVDINGVSVAKQLNIAEEYGSERAVIKKYIVPVSQ GK
GLVVRFGAVESVPILNAIRIVKEY

>SRS012902.13258-L2 *Bacteroides ovatus*

QRVTQTINDGWKFSLFEGDASTADFDISGWTDVSIPTWNAKDADDEIPGYFRGKGWYRKVVAVEELIPE
QRVYLSFEGANQETNVFVNGTFVGNHKGGSYSAFTFDVTDYVHAGRNLIAVSVDNSHNPDIAPLSADFTFF
GGIYRDVYLVYTSPVQLSTTHYASSGVYLKASKITDLQADISVKTFLSNALKSNQSLILETEILDADGNR
VALSQKKVNVKAAEKNVAFESLMAITQPKRWDVDSPLYMYKVSRLKNKKKEEVLDCVVNPLGIREYRFDAE
KGFFLNGKYRKLIGTSRHQDYKGMGNALRDEMHIRDVQLSKDMGSNFLRVAHYPDQPVVMQMCCKLGLLT
SVEIPVNAITQSKAFMDNCVEQVTEMVCQNYNPSVIIWAYMNEVLLRPPFNPENRTERADYMTFLHQI
ASAIEAQIRSLDSERYTMLPCHSTSQIYQEAGIAELPMLLGFNLNGWYGGSLSGFEELHREFPHK
PLLITEYGADVDTRIHSFSPMRFDSCFEFSIYHEHYLPEILKRDYIVGAMVWNLNDFYSEARNAMPHV
NNKGLVSMDREKRDGYLYQAYLKEAPVLHIASKSWKNRAGASRDGKSCTQPLKVYTNADKVEVFLNGKS
LGVYPVSDKVVSVDIPFINGENVDAVIEKEGREYRDQYVCNFCVNVKNGFTEVNVLLGAQRYFEDRTA
ELCWIPEQAYEKGSGWGYIGGEVAPNKTRYGSLPASDTDILGTDQDPIFQTQRVGIEAFKADVPDGVYAVY
LYWTELTSENKREALVYNLGNDDVKEEYANRVFSVDINGVSVAGQMNIAEEYGSERAVIKKYIVPVSQ GK
GLVVRFGAVESVPILNAIRIVKEY

>SRS015663.125464-L2 *Bacteroides eggerthii*

TENNPRITYSINDSWQFRLGTSDIGEDGWSMVSIPHTWNAEDCVDDFPGFYRGTGWYRKNIQVDSVMLSR
SLYLFFEGANQRTKLFVNRHRVGEHIGGYTFFCFDITPYIREGENQLVVCVDNSYDPEIPPLSADFTFFG
GIYRDVSLISVPIHISTTHYASSGVYVNTPEITEEKALVDICTMLSNTQAERSKIFVEHKIYSATGECV
ATSKKKVIVNTEREQAFNIQIVVNHPELWEIERPNLYRLQTCCLWDNEGKLLDEVSNFTGIRTCFSFAEKG
FELNGKAVKLLGTNRHQCHLGMGNALKDEMHVARDIELLHAMGGNFLRIAHYPQDEMVLAAACNRLGIVTSV
EIPVINTITMSQKFSNDCVEMMKEMIYQCFNSPSVCIIWAYMNEIMLRPLYNSESSIKKDEYLYKYLHIAE
RIENTAKEVDPGRATMLPCHGNLAAYEEANLINLSDILGMNLNGWYGGFEFGFEAILDTHQKYPDKPI
IISEYGADVDAIRHSFKPERFDVSVDYGVMYHRHYLPQIIQRKFVVGSAVWNLNDFHSEERMDAIPHVNC

KGLVTLDRIPKDTYRYYKAMLSKEPYVAIGGSDWYNRSGVADNSGVCYYPVSVYTNLKQVNLKVNGKMLA
VMKAENGIAQFNVPFINGINQLEAEGVSQGSYVKDFCRVNMKLVPSDFDNKESFTELNMLFGTARHFEDK
TASLAWQPEREYTSGGWGYVGGKAFRARTKRGSQPASNLNIYGTEQDPIFQTQRRGLNSFKADLPDGRYV
VYLYWTELSGADVVDLAYQLDNIVQKEKATERLFHVDINGMRVLSLSDVAKEVGVRRPMICKIPVTISAK
KGLTVDFIPIKGETMITAIRILKMD

>SRS023914.223128-L2 *Bacteroides stercoris*

AENNSRITYSINDSWQFRLGTSVGEDGWSIVSIPHTWNTEDCVDDSPGFHRGTGWYRRNIQVDSSMLNR
SLYIFFEGANQRTELFVNHHRVGEHIGGYTFFCFDITPYIKEGENLLAVRVDNSYSDIPPLSADFTFFG
GIYRDVSLISVSPVHISTTHYASSGVYVSTPEITKEKALVDICTMLSNTQTEREKILVEHKIFSATGECV
AASKKKVIVDTQKEQAFKTQIVVNHPELWDIEHPNLYRLQTTLWDTKGKLIDEVSNTFGIRTCFSFAEKG
FELNGKAVKLLGTNRHQCYSGMGNALKDEMHRVDIELLHEMGGNFLRIAHPQDEMVLAAACNRLGIVTSV
EIPVINAITMSQNFSDNCVEMMKEMIYQCFNSPSVCIWTYMNIEIMLRPPYNSEPTIKKDEYLYKYLHIAE
RIENTAKEIDPQRATMLPCHGNLAVYEEANLVNLPDILGMNLYNGWYGGQFSGFEAALDKIHQKYPDKPI
IVSEYGADVDAIRHSFKPERFDFSVDYGVMYHRRHYLPQIIQRKFVVGSAVWNLNDFHSEERMDAIPHINC
KGLVTLDRTPKDTYRYYKAMLGKEPYVAIGGSDWYNRSGVSDNSGVCYYPVSVYTNLKQITLKVNGKKLA
VSEVENGAAQFNVPFVNGINQLEAEGIFQNRYSIKDFCRVSMKLVPGSFDNKESTELNMLFGTARHFEDK
TAALAWQPEREYTSGSWGYVGGKAFRARTKRGSQPASNLDIYGTEQDPIYQTQRRGLDSFKADLPDGKYV
VYLYWAEELSGANTVDLAYQLDNIVQKEKATERLFHVDINGIRVLSLLDVAEEVGVRRPMICKIPVTISNK
KGLTIDFVPIKGETMVTAIRILKLD

>SRS045004.1734-L2 *Bacteroides stercoris*

QQRVVYTINDGWRFAKGSPEAHLTSCDDSSWETVHIPHTWNAKDADDDQAGFYRGPAYWRKRLFVDKSQ
EGRRAVIYFEGANQEVQVYLNQYLGAKHGGYTRFCMDVTSHLCYQENLLAVCVNNVYNPNIPPLSADF
TFFGGIYRDVYLQFMNPVHLATNDYASSGVYIRTPEVSKTTASIEVTALLTNELDKPQEVIVENVICDAS
GKEVKKHQMTCCLLAGETKTTVAKKIKMESPRLDIDDPYRYLVYTRIFDKKGALLDEVVNPLGFRWFAF
DSEKGFFLNGKHKRLVGTARHQDYFQQGNALRDELHVQDVLLKEMGGNFLRVSHYPQDPVIMEMCDKLG
IVTSVEIPLINAVTETEEFLTNSVEMAKEMVRQDFNRPSVMIWGYMNEIFLRRPYTEGKQLKDYHYTEK
VARELEATIRQEDPSRYTLMAYHNAPQFYEEAHLTEIPMILGWNLYQGWYEPDINKFQCLLDRAHQVYKG
KILMVTEYGPVDPRLHSYQSERFDFSQEYGLNYHKHYIAEMMKRPFVAGSSLWNLNDFYSESRIIDAVPH
VNNKGVVGLNREKKDITYLFYQAALCKRPVLAIGNREWKARGGVVNTSERACKQAVPVFSNAKEVELLING
RSLGVKAVENGSAIFEVFPVNGLNQLKATAVVGNSVVEDWLEVCFQLVAADLKEHGADFTMNIMLGSPR
YFEDRTAQVAWIPEQEYKPGSWGFGVGGTPYRRKTNFGFMLGSDIDIFGTDMPNIPFQTQRVGIQSFKADV
DGEYSVYLYWAEELSDNKEKEALVYNLGADYGLIAGNRSFGVSINGMQVLDDFNIARDYGYARAVVKKFI
VTVKNGEGLCVDFHKKEGEPVLNAIRLYRNY

>SRS022609.816-L2 *Bacteroidetes*

QRIVYTINDGWRFFQKGSRFEAQKQTCDDSLWEVVSIPHTWNEKDADDEVAGFYRGPAYWRKQLYIDKSQE
GRRAIYFEGANQEVQLYLNEQFVGEHKGGYTRFCFDITSKLRYGQENLLAVCVNNAYNPNIPPLSADFT
FFGGIYRDVYLQFMDPVHIATSDYASSGVYLRTPENVKTAASVEITTLTNELPQTTEVI IENI IYDADG
KEIKKTNMDVRLVSGETRANLSKKLKIDSPHLWDIDDPYRMYVYTRIFDKGKSVLLDEVITPLGFRWFEF
DPEKGFFLNGKWRKLMGTARHQDYFQKGNALRDELHVRDVLLKEMGGNFLRVSHYPQDPVIMEMCDKLG
IVTSVEIPVNAVTEETEEFLTNSVEMAKEMVRQDFNRPSVMIWGYMNEIFLRRPYTEGKQLEDYHYTEK
VARTLEATIRKEDPSRYTMMAYHNAPRFYEAHLTEIPMIQGWNLQGWYEPDINEFQRLLDRAHKMYKG
KVLIVTEYGPVDPVHSYQPERFDFSQEYGLIYHKHYLDEMMMKRPFVAGSSLWNLNDFYSES RVDAVPH
VNNKGILGLDREVKDTYLFYKTALIRRPILAIGNREWKARGGVVNTQNECVQSVPVFSNAEEVELFVNN
KSLGKKKTESGYALFVFPVNGENQLEALAVAGDTQLHDMRLIHFQLVSSQLKDKTLPFTEINVMLGSPR
YFEDRVANVAWIPEQEYKPGSWGFGVGGTPYRRKTGFGTMLGSDIDICGTDMPNPFQTQRVGIESFKADV
DGEYSVSLYWAEELSDKKKEALIYNLGADSEQTLVGNRSGISINGITVVDNFNVDYGYARAVIKKFV
VTVKDGKGLSIDFHKKEGETILNAIRIYRNY

>SRS017433.56230-L2 *Tannerella* sp. CAG:51

GVRTVYTINDRWKFCPGSPFDAQNIGADDSSWETVNIPTWNNLDGDDEIPGYRGPAYWRKSIFVDES
ENQSAVIVFEGANQVVELYVNGLYVGKHIGGYTRFNFDISRLNYGKKNVFAIKVDNGYDPNIPPLSADF
TFYGGIYRDVYLMFVDKVHLATNDYASSGVYICTPEVTEKKAVVEITSLVSNDDTSPVEI ILENKICDAT
GKEVKSRLTNLKIARTTETGIQKKIIIDNPLLWDIDNPHRYQVYTRVYDKKGDVLYDEVSNPLGLRWFE
FDPERGFFLNGKPRKLVGNRRHQDYLAAGNALRDEMHRVDILLMKQMGSNFLRVSHYPQDPVIMEMCDKL
GIVTSVEIPVNAVTDTPFEFLNSVNMVKEMIRQDFNRPSVMIWGYMNEILLRRPRTDEDFLKSYPVVE
KTARTLENTIRKEDPSRYTMMAYHNAPDLYERAHLTEIPMIMGNLYQGWYEPDIDEFQRLLDRAHDKYK

GKVLIVTEYGPVDPRLHSYKPERFDFSQEYGIYHRHYLREMMARPFIAAGFTMWNLNDFYSESRIDAVP
HVNNKGVGTGLDRERKDTYLFYEASLGNDVLRIGNREWKNRGGVVNTVENTCVQSVPVFSDYPVVELYVN
GKSIGSKKCEENIAFFDVFPVNGENVVEAVARKNGKELRDMIRIDFKLIPDDLKNGDVPFTEMNVMLGSP
RYFEDRTAEMIWIPEQPYKPGSWG YIGGEPYRRKTGFGSMLGSDIDIFGTDNNPIFQTQRTGLEAFKADV
PDGEYSVYLYWAEELDTDKKKESLVYNLGADTENGGSYKRTFGLEINGNPVMDKVDIAEDYGFGCAVIQKF
IVNVNDGKGLTVGFKRIEGEPILNAIRIYKNY

>SRS050422.73127-L2 Bacteroidetes

QQRVVYTINDGWKFTKGSPFEAQLNNCDDSSWETVNI PHTWNDKDADDETPGFYRGPWYRRQLFVDKSQ
EGRQAVIYFEGANQEVQLYLNGQFVGEHKGGYTRFCFDITSKLRYGQENLFAICVNNNTYNPNIPPLSADF
TFFGGIYRDVYLQFMNPVHIATNDYASSGVYIRTPEVNGSSASVEITLTLTNNTLRSAEVIVESLICDAD
GKEVKRTQADIKLVSGETKADISKKIKIDSPRLWDIDDPYRYMVYTRIFDKKKGALLDEVVNPLGLRWFE
FDSEKGGFFLNGKWRKLIGTARHQDYFQKGNALRDELHVQDVLMLKEMGGNFLRVSHYPQDPVIMEMCDKL
GIVTSVEIPVNAVTEETEEFLTNSVEMAKEMVRQDFNRPSVMIWGYMNEIFLRRPYTEGKQLEDYYHFTE
KVARTLEATIRKEDPSRYTMMAYHNAPRFYEAHLTEIPMIQGWNLQGWYEPDINEFQRLLDRAHKMYK
GKVLIVTEYGPIDPRVHSYQPERFDFSQEYGLIYHKHYLDEM MKRPFVAGSSLWNLNDFYSES RVDAVP
HVNNKGILGLDREVKDTYLFYKTALTRRPILAIGNREWKARGGVNTEQNECVQSVPVFSNAEEVELFVN
DKSLGKKKTENGYALFDVPFVNGENQLEALAVAGDTKLHMLRIHFQLVSSQLKDKTLPFTEINVMLGSP
RYFEDRVANVAWIPEQYKPGSWG FVGGTPYRRKTGFGTMLGSDIDICGTDMPVVFQTQRVGIESFKADV
PDGEYSVSLYWAEELSDKKKEALIYNLGADSEQTLVGNRSFGISINGITVVDFFNVARDYGYARAAIKKF
VTVKDGKGLSIDFHKKEGETILNAIRIYRNY

>SRS015217.11766-L2 Bacteroides ovatus

QQRVVYTINDGWKFTKGSPFEAQLAGCDDSSWETVNI PHTWNNKDADDEAPGFYRGPWYRKQLFVDKSQ
EGRQAVIYFEGANQEVRFYLNQFVGEHKGGYTRFCFDITSHLRYGQENLFAIYVNNVYNPNIPPLSADF
TFFGGIYRDVYLQFMNPVHIATNDYASSGVYIRTPEVNNSAASVEITLTLTNNTLQPAEIRVENVICDAD
GKEVKKTHAEIKLASGETKTDISKIKIDSPRLWDIDDPYRYMVYTRILDKKKGTLTDEVVNPLGLRWFK
FDSEKGGFFLNGKGRKLIGTARHQDYFQKGNALRDELHIQDVL LKEMGGNFLRVSHYPQDPVIMEMCDKL
GIVTSVEIPVNAVTEETEEFLQNSVEMAKEMVRQDFNRPSVMIWGYMNEIFLRRPYIEGKQLEDYYRFTE
KVARALEATIREEDPSRYTMMAYHNMPQYYEDAHLTEIPMIQGWNLQGWYEPDINEFQRLLDRAHKVYR
GKVL MVTEYGPVDPGLHSYQPERFDFSQEYGLVYHKHYLREMMKRPFIAGSSLWNLNDFYSES RVDAVP
HVNNKGVVGLDREKKDVYWFYKTALSRRPILVIGNREWKSRGGVVNTAQKECIQSVPVFSNAEEVELFVN
NKSLGKKKVEDNYALFDVPFVSGENLLEAVAVAGDSKLRDMLRIQFQLVGSQ LKDEAIPFTEINVMLGSP
RYFEDRTANVAWIPEQYKPGSWG FVGGTSYRRKTGFGSMLGSDIDILGTDMPNPIFQTQRVGIKSFKADV
PNGEYSVYLYWAELES DKEREALVYNLGADSEQTFAGNRSFGISMGNTTVLDDFN IARDYGYARAVIKKF
VTVKDGKGLSVDFHKKEGEPILNAIRIYRNY

>SRS016495.13585-L2 Bacteroides multispecies

QQRVVYTINDGWKFTKGSPFEAQLTGCDSSWETVNI PHTWNDKDADDETPGFYRGPVWYRKQLFIDKSQ
EGRRAVIYFEGANQEVRFYLNQFVGEHKGGYTRFCFDITPHLRYGQENLFAIYVNNVYNPNIPPLSADF
TFFGGIYRDVYLQFMNPVHIAANDYASSGVYIRTPEVNNSAASVEITLTLTNNDMSQPT EIRVENIICDAD
GKEVKKTQAEVKLAAGETKTDISKIKIDSPRLWDIDDPYRYMVYTRILDKRKGTLLDEVVNPLGLRWFK
FDSEKGGFFLNGKGRKLIGTARHQDYFQKGNALRDELHVQDVL LKEMGGNYLRVSHYPQDPVIMEMCDKL
GIVTSVEIPVNAVTEETEEFLHNSVEMAKEMVRQDFNRPSVMIWGYMNEIFLRRPYTEGKQLEDYYRFTE
KVARALEATIREEDPSRYTMMAYHNMPQYYEDAHLTEIPMIQGWNLQGWYEPDINEFQRLLDRAHKAYK
GKVL MVTEYGPVDPVHSYQPERFDFSQEYGLVYHKHYLNEMMKRPFIA GSSLWNLNDFYSES RVDAVP
HVNNKGVVGLNREKKDVYWFYKTALSRRPILVIGNREWKSRGGVVNTAQKECIQSVPVFSNAEEVELFVN
NKSLGKKKIENNYALFDVPFVGGENLLEAVAVTGDNKL RDMMLRIQFQLVGSQ LKDEAVPFTELNVMLGSP
RYFEDRAANVAWIPEQYKPGSWG FIGGTSYRRQTGFGTMLGSDIDIHGTDMPNPIFQTQRVGIKSFKADV
PNGEYSVYLYWAELES DKEREALVYNLGADSEQTFAGNRSFGISINGTTVSDDFN VARDYGYARAVIKKF
VITVKDGKGVSVDFHKKEGEPILNAIRIYRNY

>SRS018575.34501-L2 Bacteroides clarus

TENNPRITYSINDSWQFRLGTS DIGKEGWSTVSI PHTWNAEDCVDDFPGFYHGTGWYRKNIQVDSAMLSR
PLYIFFEGANQRTELFVNRHRVGEHIGGYTFFCFDITPYIREGENQLVVCVDNSYDSEIPPLSADFTFFG
GIYRDVSLISVPIHISTTHYASSGVYVHTPEITEEKALVDICTMLSNTQKVRSKIFVEHKIYSATGECV
AASRKKVAVDTEREQAFNTQIVVNHPELWEIGRP NLYRLQTYLWDNEGKLLDEVSNFTFGIRTCFSFAEKG
FELNGKAVKLLGTNRHQCHSGMGNALKDEMHVRDIELLHEMGGNFLRIAHPQDEMVL AACNRLGIVTSV
EIPVINAITMNQNFSDNCVEMMKEMIYQCFNSPSVCIW TYMNEIMLRPPYNSEPSIKKDEYLYKHIAE

RIENTAKEIDPGRATMLPCHGNLAAYEEANLINLSDILGMNLYNGWYGGDFSGFEATLDKIHQKYPDKPI
IVSEYGADV DARIHSFKPERFDFSV DYGVMYHRHYLPQIMQRKFVVGSAVWNLNDFHSEERMDAIPHINC
KGLVTLDRVPKDTYRYYRAMLNKEPYVAIGGSDWYNRSGVADNSGVCYYPVSVYTNLKQVSLKVNGKMLA
VSEVQNGAAQFNVPFIDGINQLEAEGTFQGRYIKDFCRVSMKLVPGSFDNKESTELNMLFGTARHFEDK
TASLAWQPEREYVSGGWGYVGGKAFRARTKRGSQPASNLDIYGTEQDPIFQTQRRGLNSFKADLPDGRYV
VYLYWAELSGADVVDLAYQLDNIVQKEKATERLFHVDINGVRVLSLSDVAKEVGVR RPMICKIPVTISDK
KGLTVDFIPIKGETMITAIRILKMD

>SRS019582.31661-L2 *Bacteroides uniformis*

QREIQTINDSWKFRKGECTAALDSTFKDDEWEDIHLPHTWNTDAYTEKDYYRGTGWYRRQIILPQDWIGK
QVFLKLDAASKATCIYINGKKVGEHTGGYTASTLNITPFLSFNSPNMLVIRVDNARMDIPPISGDFTFFG
GIYRDAWLIAPVPSQHFNLTNHGS DGI FITTPRVSEQQATLSIHGEIKNDAREKAALELVHSIYHPDGTLL
QTQKQFIQLKAGETRNFDSNISPIQLPLLWTPETPYLYRVETTLEFNCKTKSIVDQSNHHTGFRWFHFESN
KGFYLNKGPKYKLRGICRHQDQKPIGVALTDEMHRDMKMLKEMGANFIRISHYPQDDAI IEMCDKLGMLA
WEEIPVIDIVPDTPGYGDNCERSLREMIRQHYNHPSIIWGYMNEILLVTQRKYKTENELKPVIER TLAL
ANRLEQVLKEEDSTRISAMAFHGSNSYNETGLSEITDIIGWNLYQG WYGGDVTGF EKFLAEQHRNYPTHP
MIVSEYGAGSDKRLHSLNPRAFDFSIEYQQKFLEHYLPILENTPYVCGGTHWNFIDFSSALRDESMPRIN
NKGLAYS DRTPKDV FYYYYKAAWRKDIPVLHIASRDWIYRTGIQQGDSSVLLPVKIYTNLPEVELSIDGKS
LGRQQVDNYTAIFKAPFSSKEPLLLVQGN YQGT TVQDGIKVHFTPIPTNLNSTGLKDLELAVNVGSQCFY
TSDESRLTWLPDQPYTKGSWGYIGGKELSTQTEIRRTADGPLFQTLRNGIEGYRFDVPRGVYEVELLFTD
IFLQNEGIAYQLGREGEQKSRENTFGISVNGKMLEEKLS PCKESGYCQAMRKKYIITNDTDHIDIRFHFA
SGTCFLNGIKLRNIH

>SRS045713.76924-L2 *Bacteroides cellulosilyticus* CAG:158

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QVFLKLDAASKAATIYMN GRKIGEHN GGYTACTFDITPFCSF DAPNSLAIHVDNARQDIPPISADFTFFG
GIYRDVWLTA VPKQH FHLTNYGSEGIFISTPQVSEEKGTILIRGEIKNDAEQKASLELEHIIYNPDGSIA
QTQKQSIQIKGGELYSFR TETAPILSPRLWAPEAPHYLRVESILRDKKT KAILDHSNHHTGFRWFSFDGK
TGFSLNGKPKYKLRGICRHQDQKPIGVALTDEMHRDMKMLKEMGANFIRISHYPQDDALLEMC DKLGMLA
WEEIPIIDIVPDTPGYAENCENNLREMIRQHYNHPSIITWGYMNEILLVTQRRYKKEELKPV LERTLSL
ANRLEKVLKEEDSTRVSTMAFHGSNSYNEVGLGNITDIVGNLYQGWYGGDLTGFERFLTEQHKKYPSHP
MIVSEYGAGSDKRLHSLQPHAFDFSIEYQQKYLEHYLPVLEETPYICGGTHWNFIDFSSALRDESMPRIN
NKGLVYS DRSPKDV FYYYYKAAWRQDIPVLHIASRDWTHRSGVQH GKAPVPLPVKVYTNLPEVELFIDGKS
LGKQKTENYTVFQVPF SRKKHFISVQAENKESDQSI SMIEDALHINF TPI PANLNETNLRLNLELAVNVG
SNCFYTSDESQLTWLPDQPYTESWGYIGGESKNSQTQVENTHDGPLFQTLRNEIEGYCFDVPNGVYEVE
FLFTDIFRENAATVYQLGRESGVESRENIFDIVINGESVEENFSPCRESGYFHALRKRYNITNFRNKIEI
RFPVRNGKSFLNGIKLRKLY

>SRS015217.202723-L2 *Dorea longecatina*

DEITKENQQDQEISNDQNNTKDGENSNTNDGKETENDLIQGNDS DSSQEKAEENITEEESKEDNMEAQQE
EPSEETDSEEAVTQRKVLNINDGWNFSTNDTSTAGWNFP SGAGNGVVNLPHSWEYVHPRQSYIPQMNSKT
VTYEKTVNISEIKNKNLFIKFYGAARNT EVFIDGEKV GTHVGGYSAFVFDITNYVQ GKDEITIKANVTNI
DTVSIPIINVDTYQWGGIYRDVELISTDDQYISLEDYGNKG IYIDSNVNGTTADVNVKNEISNKSND DKEL
KIVTDIYDADGNKVTGT EQKV TAKANKNVQPYSTKCQIDNVHLWNGTKDPYLYTANVTVYNENDEVLDTV
SDNFGVRTYEIKNGKFYLNQGEYEIHGVGMHQDREGYGN AVPDDLKVQDMNLMQEMGVNAIRTSHYPHSQ
STYNLADERGMLVYCEIPYYLLLSNAESYKTSIKEELKEMIRQGYNHPSIMMWGIENEVYQPASAAAF GK
DFQINENTLVSFNSSVAKLAQKEDTTRYIVQAQIDSSNANKVCAKWSKNGNVDYTG VNLVVGFKSSVSSA
DDEGRKEITDTLNRKLNEYKQTYNASSMMITEYGAGANINQHATMDENFSWSSDDASKDKHYEEYQAFVL
ETYYSLIQKRKDIPVSFVWNMFDFSCYRNEGGIPRRNTKGLVCYDHTTKKDAFYFYKANWNQQDKFVYLT
SKRYTERD TKYQQIKVYSNCEGV L FVNGQSVGKGRKQQSGVFVWDDVKLDDMNTIKAVGTS DGQTY SDE
VSGIKAENKNSATS IKYQTHIQSIGWQEAKKDGQTS GTIGKGLRMEALKVKLQNKKYGGNIEYRTHIAQK
GWQDWKKNQTAGTTGEKLAMEAVRLKLTGELAEHYDIYRVHSQS YGWLGWAKNGE IAGTAGLAKRMEA
IQIKLVEKGKAPGTSEKHYSNQG VFYQSHVQTYGWQ TWKQNGETSGTSGQAKRLEAIKIKLQKMKVSG
NIEYQSHVQTYGWEKS WKKNQQLSGTSGKAKRLEAVKIRLTGEMKNKYDVYYRVHAQSYGWLGWAKNGEK
AGTEGLAKRLEAVQIVLVEKGGAAPGST EKTFIKK

>SRS049959.11650-L2 *Bacteroidetes* unconfirmed sequence

MGNYAQWQKIDIPHTWNAEDTDDETPGYFRGKCMYRKSVTIPASKSKERVFIYFEGANQETT VWVNGREA
GNHRGGYAGFGFDITEYVKFGQENQLELVVTNAYDKTIAPLSADFTFYGGVYRDVWISYLPQVHFSTRYY

GSSGIVVTTPHKDG NFRQVEVLCRLQNHEKKAKAITLKSSLVDAEGKIVATVTKRLKLKVGEDDQLCQLS
IPVMNPHLWSDVDPYIYKV NTEIVDNQTRKVYDENHTSLGFRWYRFD PQKGFFLNGKHVKLIGTSR HQDY
LKKGNALSDNIHLADVYK LKEMGGNFLRVAHY PQDPTILEVCDRLGILTSVEIPVVNAVEGSDEFLETCK
DMQMEMIYQNRNHPSVVMWGLMNEILLRIPAQYDKDRATYYPMVRRVASELDALSRRREDPERYTMMAVHN
AFERYQEAGLVNIPQIFALNLYQGWYEPDIREFEQILDKVHEALPNQSLMVTEYGAGIDPRLHSFHPERF
DFSEYGLKYHVHYLREMKKRDFVAGSSVWNLADFYSEVRGDAVPHVNSKGILGLDRCEKDAYLYYKSML
GEKPSLYIGGKNWKYRSCVSRTAEARMDVPV FVKADKVRVYCNQQLVGTFATTDGVAMASVPFTDGENRV
EAFAEVDGEKVSDAIVNM RVVPASFEGFVPVTGLHVTCGSQRYMEDKEESLCWMPEKAYEQGGWGYVGG
TVYRRAGDLLGTDADILGTDKDPIYQTQRQNI EAFKADVPDGEYIITLHFASLKEAAALVYNLSAHGADK
KDDTASVFDVVVNGEKVLEQFNAADYGVSR AVAKRIHVQAKQGQGLDVRFNPIKGKTMLNAIEIYKR

>SRS019267.93486-L2 Bacteroidetes fragment

LMGVSRHQDQYPLGIALTDEMHR RDVKMIKEMGANFLRLAHYPQDEAVLRACDELGLLVWEEVSLVDLIA
LGDEFAANSQTALVEMIRQHYNHPSVAMWGYMNEVLIQLQYRVKKNRLNVFYENTLRLARVLEETLKTED
PYRLSTMAFHGDQIYNRVGLSNITDVAGWNLYQGWYGNLNGFERFLDEEHRKYPARSLLVSEFGAGSDR
RLQSLKPEIFDFSMQWQQT FLEHYLPVIMERPYVIGGAENFIDFSSASRQESMPHINNKG L VYADRTPK
DVYYYFQAF LRKDI PVLHAAVSDWSRRTVVS DGSPVHPLKVYSNFEKVSLSVNGRPYGEKTMENCAVW
NVELQAGRN VVKASGEFLGKASEQVSEIYVDMVPEHLTEENVSGVELAVNVGSNCYFTDALSGLCWLPDQ
AYQPGSWG YVGGEVFHKSPGRIGTTAEVKGTHNNPLLQTMRS GIEGYRFDVPDGTYELELQFADLTAPSR
RLAYDLGNRNESEQKDANVFDVTVNGTVWLSEFSPSELVGGNRALVKKQLVTVTGGTIDVRF AVRKGQTF
LNAIKLHKLD

>SRS011271.20274-L2 Bacteroidetes fragment

MNCLxxxxxxxxxxxxxSLVDTKRKAID EK FHKVGFRWFSFDGEKGFSLNGKPYKL RGFNRHQDQAPV
GVAVPDEVHRRDIKIMKDLGSNI RISHYPQDDALLEACDELGLLAWEEIPIVNI VPDTPGYADNCEHNL
REMIRQHYSHPSIINWGYMNEILLVTPRPGTK EWPVYKERTVALAKRLEKVLKEEDATRKSV MAYNMTNV
YNEIGLNLVDVVGWNLYHGWYEGELNGFNRCENQHQLY PKQPMIISEWGAGSDRRLHSYQARPFDFSIE
YQQTYIEHYLPFIEEKPWISGCTYWNFIDFNVAARQESMPRVNNGVAYNDR TLKDVAYYFKSMWRKDVP
VVHIASRDWPARMGNANDIQRIKVYSNMQEVELFVNGKSVGKKKTDNCF AVFDAVLPYGISTLKAKGMSE
KWFADDFRVKGGNTEDAMNIRY TPLPD LAKGEELAINVGSNCYFISSLSHLTWLPDQAYKSGTWGYIGGE
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>SRS017433.24596-L2 Bacteroidetes

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>SRS056259.118375-L2 Bacteroidetes

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>SRS024435.10769-L2 Bacteroidetes

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>SRS015217.56118-L2 Bacteroidetes

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>SRS047014.55007-L2 unclear

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>SRS062427.9527-L2 unclear

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>SRS017433.55730-L2 unclear

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>SRS048164.208543-L2 unclear

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>SRS020233.333911-NC Bacteroidetes fragment

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>SRS053335.69470-NC Firmicutes fragment

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>SRS020328.178587-NC unclear fragment

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>SRS045713.32377-NC unclear fragment

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>SRS024435.197134-NC Bacteroidetes fragment

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>SRS023176.66822-NC Firmicutes fragment
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